# USB4

**Universal Serial Bus 4** (**USB4**), sometimes erroneously referred to as **USB 4.0**, is the most recent technical specification of the USB (Universal Serial Bus) data communication standard. The USB Implementers Forum announced USB4 in 2019.

USB4 enables multiple devices to dynamically share a single high-speed data link. USB4 defines signaling rates of 20 Gbit/s, 40 Gbit/s and 80 Gbit/s.[1][2] USB4 is only defined for USB-C connectors, and its Type-C specification[3] regulates the connector, cables, and basic power delivery features across all uses of USB-C cables, enhanced[4] with the USB Power Delivery specification.[5]

The USB4 standard mandates "functionality backward compatibility" to USB 3.x and dedicated backward compatibility with USB 2.0.[6] The dynamic sharing of bandwidth of a USB4 connection is achieved by encapsulating multiple virtual connections ("tunnels") of other protocols, such as USB 3.x, HDMI, DisplayPort and PCI Express.

USB4 is based on the Thunderbolt 3 protocol. However, it is different enough that backward compatibility to Thunderbolt 3 is optional for many device types.[7]

**USB4**

| | |
|---|---|
| **Type** | USB |
| **Production history** | |
| **Designer** | USB Promoter Group |
| **Designed** | 29 August 2019 |
| **Superseded** | USB 3.2 |
| **General specifications** | |
| **Daisy chain** | No |
| **Audio signal** | DisplayPort |
| **Video signal** | DisplayPort |
| **Pins** | 24 |
| **Connector** | USB-C |
| **Electrical** | |
| **Max. voltage** | 48 V (PD 3.1) |
| **Max. current** | 5 A (PD) |
| **Data** | |
| **Data signal** | USB or PCIe |
| **Bitrate** | 20 Gbit/s<br>40 Gbit/s<br>80 Gbit/s<br>120/40 Gbit/s asymmetric |

## History

Prior to USB4, Thunderbolt provided a way to dynamically share bandwidth between multiple DisplayPort and PCIe connections over a single cable. Thunderbolt originally used the Mini DisplayPort connector and was only backward compatible to DisplayPort connections and did not support power transfer.

The introduction of the USB Type-C connector in 2014 provided a connector that could support USB data connectivity and power transfer as well as DP connections. It also allowed the static sharing of bandwidth between DP and USB connections over the same cable.

Thunderbolt 3 switched over to using the new Type-C connector and also added backward compatibility for USB connections and power transfer features.

### USB4 Version 1.0

USB4 was announced in March 2019 by the USB Promoter Group.[8][9] The version 1.0 of the USB4 specification, released 29 August 2019, is titled "Universal Serial Bus 4 (USB4™)". Several news reports before the release of that version sometimes use the wrong terminology "USB 4.0" and "USB 4".[10][11]


USB 40Gbps, 100 W cable

USB 40Gbps port logo

In the announcement press release, the USB Promoter Group mentions that USB4 is "based on the Thunderbolt™ protocol specification recently contributed by Intel Corporation".[12] Goals stated in the USB4 specification are increasing bandwidth, helping to converge the USB-C connector ecosystem, and "minimize end-user confusion". Some of the key areas to achieve this are using a single USB-C connector type, to offer display and data transfer features, while retaining "compatibility with existing and Thunderbolt products".[13]

Version 1.0 defined 20 Gbit/s and 40 Gbit/s connections, the required support of USB 2.0 and USB 3.x connections at up to 10 Gbit/s with support for tunneling connections according to the PCIe 4.0, USB 3.2, and DP 1.4a specifications. Optional backward compatibility to Thunderbolt 3 as well as Host-to-Host networking were also defined. Compared to Thunderbolt 3, USB4 changed the raw bit rates slightly to bring them in line with other USB specifications, where the nominal bit rate matches the raw bit rate. USB4 also added support for USB3 tunnels and use of the USB2 wires for improved backward compatibility with previous USB standards and to allow for simpler USB4 devices without support for PCIe. USB4 also added support for hub topologies compared to Thunderbolt's previous restriction to daisy-chaining topology.

In July 2020 Intel announced Thunderbolt 4 as an implementation of USB4 40 Gbit/s with additional requirements, such as mandatory backward compatibility to Thunderbolt 3 and requirement for smaller notebooks to support being charged over Thunderbolt 4 ports.[14] Publications such as AnandTech described Thunderbolt 4 as "superset of TB3 and USB4" and "able to accept TB4, TB3, USB4, and USB 3/2/1 connections". Intel itself describes Thunderbolt 4 as "delivering increased minimum performance requirements, expanded capabilities and USB4 specification compliance" and as building "on the innovation of Thunderbolt 3".[15]

### USB4 Version 2.0

On 18 October 2022 the USB Promoter Group released the USB4 Version 2.0 specification.[16][17]

It added a new transmission speed that allows 80 Gbit/s symmetric connections or asymmetric connections supporting 120 Gbit/s in one direction and 40 Gbit/s in the other. The new PAM3 encoding scheme enables this over existing, passive USB 40Gbps cables. Active cables are not forward compatible in the same way, instead a new speed grade of active cables was added. It also upgraded the support of DP tunnels to DP 2.1, allowing the tunneling of DP connections with up to 80 Gbit/s (UHBR20). It also added a replacement of the previous tunneling of classic USB 3.2 connection speeds with "USB3 Gen T tunneling", which can exceed 20 Gbit/s and also removed PCIe overhead limitations.

Around the release of the new USB4 2.0 specification, USB-IF also mandated new logos and marketing names to simplify representing the maximum supported bit rates and wattages to consumers.[18]

In September 2023, Intel announced the launch of Thunderbolt 5 as an implementation of USB4, using the new abilities of 80 Gbit/s connections and updated DP support[19] Intel's own press release describes it as "built on industry standards – including USB4 V2".[20]

USB 80Gbps port logo

## Functionality of USB4 ports

USB4 ports offer multiple different connection types. Some of them can coexist and work independently or in parallel with each other. This is further complicated, by USB4 connections themselves only being a container for other connections. Some connection standards are possible, both directly from the port as well as a contained within a USB4 connection. It is even possible for the functionality provided natively and within a USB4 connection to differ.

This section describes the possible non-USB4 connections, that mostly exist for backwards compatibility. For the USB4 connection and its contents refer to USB4 protocol/connections.

The USB4 specification speaks of downstream facing ports (DFP) and upstream facing ports (UFP) rather than host and peripheral ports. Downstream facing ports includes host ports as well as any "outputs" of a USB4 hub, while upstream facing ports include anything that is connectable to a downstream facing port, like the ports of peripherals or the "input" port of a USB4 hub.[21]

Any USB4 DFP port is required to also implement USB 2.0, USB 3.2 and DP Alternative Mode support, each according to their own specifications. As such, a USB4 DFP is backward compatible to all previous USB standards and DP output.[22]

### USB 2.0 DFP features

USB 2.0 defines 3 different bit rates (Low-, Full-, High-Speed), all are required to be supported.[23] USB 2.0 abilities use separate wires and pins on the Type-C connector that are not used by USB 3.2 or USB4 or most other Alt modes.

### USB 3.x DFP features

USB 3.2, the current version, defines 3 different bit rates 5 Gbit/s, 10 Gbit/s and 20 Gbit/s. While the USB 3.2 specification[24] has been referenced by USB4 from the start, only the two lower speeds (5 Gbit/s, 10 Gbit/s) are mandatory for USB4 DFPs to support. As with any previous USB 3.x DFP, the USB 2.0 connection remains available in parallel to it.

### DP Alt Mode DFP features

The USB4 specifications make no reference to a minimum speed or features for its DP Alternative Mode functionality, but Thunderbolt 3 does. In practice, Intel's family of TB 3 controllers requires at least DisplayPort 1.2 at HBR2 speeds to support 4K60 output, but is also available with up to HBR3 speeds according to the DisplayPort 1.4a specification.[25] DP Alt Mode can always coexist with USB 2.0 connections. And it can be used in a halved mode, where half of the other high speed wires remain available for 5 or 10 Gbit/s USB 3.x connections.

### Thunderbolt 3 Alt Mode DFP/UFP features

While a very common feature, it is optional for most USB4 ports to support the TB3 connections (at 20 or 40 Gbit/s). While those TB3 connections are technically an Alt Mode, similar to DP Alt mode, the principles of it are closest to USB4 connections. See Thunderbolt 3 Compatibility for more details. It is possible for USB4 devices to support TB3 connections (to other USB4 devices or to TB3 peripherals), but not support being recognized by TB3 hosts.

### Power transfer features for DFP

The USB4 specification makes no explicit demands on power output. It outsources all requirements in terms of power to the Type-C[26] specification that underpins all USB, DP and other standards that use the USB-C connector. This requires a USB4 DFP to supply at least 7.5W Type-C current. No power consumption features (e.g., charging of a notebook) are required, but can be supported following the USB PD specification,[5] as well as supplying considerably more power. The USB PD protocol must always have support for exchanging data according to the protocol. This is separate from any functionality of PD to negotiate actual power delivery other than 5V or >15W.

### USB4 hubs & docks

USB4 hubs and docks are defined as their own category of USB4 devices that include further requirements. For example, a USB4 hub must also serve as a classic USB 3.2 hub with DP Alternative Mode passthrough with hosts that do not support USB4 connections. They are required to support Thunderbolt 3 connections. See USB4 capabilities by device type for more details.

## USB4 protocol/connections

Every USB4 port must support the USB4 protocol/connections, which is a distinct standard to establish USB4 links/connections between USB4 devices that exists in parallel to previous USB protocols. Unlike USB 2.0 and USB 3.x, it does not provide a way to transfer data directly, it is rather a mere vessel that can contain multiple virtual connections ("tunnels").

Other specifications are referenced to define the contents and internal functionality of a tunnel. USB4 defines the following tunnel types:

- USB3 connections
- DisplayPort connections
- PCIe connections
- Ethernet/network connections according to the included USB4Net and Cross-Domain specifications[27]

## General principles of USB4

USB4 forms a tree-like topology of USB4 routers, where each USB4 device includes a USB4 router to participate in this network. A tunnel can be end-to-end, where the route through the entire network of routers is preconfigured. But tunnels can also be single-hop, where it exists only for a single USB4 link (i.e., between 2 routers). In this case, the tunnel will be "unpacked" by the recipient and will use some other means specific to the tunnel type to identify where data needs to be sent next. If the next hop is another USB4 router, data will be ingested again into the next single-hop tunnel until it exits the USB4 network.[28]

Accordingly, single-hop tunnels require specific support in each USB4 router, just to support passing them through to further USB4 routers. However, end-to-end tunnels require support of a USB4 router only when the data is ingested into the tunnel and at the target, to the point where the tunnel ends.

Note: USB 2.0 is not part of any tunnel. That functionality is provided by maintaining a physically and logically separate USB 2.0 connection in parallel to a USB4 connection over the dedicated wires, just as it was handled with USB 3.x and its backward compatibility to USB 2.0.

### Protocol input/output adapters

A Protocol Input Adapter will ingest a connection according to whatever protocol it is based on and convert the contents into a USB4 tunnel. Protocol Output Adapters do the reverse. They extract a tunnel from the USB4 network and if needed recreate a regular connection from the tunnel contents.

The conversion into a tunnel typically entails removing any Phy/Electrical layer and encoding of the underlying connection standard and potentially losslessly compresses the contents; for example, by leaving out empty filler data. A USB4 tunnel itself is virtual and doesn't need to conform to any fixed bandwidth or other limitations that stem from the Phy/Electrical layer of the underlying connection standard. But since most tunnel types will eventually be converted back to a regular, physical connection again, most of those physical limitations, like max. bandwidth, are still likely to apply in the end.

## USB3 Gen X tunneling

This is a single-hop tunnel that essentially can transport any Enhanced SuperSpeed connection according to the USB 3.2 specification. USB3 Gen X follows the Enhanced SuperSpeed Hub topology, where every USB4 router with more than one USB3 endpoint must include a USB3 hub as well. It is the default way USB3 connections through USB4 are made. Supporting it at 10 Gbit/s is mandatory on every USB4 DFP. The minimum supported speed for the USB3 connection being tunneled is 10 Gbit/s as every USB4 device already has to support this speed and USB3 hubs handle converting this to 5 Gbit/s devices that may be connected.

This means that a USB4 hub will share a single upstream USB3 connection and distribute its bandwidth across all its downstream facing ports that make use of USB3 connections.

## USB3 Gen T tunneling

This is an optional alternative to USB3 Gen X tunneling that was introduced in USB4 Version 2.0. It is an end-to-end variant of USB3 Gen X tunnel.

Through this, it eschews the need for USB3 hubs in every USB4 router that can and will limit the throughput. It allows multiple separate USB3 Gen T tunnels even over shared links. Since it is an end-to-end tunnel, every USB4 hub will support passing it through. USB3 Gen T is intended as exclusively virtual, there exists no physical equivalent for it. Thus, it can only be used inside of a USB4 controller. This allows it to leave the limitations to 10 or 20 Gbit/s connections of USB 3.2 behind, while reusing most of the other parts of the Enhanced SuperSpeed protocol.[29]

No known USB4 controller implements support for Gen T tunneling to date (August 2024).

## DP tunneling

DisplayPort is also tunneled as end-to-end connection. There can be multiple independent DP tunnels, but each will be delivered to a single protocol output adapter (at which point DisplayPort MST might be used to further split each connection up).

USB4 Version 1.0 only defines how to tunnel DP connections according to the DisplayPort 1.4a specification (up to HBR3 speeds). USB4 Version 2.0 updates this support to the full DisplayPort 2.1 specification (up to UHBR20 speeds). Notably, the USB4 specification explicitly carves out needing to support the UHBR13.5 DP speed, even if UHBR20 is supported. The DP specification is not public. It is unknown if it makes similar carve-outs.

DP tunneling has great understanding of the contents of DP connections, and will efficiently skip/transmit any filler data, reducing the actually utilized bandwidth of a DP tunnel. But since DP connections have real-time requirements, bandwidth must be reserved for them. USB4 mandates that in absence of any other information, the maximum possible bandwidth for the particular DP connection (DP lanes and speed) must be reserved. This reservation only applies to other real-time tunnels though. Reserved, but unused bandwidth can be used by non-real-time tunnels such as PCIe or USB3, but the reservation may still block other DP tunnels from being established.[30]

## PCIe tunneling

Similar to USB3 Gen X tunneling, PCIe tunneling uses single-hop tunnels, requiring PCIe switches in every USB4 router that supports PCIe tunneling. USB4 has, from the start, referenced the PCI Express Specification Revision 4 and with USB4 Version 2.0 added references to PCI Express Specification Revision 5.0.

PCIe tunneling has had a significant limitation in USB4 Version 1.0 and also Thunderbolt 3: PCIe has a variable Maximum Payload Size, which applies end-to-end to a transmission. If any one component or PCIe Switch has a limited MPS, all packets passing through must be limited accordingly. Because USB4 uses a payload of up to 256 Byte per USB4 packet and a PCIe tunnel packet contains further PCIe headers and meta data, the MPS for PCIe tunnels was limited to 128

Byte. This limitation can reduce the efficiency of the PCIe connection greatly for all devices and systems that would otherwise support 256 Byte or even larger MPS.

USB4 Version 2.0 removes this bottleneck (mandatory for all implementers), by defining how a larger PCIe packet can be split across multiple USB4 packets. Support for this new feature requires every USB4 component / controller involved in the PCIe tunnel to implement USB4 Version 2.0.[31]

## USB4 signaling modes

Signaling refers to the lowest layer of the OSI Model, also called physical layer or phy. USB4 connections can be expressed with consumer facing names that are also the basis for the official logos used on packaging and products. These are the "USB 20Gbps", "USB 40Gbps", "USB 80Gbps" labels and they do not explicitly indicate how the connection is achieved on the physical layer. There are also more technical names based on the implementation and use of the USB-C cables. These usually consist of a speed per wire-pair expressed as Gen 1/2/3/4 (5 Gbit/s, 10 Gbit/s, 20 Gbit/s, 40 Gbit/s respectively) and some further information on how many wire-pairs are used in which combination.

USB commonly defines a "Lane" as a (bidirectional) connection, which for all recent transmission modes consists of one sending and one receiving wire-pair. The "Gen AxB" notation refers to B Lanes of operation mode A. Since Gen 4 modes also introduced asymmetric connections with uneven numbers of wire-pairs dedicated to sending and receiving, the Lane-notation is no longer applicable.

The USB 3.x family has had the same technical notation retroactively added in the USB 3.1 and USB 3.2 specification versions. Though this shows common principles and the same generations refer to the same nominal speeds, "Gen A" does not have the same exact meaning in both USB 3.x and USB4 specifications. The overlap in naming mainly becomes relevant for cables as shown in Cable Compatibility, which is regulated by the Type-C specification shared across all users of Type-C connector.

Comparison of signaling modes

| USB family | Signaling mode name[a] | Introduced in | Encoding | Wire-pairs sending/receiving | Raw bit rate (Gbit/s) | | Net data rate[b] (Gbit/s) | USB-IF current marketing name[32] | Logo[32] |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | per wire-pair | total (per direction) | | | |
| USB 2.x | High-Speed | USB 2.0 | NRZI with bit stuffing | 1 (shared) | 0.480 (half-duplex) | 0.480 (half-duplex) | ? | Hi-Speed USB | |
| USB 3.x | Gen 1×1 | USB 3.0 | 8b/10b | 1/1 | 5 | 5 | 4 | USB 5Gbps | |
| | Gen 2×1[c] | USB 3.1 | 128b/132b | 1/1 | 10 | 10 | ~9.7 | USB 10Gbps | |
| | Gen 1×2 | USB 3.2 | 8b/10b | 2/2 | 5 | 10 | 8 | (fallback)[d] | |
| | Gen 2×2[c] | | 128b/132b | 2/2 | 10 | 20 | ~19.39 | USB 20Gbps | |
| USB4 | Gen 2×1[c] | USB4 v1.0 | 64b/66b[e] | 1/1 | 10 | 10 | ~9.697 | (transient/fallback)[f] | |
| | Gen 2×2[c] | | | 2/2 | 10 | 20 | ~19.39 | USB 20Gbps | |
| | Gen 3×1 | | 128b/132b[e] | 1/1 | 20 | 20 | ~19.39 | (transient/fallback)[f] | |
| | Gen 3×2 | | | 2/2 | 20 | 40 | ~38.79 | USB 40Gbps | |
| | Gen 4 symmetric | USB4 v2.0 | PAM-3[33] 11b/7t | 2/2 | ~40.58[g] | ~81.15 | ~80.46 | USB 80Gbps | |
| | Gen 4 asymmetric 3:1 | | | 3/1 | | 3×: ~121.725 1×: ~40.58 | 3×: ~120.69 1×: ~40.23 | —[h] | |
| | Gen 4 asymmetric 1:3 | | | 1/3 | | | | —[h] | |
| — | TB3 Gen 2×2 | — | 64b/66b | 2/2 | 10.3125 | 20.625 | 20 | — | |
| | TB3 Gen 3×2 | | 128b/132b | 2/2 | 20.625 | 41.25 | 40 | — | |

a. Names according to the newest specifications.
b. Total data rate (1 direction) with encoding overhead removed.
c. USB4 Gen 2 is different from USB3 Gen 2. They both signify the same signal rate of 10 Gbit/s, but use different encoding and differ on the electrical layer. They also have different requirements for signal quality.
d. USB3 Gen 1×2 connection requires both sides to be USB3 "20 Gbps" / Gen 2×2 capable, but fail to establish Gen 2 / 10 Gbit/s per wire-pair connections.
e. USB4 Gen 2 & 3 can use optional Reed–Solomon forward error correction (RS FEC). In this mode, 12 × 16 B (128 bit) symbols are assembled together with 2 B (12 bit + 4 bit reserved) synchronisation bits indicating the respective symbol types and 4 B of RS FEC to allow to correct up to 1 B of errors anywhere in the total 198 B block.
f. USB4 is required to support dual-lane modes, but it uses single-lane operations during initialization of a dual-lane link; single-lane link can also be used as a fallback mode in case of a lane bonding error.
g. Per spec, lines run at 25.6 GBaud. One symbol contains 1 trit of information. Encoding transforms each group of 11 bits into 7 trits. 7 trits give 2187 different values or $log_2 3 \approx 1.585$ bits/trit. USB4 Version 2.0 Specification 2023, p84, sec. 3.2
h. Optional features of USB 80Gbps connections and devices.

Thunderbolt 3 Gen 2 and Gen 3 and the USB4 Gen 2 and Gen 3 modes use very similar signaling. However, Thunderbolt 3 runs at slightly higher speeds, called legacy speeds, compared to rounded speeds of USB4.[34] It is driven slightly faster at 10.3125 Gbit/s (for Gen 2) and 20.625 Gbit/s (for Gen 3), as required by Thunderbolt specifications.

USB4 Gen 4 is normally referred to as a speed of "40 Gbps" or 40 Gbit/s, with the full connections based on it being referred to as 80, 120/40, 40/120 Gbit/s. But since the actual signaling is no longer binary, the actual raw bit rates no longer match those numbers exactly.

# USB4 capabilities by device type

## USB4 hub

A USB4 hub is defined by having 1 USB4 UFP and one or more USB4 DFP.

## USB4-based dock

A USB4-based dock is defined as a USB4 hub that also has more specialized outputs like HDMI or DP, but still keeping some USB4 DFP.

## USB4 peripheral device

A USB4 peripheral device is defined by not having any USB4 DFP. This means devices that are colloquially called "USB-C hubs" may use USB4 to support the dynamic bandwidth sharing or higher bandwidths of USB4. But they are not USB4 hubs if they do not have any USB4 DFP. Not having any USB4 DFP allows the peripheral to only support exactly those USB4 features that it has uses for, potentially simplifying its implementation considerably.

USB4 feature support[35]

| Feature | | Host | Hub (dock) | Peripheral device |
|---|---|---|---|---|
| Type | | | | |
| USB4 connection | "20 Gbps" (Gen 2×2) | Yes | Yes | Yes |
| | "40 Gbps" (Gen 3×2) | Optional | Yes | Optional |
| | "80 Gbps" (Gen 4 symm.) | Optional | Optional | Optional |
| | "120/40 Gbps" (Gen 4 3:1) | Optional | Optional[a] | Optional |
| | "40/120 Gbps" (Gen 4 1:3) | Optional | Optional[a] | Optional |
| Tunneled | USB3 "10 Gbps" (Gen 2×1) | Yes | Yes | Optional |
| | USB3 "20 Gbps" (Gen 2×2) | Optional | Optional | Optional |
| | USB3 Gen T (variable bandwidth)[b] | Optional | Optional | Optional |
| | DisplayPort | Yes | Yes | Optional |
| | PCI Express | Optional[c] | Yes | Optional |
| | Host-to-host communications/ USB4 networking | Yes | Yes | — |
| Native | USB3 "5 Gbps" (Gen 1×1) | Yes | Yes | Optional |
| | USB3 "10 Gbps" (Gen 2×1) | Yes | Yes | Optional |
| | USB3 "20 Gbps" (Gen 2×2) | Optional | Optional | Optional |
| | USB 2.0 (Low-, Full-, High-Speed)[d] | Yes | Yes | Optional |
| | DisplayPort Alternate Mode[e] | Yes | Yes | Optional |
| | Thunderbolt Alternate Mode on DFP | Optional[c] | Yes | — |
| | Thunderbolt Alternate Mode on UFP | — | Yes (before 2024) / Optional after[f] | Optional |
| | Other alternate modes | Optional | Optional | Optional |

a. Even for "USB 80Gbps" USB4 hubs, supporting asymmetric connections (in either direction) is optional, but 80 Gbit/s support is a prerequisite for any asymmetric support.
b. USB3 Gen T tunneling has defined bandwidth options. They match the total USB4 speed numbers 10,20,40,80 and even asymmetric 40/120,120/40 connections. USB4 v2 specification, p536, tab.9-19
c. Windows HLK requires any USB4 port support PCIe tunneling and TB3 compatibility. No minimum PCIe bandwidth requirements.[36]
d. As with USB3, USB2 connection runs on separate wires from main (USB3/USB4) connection. Tunneling is not required as it runs in parallel on the cable.
e. The USB4 specification makes no requirements on the minimum speed or capabilities of any DP output.
f. The 2024 version of the USB4 specification removed the requirement for USB4 hubs to support TB3 compatibility on its UFP. The requirement to support TB3 compatibility on DFP remains unaffected. USB4 v2 Specification September 2024, p712, sec. 13

# Cable compatibility

The Type-C standard supports cable backward/downward compatibility in many situations. The compatibility typically only breaks between the different families of standards (USB 2.0, USB 3.2, USB4). There are "passive" cables, which consist mainly of plain conductors that can be used and repurposed for any manner of signals in either direction and thus have the highest forward and backward compatibility. It also contains active or hybrid active (mix of active-conductive and optical) cables which the standard mandates to have vast backward compatibility support, so as to behave as if they were regular, passive cables in the eyes of the consumer.[37] But forward compatibility is limited for active cables due the active electronics inside. Only optically isolated active cables (OIAC), which should be clearly distinguishable by price, design, cable thickness, and advertising, are allowed to strip most of the backward compatibility away.

The Gen 4 transmission mode with PAM-3 uses signalling very different from that of previous modes. Every active component needs to explicitly support this new signaling, but it stays within all signal quality requirements of existing, passive Gen 3 cables (USB4 and TB3).

The direction of signalling also poses another challenge to active cables. Most USB signalling uses predefined wire-pairs for sending and others for receiving. But alternate modes, such as DP Alt mode may require to use all of the wire-pairs in the same direction. This switching of the direction of wire pairs makes active cables more complicated and therefore support for it is optional.

## Cable naming and relation to specification versions

USB-IF intends only for the new bandwidth-based logos and names to be used with consumers.[38] For cables, the type (passive, active) and the highest supported bandwidth are usually enough to uniquely identify a cable and its supported features. Although some active types make clear distinctions where further details on the type are required. Formally, a cable type and properties are defined by a distinct specification version, which was used during the development/design of said cable model, so each cable would be a valid and possibly certified cable according to a specific set of USB specification versions, like "Type-C 2.3, USB 3.2, USB4 Version 2.0". But the standard is also designed to be interoperable, in that a newer specification version typically adds new modes of operation, new cable types, but does not restrict previously existing things. Because that would make existing things incompatible with new products. For this purpose, even the older USB logos and labels did not include a specification version, but only stated "Certified USB SuperSpeed+ 10 Gbps". This logo identified cables that could support the 10 Gbit/s connection speeds of USB3 across both the USB 3.1 and USB 3.2 version, because the requirements for the cables have not changed. Thus, a precise specification version is usually not relevant and would not make a difference.

Transmission modes such as Gen2×2 are also irrelevant to cables, as valid cables are either full-featured, having all the high speed wire-pairs for up to dual-lane connections at the stated speed or they are USB2-only or some other specific and restrictive type, as listed below.

### USB4 cable compatibility

Overview of passive[39][40] and active Type-C cables[41] and their USB4 support

| Cable type | | Speed | Marketing names | Max. USB4 bit rate | Expected max. cable length[a] | Other support | | | | Pow |
|---|---|---|---|---|---|---|---|---|---|---|
| | Remarks | | | | | USB2 | USB3 | TB3 | DP | |
| USB2 | — | | Hi-Speed USB | ✘ | ≤ 4m | Yes | No | No | No | USB PD: 60W or 100W or 240W |
| Full-Featured passive | — | Gen 1 | USB 5Gbps | 20 Gbit/s[b] | ≤ 2m | Yes | 5 Gbit/s | 20 Gbit/s | Yes[c] | |
| | | Gen 2 | USB 20Gbps (USB 10Gbps deprecated) | 20 Gbit/s | ≤ 1m | Yes | Yes | 20 Gbit/s | | |
| | (incl. passive TB4 & TB5) | Gen 3 & Gen 4 | USB 40Gbps USB 80Gbps | 80 Gbit/s (or asymm.) | ≤ 0.8m | Yes | Yes | Yes[d] | Yes[c][e] | |
| Full-Featured active (also optical hybrid) | — | Gen 2 | USB 20Gbps (USB 10Gbps deprecated) | 20 Gbit/s | < 5m | Yes | Yes | Yes | Optional[f] | |
| | (incl. active TB4) | Gen 3 | USB 40Gbps | 40 Gbit/s | | Yes | Yes | Yes[d] | Optional[f] TB up to 2m[e] | |
| | (incl. active TB5) | Gen 4 | USB 80Gbps | 80 Gbit/s (or asymm.) | | Yes | Yes | Yes[d] | | |
| | USB3 active | Gen 2 | ? | ✘ | | Yes | Yes | No | Optional | |
| OIAC | USB3 | Gen 2 | ? | ✘ | ≤ 50m | only if optical | Gen 2 only (10 / 20 Gbit/s) | No | Optional | — |
| | USB4 | Gen 3 | ? | 40 Gbit/s | | | | Optional | | |
| | | Gen 4 | ? | 80 Gbit/s (asymm. optional) | | | | | | |
| Thunderbolt 3 | passive | Gen 2 | TB Logo without "3" | 20 Gbit/s | ≤ 2m | Yes | only 5 Gbit/s when > 1m[42] | 20 Gbit/s | Yes[c] | USB PD: 60W or 100W |
| | | Gen 3 | TB Logo + "3" | 80 Gbit/s (or asymm.)[43] | ≤ 0.8m | Yes | Yes | Yes | | |
| | active | Gen 3 | TB Logo + "3" | ✘[g] | (longest available: 3m) | Yes | (mostly no)[44] | Yes | (mostly no)[45] | |
| | optical[46] | Gen 3 | TB Logo + "3" | ✘ | ? | No | No | Yes | No | — |

a. Maximum cable lengths are not normative, but simply estimates of the USB specification, based on the expected physical limits of conventional copper cables.
b. USB4 Gen 2 signaling has less strict signal requirements than USB3 Gen 2 signaling. Spec compliant Gen 1 cables support USB4 Gen 2 / 20 Gbit/s connections
c. No specific max. DP speed guaranteed by Type-C specification
d. Old TB3 equipment may require an optional identification of the cable as "TB3 40 Gbit/s", because they predate the USB-C definition of Gen 3. Without this additional identification, these cables may only be seen as "20 Gbps" / Gen 2 by such equipment. Type-C Cable and Connector Specification 2023, p393
e. TB4 & TB5 cables up to 2m length (active & passive) are "universal cables", including DP support. DP guarantees may only include the highest speeds covered by DP 1.4 for TB4 (HBR3) or DP 2.1 for TB5 (UHBR20).
f. No specific max. DP speed guaranteed by Type-C specification. There are different implementations of active cable implementations that may behave differently.
g. The Apple TB3 Pro cable is one of the few active TB3 cables that supports DP and USB3. It is unclear if that is special behavior or the cable would be compatible to USB4 as well. "Apple now sells a $129 Thunderbolt 3 Pro cable" (https://www.theverge.com/2020/7/27/21339861/apple-thunderbolt-3-pro-cable-specs-price-available). *theverge.com*. The Verge. 2020-07-27. Retrieved 2024-08-09.

### DP Alt Mode support for USB4 cables

The Type-C specification does not name specific DP speeds that it considers supported for passive cables where support is optional for active cables. The USB-C presentation on DP Alt mode[47] calls out passive full-featured USB-C cables for their DisplayPort support and headroom for future DP speed increases. HBR3 was the highest available DP speed at the time.

Active cables may have additional complications, because the active electronics do not need to operate all high speed wire-pairs in the same direction for normal USB operations (but "USB 80Gbps" cables are mandated to support asymmetric connections, which includes at least some of the wire-pairs operating in either direction). Active cables can have further limitations, since the active electronics may only support specific signaling modes. There are 2 variants of active electronics. Linear ReDrivers only amplify the signal without any particular signaling mode or encoding in mind. ReTimers explicitly reconstruct the incoming signal for a higher quality result.

TB4 cables, even active ones, at least up to 2m in length, are guaranteed to support DP Alt mode. A specific maximum speed is also not mentioned, but the other requirements for TB4 all refer to DP 1.4 and its maximum speed of HBR3.[48] TB5 renews the same guarantee[49] for USB 80Gbps cables while referencing the DP 2.1 specification (up to UHBR20 speeds).

DP 2.1 aligned itself to the USB4 PHY layer, according to VESA, the creator of DisplayPort.[50] It is unclear how complete this alignment is. However, the UHBR10 DP speed matches USB4 Gen 2 in bit rate and encoding, whereas the UHBR20 DP speed matches USB4 Gen 3 in bit rate and encoding. A USB and DP certification service lists USB 5Gbps cables (USB 3.2 Gen 1) as supporting UHBR10 speeds, which would fit for having the same requirements as USB 20Gbps (USB4 Gen 2x2) connections.[51]

Anandtech reports[52] that "this also means that DP Alt Mode 2.0 should largely work with USB4-compliant cables, although VESA is being careful to avoid promising compatibility with all cables".

There are linear redrivers[53] and retimers[54] available that are advertised for USB4 Gen 3 speeds and all current DP speeds up to UHBR20 and including UHBR13.5.

## Thunderbolt compatibility

### Thunderbolt 3

The USB4 specification states that a design goal is to "Retain compatibility with existing ecosystem of USB and Thunderbolt products." Compatibility with Thunderbolt 3 is required for USB4 hubs, where this is optional for USB4 hosts and USB4 peripheral devices.[55] Compatible products need to implement 40 Gbit/s mode, at least 15W of supplied power and a different clock. Implementers need to sign the license agreement and register a Vendor ID with Intel.[56]

The USB4 protocol is based on and related to the operating principles of Thunderbolt 3. The USB4 specification simply defines which features to disable, downgrade and which parameters to change to get to an implementation compatible with Thunderbolt 3.[57] This includes, for example: limitation to daisy-chain topology (a hub must expose at most one USB4 DFP), downgrade of DP capabilities to DP 1.2, disabling any USB3 tunnel or the parallel USB 2.0 connection and switching back to the previous, slightly higher signaling rate of TB3. Any USB 3.x or USB 2.0 connection - if needed at all - is handled by PCIe-USB controllers and via the PCIe tunnel.

The TB3 connection is formally an Alt Mode and not a native USB operating mode. For this the extensible "vendor defined" mechanisms of USB PD are used to query cables and devices for support and to initiate the connection.

### Thunderbolt 4

During CES 2020, USB-IF and Intel stated their intention to allow USB4 products that optionally support any or all of the same functionality as Thunderbolt 4 products. The first products compatible with USB4 were Intel's Tiger Lake processors, with more devices appearing around the end of 2020.[58][59]

Thunderbolt 4 is a "solution brand"[60] around USB 40Gbps. Thunderbolt 4 mandates some features that are optional in USB4, including backward compatibility to Thunderbolt 3, minimum PCIe ("32 Gbps") and DP capabilities (2 DP tunnels, "4K60 each", HBR3+DSC).[61]

For cables, it similarly mandates above-minimum features of at least 100W support and the less well defined "universality" for up to 2 meters.

Thunderbolt 4 even requires certain notebooks to support charging via Thunderbolt ports, which is otherwise independent of any USB4 support.

### Thunderbolt 5

Thunderbolt 5 is an upgrade over Thunderbolt 4 and likewise a solution brand around USB 80Gbps. It mandates even higher minimum PCIe ("64 Gbps") and DP capabilities (2 DP tunnels, "6K60 each", unclear min. DP speed). It also mandates support for asymmetric 120/40 Gbit/s connections from hosts to docks, but does not mention the reverse.[62]

For cables, Thunderbolt 5 mandates 240W support and "universality" up to 2 meters.

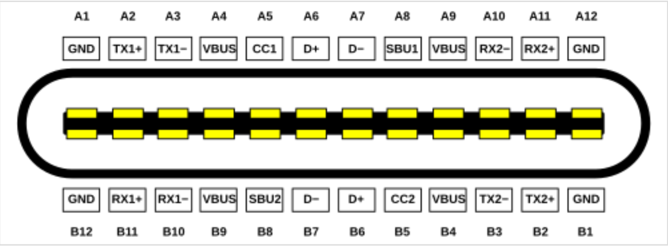
Notably, other features that are part of the advertising of Thunderbolt 5, like USB3 20Gbit/s or support for 3 DP tunnels at speeds of up to UHBR20 are optional even for Thunderbolt 5. While Intel's own controllers widely support these features and some are even available from newer Thunderbolt 4 controllers, not every Thunderbolt 5 device or controller does. See Known USB4 Controllers.

## Pinout

USB4 uses the Type-C connector with its 24 pins in a symmetrical shell. Type-C has 12 A pins on the top and 12 B pins on the bottom.[63]

The CC configuration channel is used per the Type-C standard to detect connections, plug orientation (due to the reversible USB Type-C shell) and establishing USB PD communication to negotiate power and query the cable (eMarker) and opposing side for USB4 or TB3 capability.[64][65] If all participants support USB4, the entering of USB4 mode is initiated. If not, the port may fall back on other alternative modes such as Thunderbolt 3 Alt mode, DP Alt mode or USB3 connections.

Once USB4 connection has been requested, USB4 uses the SBU channel to manage the overall connection, initialization, disconnect.[66] All SuperSpeed differential pairs are used for the main USB4 link. For Gen 2 and Gen 3 connections, TX1+, TX1−, RX1+, RX1− form lane one and TX2+, TX2−, RX2+, RX2− form lane two, each in sending and receiving direction respectively. Gen 4 connections can also use three pairs in one direction for the asymmetric modes. The differential D+ and D− pair is used for USB 2.0 transfer that runs independent of the other USB4 communication.[67] While USB3 communication happens via a "tunnel" inside the main USB4 link, any USB2 communication with potential USB2 devices continues to happen over the same wires as it would without USB4.



Type-C receptacle pinout

USB4 only requires that PD communication occurs and that the participants have agreed upon an initial power contract, before proceeding to USB4 negotiation. They are otherwise independent and power can be renegotiated at any point (including direction changes) without disturbing the USB4 link and is only guided by the Type-C and PD specifications, not by USB4 itself.

Type-C receptacle A pin layout

| Pin | Name | Description |
|---|---|---|
| **A1** | GND | Ground return |
| **A2** | SSTXp1 ("TX1+") | SuperSpeed differential pair #1, TX, positive |
| **A3** | SSTXn1 ("TX1-") | SuperSpeed differential pair #1, TX, negative |
| **A4** | $V_{BUS}$ | Bus power |
| **A5** | CC1 | Configuration channel |
| **A6** | Dp1 | USB 2.0 differential pair, position 1, positive |
| **A7** | Dn1 | USB 2.0 differential pair, position 1, negative |
| **A8** | SBU1 | Sideband use (SBU) |
| **A9** | $V_{BUS}$ | Bus power |
| **A10** | SSRXn2 ("RX2-") | SuperSpeed differential pair #4, RX, negative |
| **A11** | SSRXp2 ("RX2+") | SuperSpeed differential pair #4, RX, positive |
| **A12** | GND | Ground return |

Type-C receptacle B pin layout

| Pin | Name | Description |
|---|---|---|
| **B12** | GND | Ground return |
| **B11** | SSRXp1 | SuperSpeed differential pair #2, RX, positive |
| **B10** | SSRXn1 | SuperSpeed differential pair #2, RX, negative |
| **B9** | $V_{BUS}$ | Bus power |
| **B8** | SBU2 | Sideband use (SBU) |
| **B7** | Dn2 | USB 2.0 differential pair, position 2, negative[a] |
| **B6** | Dp2 | USB 2.0 differential pair, position 2, positive[a] |
| **B5** | CC2 | Configuration channel |
| **B4** | $V_{BUS}$ | Bus power |
| **B3** | SSTXn2 | SuperSpeed differential pair #3, TX, negative |
| **B2** | SSTXp2 | SuperSpeed differential pair #3, TX, positive |
| **B1** | GND | Ground return |

a. There is only a single non-SuperSpeed differential pair in the cable. This pin is not connected in the plug/cable.

## Software support

USB4 is supported by:

- Linux kernel 5.6, released on 29 March 2020[68]
- macOS Big Sur (11.0), released on 12 November 2020[69]
- Windows 11, released with support for USB4 Version 1.0 on 5 October 2021[70]
  - upgraded to USB4 Version 2.0 support including 80 Gbit/s around March 2024[71]

### Connection manager

Connection manager is the part of a USB4 host that manages connections across the entire USB4 topology, establishing of tunnels, handling any bandwidth reservations and data prioritization, like which DP tunnels can be established and at what speed. The USB4 driver in Windows 11 implements native OS support of USB4, where the connection manager is part of a driver that only works with matching controllers. Older controllers had the connection manager implemented inside their firmware and thus required far less support from the OS.

On Linux, the USB4 and Thunderbolt driver (named "thunderbolt") supports both firmware-managed and also OS-managed controllers via the same tools.

## Hardware support

Brad Saunders, CEO of the USB Promoter Group, anticipates that most PCs with USB4 will support Thunderbolt 3, but for phones the manufacturers are less likely to implement Thunderbolt 3 support.[72]

On 3 March 2020, Cypress Semiconductor announced new Type-C power (PD) controllers supporting USB4, CCG6DF as dual port and CCG6SF as single port.[73]

In November 2020, Apple unveiled MacBook Air (M1, 2020), MacBook Pro (13-inch, M1, 2020) and Mac mini (M1, 2020), featuring two USB 40Gbps (USB4 Gen 3) ports.

AMD also stated that Zen 3+ (Rembrandt) processors will support USB4[74] and released products do have this feature after a chipset driver update.[75] However, AMD has only announced support for USB 20Gbps (USB 3.2 Gen 2×2) in Zen 4 processors that were released in September 2022.[76][77] Intel supported Thunderbolt 3 and USB-C with their 8th generation mobile processors in 2018. For example, the Lenovo P52 has dual TB3 ports on the rear.

TB/USB has evolved as Intel is able to refine logic design.

## Known USB4 controllers

The following table of USB4 controllers lists what features the controllers could support. The boards or products in which they are used can often influence the specific supported features, for example by not connecting all inputs, configuring the controller differently for compatibility or power saving or by combining it with other, lower tier components.

For example, many USB4 controllers Intel builds into their CPUs require further external components such as ReTimers to reach the full 40 Gbit/s speeds. They can also be used in simpler configurations for only 20 Gbit/s.[78]

Features of alternate connections, such as the native DP or USB3 speeds - which can differ from the speeds supported for DP and USB3 tunnels inside USB4 connections - may also depend on other components, such as the ReTimers used.[79]

Known USB4 Controllers

| Manufacturer | Controller | Family | Type | native managed[a] | USB4 ver. | Speed Gbit/s | USB4 Ports (up / down) | DP In / out protocol adapters[b] | USB3 Gbit/s | |
|---|---|---|---|---|---|---|---|---|---|---|
| Intel | JHL8540[80] | Maple Ridge | Host | No | 1 | 40 | 2 dn | 2 in<br>DP 1.4 (up to HBR3) | 10<br>(integ. ctrl.) | x4 Gen 3 |
| | JHL8440[81] | Goshen Ridge | Hub / Peri. | — | | | 1 up,<br>3 dn | 2 out<br>DP 1.4 (up to HBR3) | 10<br>(hub, integ. ctrl.) | (32 Gbit/ |
| | JHL8140 | Hoover Ridge | Peri. | | | | 1 up | 2 out<br>DP 1.4 (up to HBR3) | 10<br>(hub, integ ctrl.) | |
| | JHL9540[82] | Barlow Ridge | Host | Yes | 2 | | 2 dn | 3 in<br>DP 2.1 (up to UHBR10, +UHBR20)[d] | 20<br>(integ. ctrl.) | x4 Gen 4 |
| | JHL9440[83] | | Hub / Peri. | — | | | 1 up,<br>3 dn | 3 out<br>2 in DP 2.1 (up to UHBR10, +UHBR20)[d] | 20<br>(hub, integ. ctrl.) | x4 Gen 4 |
| | JHL9580[84][85] | | Host | Yes | | 80 | 2 dn | 3 in<br>DP 2.1 (up to UHBR10, +UHBR20)[d] | 20<br>(integ. ctrl.) | x4 Gen 4 |
| | JHL9480[86] | | Hub / Per. | — | | | 1 up,<br>3 dn | 3 out<br>2 in DP 2.1 (up to UHBR10, +UHBR20)[d] | 20<br>(hub, integ. ctrl.) | x4 Gen 4 |
| | Tiger Lake[87] | CPU-integrated | Host | No | 1 | 40 | up to<br>2 + 2 dn[f] | 2 in/group:<br>up to HBR3 | 10 | (32 Gbit/ |
| | Alder Lake[88] | | | Yes | | | | | 10[g] | |
| | Raptor Lake[89] | | | | | | | | 10<br>(20 internal)[g] | |
| | Meteor Lake[90] | | | | | | | | 20[g] | |
| | Lunar Lake[91] | | | | | | up to<br>2 +1 dn[f] | | | |
| | Arrow Lake[92] | | | | | | 2 dn | | | |
| AMD | AMD Renoir, Phoenix[93] | | | | | | up to 1+1 dn[f] | 2 in/port<br>up to HBR3 | 10 | |
| | AMD Strix Point[94] | | | | | | | | | |
| | AMD Strix Halo[97] | | | | | | | | | |
| Apple | M1/2 | Apple USB4/TB3 | Host | ? | 1 | 40 | ? | 1 in/port<br>up to HBR3 | 10 | |
| | M1-3 Pro, Max, M4 | Apple TB4 | | | | | | 2 in/port<br>up to HBR3 | | |
| | M3 Ultra,<br>M4 Pro/Max | Apple TB5 (1st gen) | | | 2 | 80 | | 2 in/port<br>up to UHBR10, +UHBR20[d] | | |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | M5 Pro/Max[98] | Apple TB5 (2nd gen) | | | | | | 3 in/port up to UHBR10, +UHBR20 ?[d] | 10 ? | |
| Via | VL830[99] | — | Peri. | — | 1 | 40 | 1 up | 1 out: DP 1.4 (up to HBR3) | 10 (hub) | |
| | VL832[100] | | | | | | | | | |
| Asmedia | ASM2464PD[101] | | Peri. / NVMe | | | | 1 up | — | 20 (peripheral) | x4 Gen 4 |
| | ASM2464PDX[102] | | Peri. / NVMe | | | | 1 up | | 20 (peripheral) | x4 Gen 4 (bifurca |
| | ASM4242[103] | | Host | Yes | | | 2 dn | 2 in: DP 1.4 (up to HBR3) | 20 (integ. ctrl.) | x4 Gen 4 |
| Realtek | RTX5490[104][105] | | Peri. / Hub | — | 1 | 40 | 2 dn | USB4 up/dn: UHBR20, ? | 20 (hub, integ. ctrl.) | |

a. Host controller is not managing itself with onboard firmware, but via ACPI and USB4 standard with generic OS drivers, such as Windows 11 USB4 drivers. Only applicable to host controllers. All other controllers have always been managed by the host's connection manager, no matter if that is implemented in firmware or by the OS.
b. Protocol adapters are what converts between a tunnel and a connection external to USB4. DP In adapters map directly to some input from a GPU. DP Out adapters may be shared across multiple physical outputs, in which case they limit how many can be used at the same time.
c. Main PCIe port of the controller. Not applicable for CPU-integrated host controllers. If PCIe is only used internal to the controller, PCIe throughput specifications are rarely given.
d. Supports UHBR10 and all lower speeds. And UHBR20, while not supporting UHBR13.5 speeds. UHBR10 and UHBR20 are aligned to USB4 Gen 2 and Gen 3 respectively, UHBR13.5 is not aligned to any existing USB4 signaling.
e. Unclear how much the output can do / be used for yet. Per leak "side port", may only be a passthrough of the third DP input
f. "2+2" indicates 2 separate dual-port USB4 host routers. But because integrated into the CPU, can share functionality that is not technically part of USB4 host router, like USB3. Systems may share a common USB3 controller, supplying 1 USB3 root port per USB4 port, similar with PCIe ports.
g. USB3 controller shared across all integrated USB4 ports
h. What the respective USB4 ports can output directly, instead of tunneled through USB4. Can be higher than tunneled support, as not limited by the USB4 specification and technically independent of USB4 operations/connections.
i. Unclear, whether this includes UHBR13.5 support, unlike competing Intel products

## References


1. USB4 Version 2.0 Specification 2023
2. "USB-IF Announces Publication of New USB4 Specification to Enable USB 80Gbps Performance" (https://www.usb.org/sites/default/files/2022-10/USB-IF%20USB%2080Gbps%20Announcement_FINAL_v2.pdf) (PDF). 2022-10-18.
3. Type-C Cable and Connector Specification 2023
4. Type-C Cable and Connector Specification, p216f, sec. 4.6
5. USB Power Delivery Specification 2023
6. USB4 Version 2.0 Specification 2023, p1, sec. 1.5, p12, sec. 2
7. USB4 Version 2.0 Specification 2023, p15, sec. 2.1.5
8. Hill, Brandon (2019-03-04). "USB4 Leverages Thunderbolt 3 Protocol Doubling Speeds To 40Gbps" (https://hothardware.com/news/usb4-40-gbps-royalty-free-intel-thunderbolt-3). *HotHardware*. Archived (https://web.archive.org/web/20211023152301/https://hothardware.com/news/usb4-40-gbps-royalty-free-intel-thunderbolt-3) from the original on 2021-10-23. Retrieved 2020-04-28.
9. "USB4 announced with 40Gbps bandwidth, it's based on Thunderbolt 3" (https://www.gsmarena.com/usb4_announced_with_40gbps_bandwidth_its_based_on_thunderbolt_3-news-35841.php). *GSMArena.com*. Archived (https://web.archive.org/web/20220115155541/https://www.gsmarena.com/usb4_announced_with_40gbps_bandwidth_its_based_on_thunderbolt_3-news-35841.php) from the original on 2022-01-15. Retrieved 2020-04-29.
10. "With USB 4, Thunderbolt and USB will converge" (https://techcrunch.com/2019/03/04/with-usb-4-thunderbolt-and-usb-will-converge/). 2019-03-04. Archived (https://web.archive.org/web/20220910201245/https://techcrunch.com/2019/03/04/with-usb-4-thunderbolt-and-usb-will-converge/) from the original on 2022-09-10. Retrieved 2020-05-01.
11. Hagedoorn, Hilbert (2019-06-13). "USB 4.0 Will Arrive in Late 2020" (https://www.guru3d.com/news-story/usb-4-will-arrive-in-late-2020.html). *Guru3D*. Archived (https://web.archive.org/web/20211026182556/https://www.guru3d.com/news-story/usb-4-will-arrive-in-late-2020.html) from the original on 2021-10-26. Retrieved 2020-04-30.
12. USB Promoter Group (2019-03-04). "USB Promoter Group Announces USB4 Specification" (https://usb.org/sites/default/files/2019-03/USB_PG_USB4_DevUpdate_Announcement_FINAL_20190226.pdf) (PDF). *usb.org*. Retrieved 2024-12-21.
13. "USB4 Specification" (https://web.archive.org/web/20210419213328/https://usb.org/document-library/usb4tm-specification). p1, sec 1.4 Design Goals. Archived from the original (https://usb.org/document-library/usb4tm-specification) on 2021-04-19.
14. Dr. Cutress, Ian (2020-07-08). "Intel Thunderbolt 4 Update: Controllers and Tiger Lake in 2020" (https://web.archive.org/web/20200708152158/https://www.anandtech.com/show/15902/intel-thunderbolt-4-update-controllers-and-tiger-lake-in-2020). Archived from the original (https://www.anandtech.com/show/15902/intel-thunderbolt-4-update-controllers-and-tiger-lake-in-2020) on 2020-07-08. Retrieved 2024-12-21.
15. "Introducing Thunderbolt 4: Universal Cable Connectivity for Everyone" (https://www.intel.com/content/www/us/en/newsroom/news/thunderbolt-4-universal-cable-connectivity.html). 2020-07-08. Retrieved 2024-12-21.
16. "USB Promoter Group Announces USB4 Version 2.0" (https://www.businesswire.com/news/home/20220901005211/en/USB-Promoter-Group-Announces-USB4%C2%AE-Version-2.0). *www.businesswire.com*. 2022-09-01. Archived (https://web.archive.org/web/20220902075428/https://www.businesswire.com/news/home/20220901005211/en/USB-Promoter-Group-Announces-USB4%C2%AE-Version-2.0) from the original on 2022-09-02. Retrieved 2022-09-02.
17. "USB-IF Announces Publication of New USB4 Specification to Enable USB 80Gbps Performance" (https://usb.org/sites/default/files/2022-10/USB-IF%20USB%2080Gbps%20Announcement_FINAL_v2.pdf) (PDF). *USB-IF*. USB Implementers Forum. 2022-10-18. Retrieved 2023-01-19.
18. Porter, Jon (2022-09-30). "USB kills off SuperSpeed branding as it tries to simplify its ubiquitous connector" (https://www.theverge.com/2022/9/30/23378231/usb-rebranding-2022-logos-gbps-wattage-charging-transfer-speeds-simplification-usb4-superspeed). *The Verge*. Retrieved 2024-08-05.

19. Ganesh, T S (2023-10-12). "Intel Unveils Barlow Ridge Thunderbolt 5 Controllers - TB5 Launching In 2024" (https://web.archive.org/web/20230912135807/https://www.anandtech.com/show/20050/intel-unveils-barlow-ridge-thunderbolt-5-controllers). Archived from the original (https://www.anandtech.com/show/20050/intel-unveils-barlow-ridge-thunderbolt-5-controllers) on 2023-09-12. Retrieved 2024-12-21.
20. "Intel Introduces Thunderbolt 5 Connectivity Standard" (https://www.intel.com/content/www/us/en/newsroom/news/intel-introduces-thunderbolt-5-standard.html). 2022-10-12. Retrieved 2024-12-21.
21. USB4 Version 2.0 Specification 2023, p13ff, fig. 2-1 USB4/USB3.2 Dual Bus System Architecture
22. USB4 Version 2.0 Specification 2023, p12ff, sec. 2 Architectural Overview
23. USB 2.0 Specification 2024
24. USB 3.2 Specification 2022
25. "Intel® JHL8540 Thunderbolt™ 4 Controller Product Specifications" (https://www.intel.com/content/www/us/en/products/sku/193684/intel-jhl8540-thunderbolt-4-controller.html). *www.intel.com*. Retrieved 2024-08-06.
26. Type-C Cable and Connector Specification 2023, p244 sec. 5.3
27. USB4 Version 2.0 Specification Inter-Domain Service 2023
28. USB4 Version 2.0 Specification 2023, p13
29. USB4 Version 2.0 Specification 2023, p487
30. USB4 Version 2.0 Specification Connection Manager Guide 2023, p59, sec. 6.1
31. USB4 Version 2.0 Specification 2023, p660, sec. 11.1.1.1.3
32. USB Trademark Requirements Chart from USB-IF (https://usb.org/sites/default/files/trademark_license_agreement_licensed_mark_requirements_final_as_of_september_20_2023.pdf)
33. Team GraniteRiverLabs (2023-01-17). "Welcome to the 80Gpbs Ultra-High Speed Era of USB4" (https://www.graniteriverlabs.com/en-us/technical-blog/usb4-80-cio80). *www.graniteriverlabs.com*. GraniteRiverLabs Taiwan. Archived (https://web.archive.org/web/20230221162539/https://www.graniteriverlabs.com/en-us/technical-blog/usb4-80-cio80) from the original on 2023-02-21. Retrieved 2023-02-21.
34. "How to Test and Troubleshoot USB4" (https://files.keysightevent.com/files/20211215/5992-3142.pdf) (PDF). Archived (https://web.archive.org/web/20220910201247/https://files.keysightevent.com/files/20211215/5992-3142.pdf) (PDF) from the original on 2022-09-10. Retrieved 2022-07-25.
35. USB4 Version 2.0 Specification 2023, p17ff, sec. 2.1.1.4
36. windows-driver-content (2022-05-18). "USB4 Systems PCIe Tunneling Support" (https://learn.microsoft.com/en-us/windows-hardware/test/hlk/testref/7d627bf0-25f3-4564-b554-b2a3450e2bcf). *learn.microsoft.com*. Retrieved 2024-08-05.
37. Type-C Cable and Connector Specification 2023, p261 para. 3
38. "USB Branding Session 2019" (https://usb.org/sites/default/files/D1T2-1%20-%20USB%20Branding%20Session.pdf) (PDF). *usb.org*. 2020-02-07. p16. Retrieved 2024-07-08.
39. Type-C Cable and Connector Specification 2023, p42, Tab 3-1
40. "How to Beat the Maximum USB Cable Length Limit" (https://blog.tripplite.com/usb-cable-max-length). *blog.tripplite.com*. Retrieved 2024-03-14.
41. Type-C Cable and Connector Specification 2023, p261 sec. 6
42. "Cable Matters 107002 40 Gbps / 20 Gbps Thunderbolt 3 Cable" (https://kb.cablematters.com/index.php?View=entry&EntryID=76). *kb.cablematters.com*. 2024-10-09. Retrieved 2024-08-12. "When connected to USB-C devices, the data rate can reach 10 Gbps (0.5m & 1m) and 5 Gbps (2m)."
43. Type-C Cable and Connector Specification 2023, p246, fig. 5-1
44. "USB-C Cables, Thunderbolt 3 Cables - How to tell them apart" (https://archive.caldigit.com/Thunderbolt-3-USB-C-Type-C-Charge-Cable/). *archive.caldigit.com*. CalDigit. Retrieved 2024-08-08.
45. "How to select a Thunderbolt 3 cable?" (https://www.chargerlab.com/how-to-select-a-thunderbolt-3-cable-2). chargerlab.com. 2019-05-11. Retrieved 2024-08-08.
46. "Corning Thunderbolt™ 3 Optical Cable (40 Gb/s)" (https://www.corning.com/microsites/coc/ocbc/Documents/OEM-062-A4-AEN.pdf) (PDF). Corning. Retrieved 2024-08-08.
47. "VESA – DisplayPortTM Alternate Mode on USB-C®" (https://www.usb.org/sites/default/files/D2T1-4%20-%20VESA%20DP%20Alt%20Mode%20over%20USB%20Type-C.pdf) (PDF). 2019-11-19. p23. Retrieved 2024-08-08.
48. "Intel Thunderbolt 4™ Announcement Press Deck" (https://www.thunderbolttechnology.net/sites/default/files/intel-thunderbolt4-announcement-press-deck.pdf) (PDF). Intel. p9, p13. Retrieved 2024-08-08.
49. "Thunderbolt 5™ Press Deck" (https://www.thunderbolttechnology.net/sites/default/files/TBT5%20Press%20Deck%20v3%20Final.pdf) (PDF). *thunderbolttechnology.net*. Intel. Retrieved 2024-08-09.
50. "VESA Releases DisplayPort 2.1 Specification" (https://www.displayport.org/pr/vesa-releases-displayport-2-1-specification/). Retrieved 2024-08-06.
51. "Specification and Test Overview of DisplayPort™ 2.1" (https://www.graniteriverlabs.com/en-us/technical-blog/displayport-dp-2-1-overview). Granite River Labs. Ultra High Bit Rate in DisplayPort™ 2.1. Retrieved 2024-08-08.
52. "DisplayPort Alt Mode 2.0 Spec Released: Defining Alt Mode for USB4" (https://web.archive.org/web/20200430080010/https://www.anandtech.com/show/15752/displayport-alt-mode-20-spec-released). Anandtech. Archived from the original (https://www.anandtech.com/show/15752/displayport-alt-mode-20-spec-released) on 2020-04-30. Retrieved 2024-08-08.
53. "PI2DPX2020" (https://www.diodes.com/part/view/PI2DPX2020?BackID=237). *Diodes*. 2011-03-02. Retrieved 2024-08-08.
54. "KM864742" (https://www.nuvoton.com/products/high-speed-interface/usb4-re-timer/cable/km864742/). Nuvoton. Retrieved 2024-08-09.
55. USB4 Specification V1.0 August 2019 Chapter 13: "A USB4 host and USB4 peripheral device may optionally support TBT3-Compatibility. If a USB4 host or USB4 peripheral device supports TBT3-Compatibility, it shall do so as defined in this chapter".
56. "USB4 Thunderbolt3 Compatibility Requirements Specification – USB-F" (https://usb.org/document-library/usb4tm-thunderbolt3tm-compatibility-requirements-specification). Archived (https://web.archive.org/web/20211124133132/https://www.usb.org/document-library/usb4tm-thunderbolt3tm-compatibility-requirements-specification) from the original on 2021-11-24. Retrieved 2021-11-13.
57. USB4 Version 2.0 Specification 2023, p709ff, sec. 13 Interoperability with Thunderbolt 3 (TBT3) Systems
58. "USB4 devices are clear to roll out next year" (https://www.engadget.com/2019-09-03-usb4-devices-next-year.html). *Engadget*. 2019-09-03. Archived (https://web.archive.org/web/20211124132956/https://www.engadget.com/2019-09-03-usb4-devices-next-year.html) from the original on 2021-11-24. Retrieved 2020-04-28.
59. Maislinger, Florian (2019-06-14). "First USB 4 devices to be launched at the end of 2020" (https://www.pcbuildersclub.com/en/2019/06/first-usb-4-devices-to-be-launched-at-the-end-of-2020/). Archived (https://web.archive.org/web/20211124151515/https://www.pcbuildersclub.com/en/2019/06/first-usb-4-devices-to-be-launched-at-the-end-of-2020/) from the original on 2021-11-24. Retrieved 2020-04-28.
60. "Intel® Core™ Ultra 200H and 200U Series Processors USB Type-C Sub System" (https://edc.intel.com/content/www/us/en/design/products-and-solutions/processors-and-chipsets/core-ultra-200h-and-200u-series-processors-datasheet-volume-1-of-2/usb-type-c-sub-system/). *Intel.com*. Retrieved 2026-01-10. "Thunderbolt™ 4 is a USB Type-C solution brand which requires the following elements..."
61. "Intel Thunderbolt 4 Announcement Press Deck" (https://www.thunderbolttechnology.net/sites/default/files/intel-thunderbolt4-announcement-press-deck.pdf) (PDF). *thunderbolttechnology.net*. Intel. p4. Retrieved 2024-08-08.
62. "Thunderbolt 5™ Press Deck" (https://www.thunderbolttechnology.net/sites/default/files/TBT5%20Press%20Deck%20v3%20Final.pdf) (PDF). *thunderbolttechnology.net*. Intel. p19. Retrieved 2024-08-09.
63. "The Relationship Between USB4 and the USB Type-C Connector" (https://www.totalphase.com/blog/2020/02/usb4-and-usb-type-c-connector/). *Total Phase Blog*. 2020-02-18. Archived (https://web.archive.org/web/20220910201251/https://www.totalphase.com/blog/2020/02/usb4-and-usb-type-c-connector/) from the original on 2022-09-10. Retrieved 2022-04-05.
64. Type-C Cable and Connector Specification 2023, p260, fig. 5-1 USB4 Discovery and Entry Flow Model
65. Leung, Benson (2018-11-19). "USB Type-C's Configuration Channel" (https://medium.com/@leung.benson/usb-type-c-s-configuration-channel-31e08047677d). *Medium*. Archived (https://web.archive.org/web/20220405163245/https://medium.com/@leung.benson/usb-type-c-s-configuration-channel-31e08047677d) from the original on 2022-04-05. Retrieved 2022-04-05.
66. USB4 Version 2.0 Specification 2023, p25

67. "USB4 Specification" (https://www.usb.org/document-library/usb4r-specification). *www.usb.org*. Archived (https://web.archive.org/web/20220414181705/https://www.usb.org/document-library/usb4r-specification) from the original on 2022-04-14. Retrieved 2022-04-05.
68. "Linux 5.6 Kernel Released With WireGuard, USB4, New AMD + Intel Hardware Support – Phoronix" (https://www.phoronix.com/scan.php?page=news_item&px=Linux-5.6-Released). *Phoronix.com*. Archived (https://web.archive.org/web/20211102223627/https://www.phoronix.com/scan.php?page=news_item&px=Linux-5.6-Released) from the original on 2021-11-02. Retrieved 2020-04-28.
69. "Introducing the next generation of Mac" (https://www.apple.com/newsroom/2020/11/introducing-the-next-generation-of-mac/). *apple.com*. 2020-11-10. Archived (https://web.archive.org/web/20210301173940/https://www.apple.com/newsroom/2020/11/introducing-the-next-generation-of-mac/) from the original on 2021-03-01. Retrieved 2020-11-13.
70. "Introduction to the USB4 connection manager in Windows" (https://docs.microsoft.com/en-us/windows-hardware/design/component-guidelines/usb4-intro-to-connection-manager). *docs.microsoft.com*. 2021-09-03. Archived (https://web.archive.org/web/20211103200955/https://docs.microsoft.com/en-us/windows-hardware/design/component-guidelines/usb4-intro-to-connection-manager) from the original on 2021-11-03. Retrieved 2021-11-03.
71. "February 29, 2024—KB5034848 (OS Builds 22621.3235 and 22631.3235) Preview - Microsoft Support" (https://support.microsoft.com/en-us/topic/february-29-2024-kb5034848-os-builds-22621-3235-and-22631-3235-preview-5b718126-82b1-4e14-bf01-da3500a3fc88). *support.microsoft.com*. Retrieved 2024-08-05.
72. Piltch, Avram (2021-04-20). "USB 4: Everything We Know So Far" (https://www.tomshardware.com/news/usb-4-faq,38766.html). *Tom's Hardware*. Archived (https://web.archive.org/web/20210630140440/https://www.tomshardware.com/news/usb-4-faq,38766.html) from the original on 2021-06-30. Retrieved 2020-04-30.
73. Shilov, Anton. "Cypress Announces USB 3.2 & USB4-Ready Controllers: EZ-PD CCG6DF & CCG6SF" (https://web.archive.org/web/20211206152338/https://www.anandtech.com/show/15567/cypress-announces-usb-32-usb4-controllers-ezpd-ccg6df-ccg6sf). *www.anandtech.com*. Archived from the original (https://www.anandtech.com/show/15567/cypress-announces-usb-32-usb4-controllers-ezpd-ccg6df-ccg6sf) on 2021-12-06. Retrieved 2020-04-28.
74. Cutress, Ian (2022-01-04). "AMD Announces Ryzen 6000 Mobile CPUs for Laptops: Zen3+ on 6nm with RDNA2 Graphics" (https://web.archive.org/web/20220625234326/https://www.anandtech.com/show/17166/amd-announces-ryzen-6000-mobile-cpus-for-laptops-zen3-on-6nm-with-rdna2-graphics/). Anandtech. Archived from the original (https://www.anandtech.com/show/17166/amd-announces-ryzen-6000-mobile-cpus-for-laptops-zen3-on-6nm-with-rdna2-graphics) on 2022-06-25. Retrieved 2022-07-26.
75. Klotz, Aaron (2022-06-28). "AMD Introduces USB 4 Support in Chipset Update for Ryzen 6000 Mobile (Updated)" (https://www.tomshardware.com/news/amd-usb4-chipset-driver-update-for-ryzen-6000). *Tom's Hardware*.
76. Bonshor, Gavin (2022-05-23). "AMD Ryzen 7000 Announced: 16 Cores of Zen 4, Plus PCIe 5 and DDR5 for Socket AM5, Coming This Fall" (https://web.archive.org/web/20220726130535/https://www.anandtech.com/show/17399/amd-ryzen-7000-announced-zen4-pcie5-ddr5-am5-coming-fall). Anandtech. Archived from the original (https://www.anandtech.com/show/17399/amd-ryzen-7000-announced-zen4-pcie5-ddr5-am5-coming-fall) on 2022-07-26. Retrieved 2022-07-26.
77. "AMD confirms Zen4 & Ryzen 7000 series lineup: Raphael in 2022, Dragon Range and Phoenix in 2023" (https://videocardz.com/newz/amd-confirms-zen4-ryzen-7000-series-lineup-raphael-in-2022-dragon-range-and-phoenix-in-2023). *VideoCardz.com*. Retrieved 2022-11-29.
78. "13th Generation Intel® Core™ Datasheed, Volume 1 of 2" (https://edc.intel.com/content/www/us/en/design/products/platforms/details/raptor-lake-s/13th-generation-core-processors-datasheet-volume-1-of-2/usb-c-sub-system/). *intel.com*. Retrieved 2026-01-09.
79. "13th Generation Intel® Core™ Datasheed, Volume 1 of 2" (https://edc.intel.com/content/www/us/en/design/products/platforms/details/raptor-lake-s/13th-generation-core-processors-datasheet-volume-1-of-2/introduction/#IMAGE_D6D165F6C0F446F1A3BCC8AA81F4ABBE). *intel.com*. P/H/H Refresh Processor Line Platform Diagram, footnote 2. Retrieved 2026-01-09.
80. "Intel JHL8540 USB4 Controller Specifications" (https://www.intel.com/content/www/us/en/products/sku/193684/intel-jhl8540-thunderbolt-4-controller/specifications.html).
81. "Intel JHL8440 USB4 Controller Specifications" (https://www.intel.com/content/www/us/en/products/sku/189982/intel-jhl8440-thunderbolt-4-controller/specifications.html).
82. "Intel JHL9540 USB4 Controller Specifications" (https://www.intel.com/content/www/us/en/products/sku/225917/intel-jhl9540-thunderbolt-4-controller/specifications.html).
83. "Intel JHL9440 USB4 Controller Specifications" (https://www.intel.com/content/www/us/en/products/sku/225918/intel-jhl9440-thunderbolt-4-accessory-controller/specifications.html).
84. "Intel JHL9580 USB4 Controller Specifications" (https://www.intel.com/content/www/us/en/products/sku/225921/intel-jhl9580-thunderbolt-5-controller/specifications.html).
85. "Barlow Ridge Leak" (https://videocardz.com/newz/intel-barlow-ridge-thunderbolt-5-controller-for-arrow-lake-s-cpus-detailed-enabling-pcie-gen4x4-for-external-gpus).
86. "Intel JHL9480 USB4 Controller Specifications" (https://www.intel.com/content/www/us/en/products/sku/225919/intel-jhl9480-thunderbolt-5-accessory-controller/specifications.html).
87. "Intel Tiger Lake CPU Specifications" (https://www.intel.com/content/www/us/en/content-details/631121/11th-generation-intel-core-processor-datasheet-volume-1-of-2.html).
88. "Intel Alder Lake CPU Specifications" (https://edc.intel.com/content/www/us/en/design/ipla/software-development-platforms/client/platforms/alder-lake-desktop/12th-generation-intel-core-processors-datasheet-volume-1-of-2/usb-c-sub-system/).
89. "Intel Raptor Lake CPU Specifications" (https://www.intel.com/content/www/us/en/content-details/743844/13th-generation-intel-core-and-intel-core-14th-generation-processors-datasheet-volume-1-of-2.html).
90. "Intel Meteor Lake CPU Specifications" (https://edc.intel.com/content/www/us/en/design/products/platforms/details/meteor-lake-u-p/core-ultra-processor-datasheet-volume-1-of-2/).
91. "Intel Lunar Lake CPU Specifications" (https://www.intel.com/content/www/us/en/content-details/829568/intel-core-ultra-200v-series-processors-datasheet-volume-1-of-2.html).
92. "Intel Arrow Lake CPU Specifications" (https://edc.intel.com/content/www/us/en/design/products/platforms/details/arrow-lake-s/core-ultra-200s-series-processors-datasheet-volume-1-of-2/).
93. "AMD Ryzen™ 7 7840HS specifications" (https://www.amd.com/en/products/processors/laptop/ryzen/7000-series/amd-ryzen-7-7840hs.html). *amd.com*. Retrieved 2026-01-09.
94. "AMD Ryzen™ AI 9 HX 370 specifications" (https://www.amd.com/en/products/processors/laptop/ryzen/ai-300-series/amd-ryzen-ai-9-hx-370.html). *amd.com*. Retrieved 2026-01-09.
95. "HP Elitebook X G1a 14 Notebookcheck Review" (https://www.notebookcheck.net/HP-EliteBook-X-G1a-14-AI-review-Supercharged-business-laptop-with-AMD-Ryzen-AI-9-HX.963360.0.html). Retrieved 2026-01-09.
96. "HP ZBook G1a 14 Notebookcheck Review" (https://www.notebookcheck.net/HP-ZBook-Ultra-G1a-14-review-Powerful-MacBook-Pro-alternative-for-work-and-game.994758.0.html). Retrieved 2026-01-09.
97. "AMD Ryzen™ AI Max+ 395 specifications" (https://www.amd.com/en/products/processors/laptop/ryzen/ai-300-series/amd-ryzen-ai-max-plus-395.html). *amd.com*. Retrieved 2026-01-09.
98. "Macbook Pro M5 External Display Support" (https://www.apple.com/macbook-pro/specs/). Retrieved 2026-04-06.
99. "Via VL830 USB4 Controller Specifications" (https://www.via-labs.com/product_show.php?id=114).
100. "Via VL832 USB4 Controller Specifications" (https://www.via-labs.com/product_show.php?id=119).
101. "ASM2464PD USB4 Controller Specifications" (https://www.asmedia.com.tw/product/802zX91Yw3tsFgm4/C64ZX59yu4sY1GW5.html).
102. "ASM2464PDX USB4 Controller Specifications" (https://www.asmedia.com.tw/product/bDFzXa0ip1YI7Wj1/C64ZX59yu4sY1GW5.html).
103. "ASM4242 USB4 Controller Specifications" (https://www.asmedia.com.tw/product/e20zx49yU0SZBUH5/363Zx80yu6sY3XH2.html).
104. "Realtek Launched Worldwide First USB-IF Fully Certified (TID: 11930) Multi-Port USB4 Hub Controller (RTS5490), Integrating USB Type C/PD Functionality" (https://www.realtek.com/Article/NewsDetail?id=4609&app_id=18).
105. "RealTek RTS5490 USB4 hub controller port analysis" (https://dancharblog.wordpress.com/2023/06/05/realtek-rts5490-controller-downstream-port-analysis/).

## Specification References


- "USB Type-C® Cable and Connector Specification Release 2.3 | USB-IF" (https://usb.org/document-library/usb-type-cr-cable-and-connector-specification-release-23). *usb.org*. USB-IF. Retrieved 2024-08-05.
- "USB4® Specification v2.0 | USB-IF" (https://usb.org/document-library/usb4r-specification-v20). *usb.org*. USB-IF. Retrieved 2024-08-05.
  - "USB4® Specification v2.0, Adopters Agreement | USB-IF" (https://usb.org/document-library/usb4r-specification-v20). *usb.org*. USB-IF. Retrieved 2024-08-05.
  - "USB4® Specification v2.0, Connection Manager Guide | USB-IF" (https://usb.org/document-library/usb4r-specification-v20). *usb.org*. USB-IF. Retrieved 2024-08-05.
  - "USB4® Specification v2.0 Inter-Domain Service | USB-IF" (https://usb.org/document-library/usb4r-specification-v20). *usb.org*. USB-IF. Retrieved 2024-08-05.
- "USB4® Specification v2.0 with Errata and ECN through September 25, 2024 | USB-IF" (https://usb.org/document-library/usb4r-specification-v20). *usb.org*. USB-IF. Retrieved 2026-03-25.
- "USB Power Delivery | USB-IF" (https://usb.org/document-library/usb-power-delivery). *usb.org*. USB-IF. Retrieved 2024-08-05.
- "USB 2.0 Specification | USB-IF" (https://usb.org/document-library/usb-20-specification). *usb.org*. Retrieved 2024-08-05.
- "USB 3.2 Revision 1.1 - June 2022 | USB-IF" (https://usb.org/document-library/usb-32-revision-11-june-2022). *usb.org*. USB-IF. Retrieved 2024-08-05.


## External links

- USB4 | USB-IF (https://www.usb.org/usb4)
- USB4 Taxonomy | USB-IF (https://www.usb.org/taxonomy/term/61)
- Current USB specifications (https://www.usb.org/documents)
- Podcast with Jit Lim (https://www.youtube.com/watch?v=Nax50U18iyo) from Keysight, 2019-11-21

# Thunderbolt (interface)

**Thunderbolt** is the brand name of a hardware interface for the connection of external peripherals to a computer. It was developed by Intel in collaboration with Apple.[7][8] It was initially marketed under the name **Light Peak**, and first sold as part of an end-user product on 24 February 2011.[1]

Thunderbolt combines PCI Express (PCIe) and DisplayPort (DP) into two serial signals[9][10] and provides DC power via a single cable. Up to six peripherals may be supported by a single connector through various topologies. Thunderbolt 1 and 2 use the same connector as Mini DisplayPort (MDP), whereas Thunderbolt 3, 4, and 5 use the USB-C connector and support USB devices.

**Thunderbolt**

| | |
|---|---|
| **Production history** | |
| **Designer** | Apple<br>Intel |
| **Manufacturer** | Various |
| **Produced** | Since 24 February 2011[1] |
| **Superseded** | IEEE 1394 (FireWire)<br>ExpressCard |
| **General specifications** | |
| **Length** | Up to 3 meters (10 ft) (copper)<br>Up to 60 meters (200 ft) (optical)[2] |
| **Width** | 7.4 mm plug (8.3 mm receptacle) |
| **Height** | 4.5 mm plug (5.4 mm receptacle) |
| **Hot pluggable** | Yes |
| **Daisy chain** | Up to 6 devices[2]<br>Thunderbolt 4: Hub support[3][4] |
| **External** | Yes |
| **Audio signal** | Via DisplayPort protocol or USB-based external audio cards. Supports audio through HDMI converters. |
| **Video signal** | Via DisplayPort protocol |
| **Pins** | Thunderbolt 1 and 2: 20<br>Thunderbolt 3, 4 and 5: 24 |
| **Connector** | Thunderbolt 1 and 2: Mini DisplayPort<br>Thunderbolt 3, 4, and 5: USB-C |
| **Electrical** | |
| **Max. voltage** | Thunderbolt 1 and 2: 18 V (bus power)<br>Thunderbolt 3, 4, and 5: 5 V |
| **Max. current** | Thunderbolt 1 and 2: 550 mA (9.9 W max.)<br>Thunderbolt 3, 4, and 5: 3000 mA (15 W) |
| **Data** | |

## Description

Thunderbolt controllers multiplex one or more individual data lanes from connected PCIe and DisplayPort devices for transmission via two duplex Thunderbolt lanes, then de-multiplex them for use by PCIe and DisplayPort devices on the other end.[2] A single Thunderbolt port supports up to six Thunderbolt devices via hubs or daisy chains; as many of these as the host has DP sources may be Thunderbolt monitors.[11] A device with fewer than four PCIe lanes still qualifies as a Thunderbolt device, but will not support the full capabilities and speeds of Thunderbolt.

A single Mini DisplayPort monitor or other device of any kind may be connected directly or at the very end of the chain. Thunderbolt is interoperable with DP-1.1a compatible devices. When connected to a DP-compatible device, the Thunderbolt port can provide a native DisplayPort signal with four lanes of output data at no more than 5.4 Gbit/s per Thunderbolt lane. When connected to a Thunderbolt device, the per-lane data rate achieves 10 Gbit/s and the four Thunderbolt lanes are configured as two duplex lanes, each 10 Gbit/s, comprising one lane of input and one lane of output.[2]

Thunderbolt can be implemented on PCIe graphics cards, which have access to DisplayPort data and PCIe connectivity, or on the motherboard of computers with onboard video, such as the MacBook Air.[11][12][13]

The interface was originally intended to run exclusively on an optical physical layer using components and flexible optical fiber cabling developed by Intel partners and at Intel's Silicon Photonics lab. It was initially marketed under the name Light Peak[14] and, after 2011, as Silicon Photonics Link.[15] However, it was eventually discovered that conventional copper wiring could furnish the desired 10 Gbit/s per channel at lower cost.

This copper-based version of the Light Peak concept was co-developed by Apple and Intel. Apple registered *Thunderbolt* as a trademark, but later transferred the mark to Intel, which held overriding intellectual property rights.[16] Thunderbolt was commercially introduced on Apple's 2011 MacBook Pro, using the same Apple-developed connector as Mini DisplayPort.

In January 2013, Sumitomo Electric Industries began selling optical Thunderbolt cables of up to 30 m (100 ft) in length in Japan,[17] while in late September 2013, Corning Inc. started selling fiber-optic cables of up to 60 m (200 ft) in length in the United States.[18]

| | |
|---|---|
| **Data signal** | Yes |
| **Bitrate** | Thunderbolt 1: d2 channels, 10 Gbit/s each (20 Gbit/s in total)[5]<br>Thunderbolt 2: 20 Gbit/s in total<br>Thunderbolt 3 and 4: 40 Gbit/s bidirectional<br>Thunderbolt 5: 80 Gbit/s bidirectional |
| **Protocol** | Thunderbolt 1: PCI Express 2.0 × 4,[5] DisplayPort 1.1a[2]<br>Thunderbolt 2: PCI Express 2.0 × 4, DisplayPort 1.2<br>Thunderbolt 3: PCI Express 3.0 × 4, DisplayPort 1.2,[6] USB 3.1 Gen 2<br>Thunderbolt 4: PCI Express 3.0 × 4, DisplayPort 2.0, USB4<br>Thunderbolt 5: PCI Express 4.0 × 4, DisplayPort 2.1, USB4 |

**Pinout**

| | | |
|---|---|---|
| **Pin 1** | GND | Ground |
| **Pin 2** | 'HPD' | Hot plug detect |
| **Pin 3** | HS0TX(P) | HighSpeed transmit 0 (positive) |
| **Pin 4** | HS0RX(P) | HighSpeed receive 0 (positive) |
| **Pin 5** | HS0TX(N) | HighSpeed transmit 0 (negative) |
| **Pin 6** | HS0RX(N) | HighSpeed receive 0 (negative) |
| **Pin 7** | GND | Ground |
| **Pin 8** | GND | Ground |
| **Pin 9** | LSR2P TX | LowSpeed transmit |
| **Pin 10** | GND | Ground (reserved) |
| **Pin 11** | LSP2R RX | LowSpeed receive |
| **Pin 12** | GND | Ground (reserved) |
| **Pin 13** | GND | Ground |
| **Pin 14** | GND | Ground |
| **Pin 15** | HS1TX(P) | HighSpeed transmit 1 (positive) |
| **Pin 16** | HS1RX(P) | HighSpeed receive 1 (positive) |
| **Pin 17** | HS1TX(N) | HighSpeed transmit 1 (negative) |
| **Pin 18** | HS1RX(N) | HighSpeed receive 1 (negative) |
| **Pin 19** | GND | Ground |
| **Pin 20** | DPPWR | Power |

This is the pinout for both sides of the connector, the source side and the sink side. The cable is actually a crossover cable. It swaps all receive and transmit lanes; e.g., HS1TX(P) of the source is connected to HS1RX(P) of the sink.

# History

## Introduction (2009)

Intel introduced Light Peak at the 2009 Intel Developer Forum (IDF), using a prototype Mac Pro logic board to run two 1080p video streams plus LAN and storage devices over a single 30-meter optical cable with modified USB ends.[19] The system was driven by a prototype PCI Express card, with two optical buses powering four ports.[20] Jason Ziller, head of Intel's Optical I/O Program Office, presented the internal components of the technology under a microscope and the delivery of data through an oscilloscope.[21] The technology was described as having an initial speed of 10 Gbit/s over plastic optical cables, alongside promising a final speed of 100 Gbit/s.[22] At the show, Intel announced Light Peak-equipped systems would begin to appear in 2010, and posted a YouTube video showing Light Peak-connected HD cameras, laptops, docking stations, and HD monitors.[23]

Symbol used on Thunderbolt ports

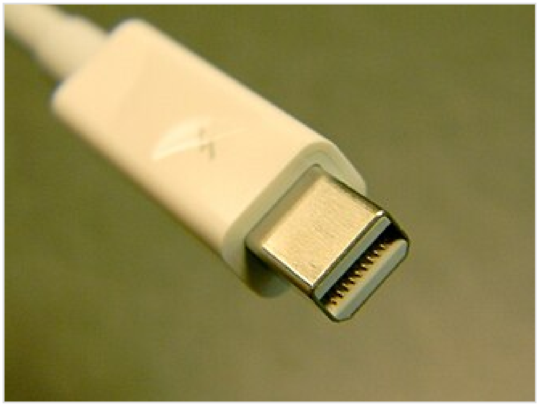
On 4 May 2010, in Brussels, Intel demonstrated a laptop with a Light Peak connector, indicating that the technology had shrunk enough to fit inside such a device, and had the laptop send two simultaneous HD video streams down the connection, indicating that at least some fraction of the software/firmware stacks and protocols were functional. At the same demonstration, Intel officials stated they expected hardware manufacturing to begin around the end of 2010.[24]

In September 2010, some early commercial prototypes from manufacturers were demonstrated at the Intel Developer Forum 2010.[25]

Former Thunderbolt logo, used until September 2023

## Thunderbolt 1 (2011)

CNET's Brooke Crothers said it was rumored that the early-2011 MacBook Pro update would include some sort of new data port, and he speculated it would be Light Peak (Thunderbolt).[26] At the time, there were no details on the physical implementation, and mock-ups appeared showing a system similar to the earlier Intel demos that utilizes a combined USB/Light Peak port.[27] Shortly before the release of the new machines, the USB Implementers Forum (USB-IF) announced it would not allow such a combination port, and that USB was not open to modification in that way.



Thunderbolt 1 or 2 connector

Other implementations of the technology began in 2012, with desktop boards offering the interconnection now available.[28]

Apple stated in February 2011 that the port was based on Mini DisplayPort, not USB. As the system was described, Intel's solution to the display connection problem became clear: Thunderbolt controllers multiplex data from existing DP systems with data from the PCIe port into a single cable. Older displays using DP 1.1a or earlier must be located at the end of a Thunderbolt device chain, but native displays can be anywhere along the line.[12] Thunderbolt devices can go anywhere on the chain. In that respect, Thunderbolt shares a relationship with the older ACCESS.bus system, which used the display connector to support a low-speed bus.

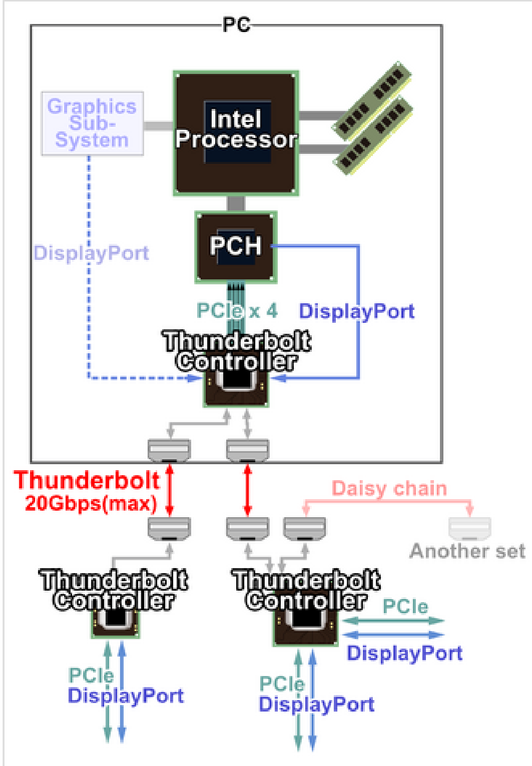
Apple states that up to six daisy-chained peripherals are supported per Thunderbolt port[29] and that the display should be attached at the end of the chain if it does not support daisy-chaining.

In February 2011, Apple introduced MacBook Pro (13-inch, Early 2011),[30] MacBook Pro (15-inch, Early 2011),[31] and MacBook Pro (17-inch, Early 2011)[32] featuring one Thunderbolt port. In May 2011, Apple introduced the iMac (21.5-inch, Mid 2011),[33] featuring one Thunderbolt port, and the iMac (27-inch, Mid 2011),[34] featuring two Thunderbolt ports. In July 2011, Apple introduced Mac Mini (Mid 2011),[35] MacBook Air (11-inch, Mid 2011),[36] MacBook Air (13-inch, Mid 2011),[37] and Apple Thunderbolt Display,[38] featuring one Thunderbolt port for daisy-chaining or other devices.

In May 2011, Apple announced a new line of iMacs that includes the Thunderbolt interface.[39]

The Thunderbolt port on the new Macs is in the same location relative to other ports and maintains the same physical dimensions and pinout as the prior MDP connector. The main visible difference on Thunderbolt-equipped Macs is a Thunderbolt symbol next to the port.[11]

The DisplayPort standard is partially compatible with Thunderbolt, as the two share Apple's physically compatible MDP connector. The Target Display mode on iMacs requires a Thunderbolt cable to accept a video-in signal from another Thunderbolt-capable computer.[40] A DP monitor must be the last (or only) device in a chain of Thunderbolt devices.



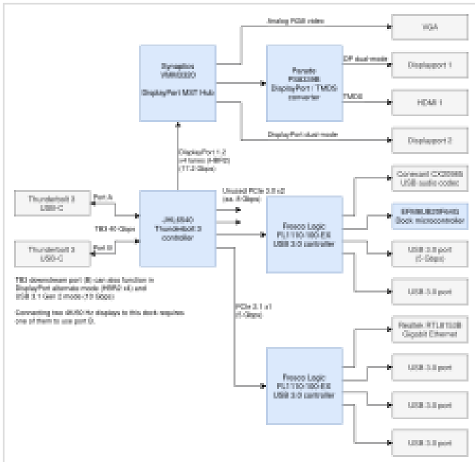

**Thunderbolt link connections**

Intel provides two types of Thunderbolt controllers: a two-port and a one-port type. Both peripherals and computers require a controller.

Intel announced it would release a developer kit in the second quarter of 2011,[41] while manufacturers of hardware development equipment have indicated they will add support for testing and development of Thunderbolt devices.[42] The developer kit is being provided only on request.[43]

In July 2011, Sony released its Vaio Z21 line of notebook computers that had a "Power Media Dock" that uses optical Thunderbolt (Light Peak) to connect to an external graphics card using a combination port that behaves like USB electrically, but that also includes the optical interconnect required for Thunderbolt.

Thunderbolt 1 ran at 10 Gbit/s, making it faster than USB at the time.[44]



Block diagram of a typical Thunderbolt 3 dock

## Thunderbolt 2 (2013)

In June 2013, Intel announced that the next version of Thunderbolt, based on the controller code-named "Falcon Ridge" (running at 20 Gbit/s), is officially named "Thunderbolt 2" and entered production in 2013.[45] The data rate of 20 Gbit/s is made possible by joining the two existing 10 Gbit/s-channels, which does not change the maximum bandwidth, but makes its usage more flexible.

Thunderbolt 2 port on a MacBook Pro

In June 2013, Apple announced the Mac Pro (Late 2013)[46] featuring six Thunderbolt 2 ports. In October 2013, Apple announced MacBook Pro (Retina, 13-inch, Late 2013),[47] and MacBook Pro (Retina, 15-inch, Late 2013)[48] featuring two Thunderbolt 2 ports. In October 2014, Apple announced Mac Mini (Late 2014),[49] and iMac (Retina 5K, 27-inch, Late 2014)[50] featuring two Thunderbolt 2 ports. In March 2015, Apple announced MacBook Air (11-inch, Early 2015),[51] and MacBook Air (13-inch, Early 2015)[52] featuring one Thunderbolt 2 port.

At the physical level, the bandwidth of Thunderbolt 1 is identical to that of Thunderbolt 2, which means Thunderbolt 1 cabling is compatible with Thunderbolt 2 interfaces. At the logical level, Thunderbolt 2 enables channel aggregation, whereby the two previously separate 10 Gbit/s channels can be combined into a single logical 20 Gbit/s channel.[53]

Intel says Thunderbolt 2 will be able to transfer a 4K video while simultaneously displaying it on a discrete monitor.[54]

Thunderbolt 2 incorporates DisplayPort 1.2 support, which allows for video streaming to a single 4K video monitor or dual QHD monitors. Thunderbolt 2 is backward compatible, which means that all Thunderbolt cables and connectors are compatible with Thunderbolt 1.

The first Thunderbolt 2 product for the consumer market was Asus's Z87-Deluxe/Quad motherboard, announced on 19 August 2013,[55] and the first system released with Thunderbolt 2 was Apple's late 2013 Retina MacBook Pro, on 22 October 2013.[56]

## Thunderbolt 3 (2015)

Thunderbolt 3, 4, or 5 ports

The USB Full-Featured Type-C (USB-C) plug, the plug for Thunderbolt 3, 4, and 5

Thunderbolt 3 is a hardware interface developed by Intel.[57] It shares USB-C connectors with USB, supports USB 3.1 Gen 2,[58][59][60] and can require special *active* cables for maximum performance for cable lengths over 0.5 meters (1.5 feet). Compared to Thunderbolt 2, it doubles the bandwidth to 40 Gbit/s (5 GB/s). It allows up to four lanes of PCI Express 3.0 (32.4 Gbit/s) for general-purpose data transfer, and four lanes of DisplayPort 1.4 HBR3 (32.40 Gbit/s before 8/10 encoding removal, and 25.92 Gbit/s after) for video,[61] but the maximum combined data rate cannot exceed 40 Gbit/s; video data is prioritized and so uses all needed bandwidth, the remainder being available to PCIe. DP 1.2 support is mandatory, while DP 1.4 is optional. Other overheads are possible on PCIe data (1.5% of 128b/130b is also removed) and Thunderbolt 3 protocol (either optimise for speed or for latency), the last one gives only 21.6 Gbit/s to 25 Gbit/s.[62] Thunderbolt 3 uses 64b/66b encoding after that, which means the real rate is bigger than 40 Gbit/s, 2 times 20.625 Gbit/s.

Intel's Thunderbolt 3 controller (codenamed *Alpine Ridge*, or the new *Titan Ridge*) halves power consumption, and simultaneously drives two external 4K displays at 60 Hz (or a single external 4K display at 120 Hz, or a 5K display at 60 Hz when using Apple's implementation for the late-2016 MacBook Pros) instead of just the single display previous controllers can drive. The new controller supports PCIe 3.0 and other protocols, including DisplayPort 1.2 (allowing for 4K resolutions at 60 Hz).[63] Thunderbolt 3 has up to 15 watts of power delivery on copper cables and no power delivery capability on optical cables. Using USB-C on copper cables, it can incorporate USB power delivery, allowing the ports to source or sink up to 100 watts of power. This eliminates the need for a separate power supply from some devices. Thunderbolt 3 allows backwards compatibility with the first two versions by the use of adapters or transitional cables.[64][65][66]

Intel offers three varieties for each of the controllers:[67]

- Double Port (DP) uses a PCIe 3.0 ×4 link to provide two Thunderbolt 3 ports (DSL6540, JHL6540, JHL7540)
- Single Port (SP) uses a PCIe 3.0 ×4 link to provide one Thunderbolt 3 port (DSL6340, JHL6340, JHL7340)
- Low Power (LP) uses a PCIe 3.0 ×2 link to provide one Thunderbolt 3 port (JHL6240).

This follows previous practice, where higher-end devices such as the second-generation Mac Pro, iMac, Retina MacBook Pro, and Mac Mini use two-port controllers; while lower-end, lower-power devices, like the MacBook Air, use the one-port version.

Support was added to Intel's Skylake architecture chipsets, shipping during late 2015 into early 2016.[64][65][66]

Devices with Thunderbolt 3 ports began shipping at the beginning of December 2015, including notebooks running Microsoft Windows (from Acer, Asus, Clevo, HP, Dell, Dell Alienware, Lenovo, MSI, Razer, and Sony), as well as motherboards (from Gigabyte Technology), and a 0.5 m Thunderbolt 3 passive USB-C cable (from Lintes Technology).[68]

In October 2016, Apple announced MacBook Pro (13-inch, 2016, 2 Thunderbolt 3 Ports),[69] which, as the name indicates, features two Thunderbolt 3 ports, MacBook Pro (13-inch, 2016, 4 Thunderbolt 3 Ports),[70] and MacBook Pro (15-inch, 2016),[71] which features four Thunderbolt 3 ports. In June 2017, Apple announced iMac (21.5-inch, 2017),[72] iMac (Retina 4K, 21.5-inch, 2017),[73] iMac (Retina 5K, 27-inch, 2017)[74] which feature two Thunderbolt 3 ports, as well as the iMac Pro,[75] which featured four Thunderbolt 3 ports and was released in December 2017. In October 2018, Apple announced MacBook Air (Retina, 13-inch, 2018),[76] featuring 2 Thunderbolt 3 ports and Mac mini (2018)[77] featuring four Thunderbolt 3 ports. In June 2019, Apple unveiled Mac Pro (2019)[78] and Mac Pro (Rack, 2019)[79] featuring up to twelve Thunderbolt 3 ports, and Pro Display XDR,[80] which features one Thunderbolt 3 port, both released in December 2019. In March 2022, Apple released Studio Display[81] featuring one Thunderbolt 3 port.

On 8 January 2018, Intel announced a product refresh (codenamed Titan Ridge) with "enhanced robustness" and support for DisplayPort 1.4. Intel offers a single-port (JHL7340) and double-port (JHL7540) version of this host controller and a peripheral controller supporting two Thunderbolt 3 ports (JHL7440). The new peripheral controller can now act as a USB sink (compatible with regular USB-C ports).[82]

The Apple Pro Display XDR, which macOS allows to connect using two HBR3 connections to a Mac, doesn't support Display Stream Compression (DSC). That would be 51.84 Gbit/s, impossible for Thunderbolt 3, but it works because the two 3008×3384 10bpc 60 Hz 648.91 MHz signals of the XDR display only require 38.9 Gbit/s total, and Thunderbolt does not transmit the DisplayPort stuffing symbols used to fill the HBR3 bandwidth.

### USB4 (2019)

The USB4 specification was released on 29 August 2019 by the USB Implementers Forum,[83] based on the Thunderbolt 3 protocol specification.[84]

It supports 40 Gbit/s (5 GB/s) throughput, is *optionally* compatible with Thunderbolt 3, and is backward compatible with USB 3.2 and USB 2.0.[85][86] The architecture defines a method to share a single high-speed link with multiple end device types dynamically that best serves the transfer of data by type and application.

USB4 supports DisplayPort 2.0 over its alternative mode.[87]

DisplayPort 2.0 can support higher than 8K resolution at 60 Hz losslessly due to new UHBR 10, 13.5, and 20 signaling standards (DSC 1.2 used in DisplayPort 1.4 for that resolution is not lossless) in 8 bit and 8K 60 Hz with 10 bit color and use up to 80 Gbit/s (effective bandwidth 77.37 Gbit/s), which is double the amount available to USB data, because (just as previously in DisplayPort 1.4) it sends almost all the data in one direction (to the monitor) and can thus use all four data lanes at once.[88] Resolutions up to 16K (15360×8640) 60 Hz display with 10-bit Y'CbCr 4:4:4 or RGB are possible.[89]

In November 2020, Apple announced MacBook Air (M1, 2020),[90] MacBook Pro (13-inch, M1, 2020),[91] and Mac mini (M1, 2020)[92] featuring USB4 ports.

### USB4 PCIe Mode

USB4 makes the PCIe aspects of Thunderbolt "open source" – PCIe USB devices can be released without Thunderbolt certification. But notably, those devices will not be allowed to use Thunderbolt branding. However, Thunderbolt 4 devices use PCIe Mode with added certification labeling, and promoting backwards compatibility. This means multiple rival devices may use different brandings to accomplish the same task. USB4 PCIe devices can be backwards compatible with Thunderbolt 1–3, but this is not required. USB4 PCIe Mode is not an Alternate Mode, like DisplayPort Alternate Mode, and Microsoft requires devices with USB4 to include PCIe support in order to be WHQL/Windows certified PCs.[93][94][95][96][97]

## Thunderbolt 4 (2020)

Thunderbolt 4 was announced at CES 2020[98] and the final specification was released in July 2020.[99] The key differences between Thunderbolt 4 and Thunderbolt 3 are a minimum bandwidth requirement of 32 Gbit/s for PCIe link, support for dual 4K displays (DisplayPort 1.4),[100][101] and Intel VT-d-based direct memory access protection to prevent physical DMA attacks.

Another major improvement is that Thunderbolt 4 supports Thunderbolt Alternate Mode USB hubs ("Multi-port Accessory Architecture"), and not just daisy chaining.[3][4] Those hubs are backward compatible with Thunderbolt 3 devices and can be backward compatible with Thunderbolt 3 hosts (Titan Ridge only; with Alpine Ridge, the additional downstream ports get downgraded to USB 3).[102][103]

The maximum bandwidth remains at 40 Gbit/s, the same as Thunderbolt 3 and four times as fast as USB 3.2 Gen 2x1.[104][99] Supporting products began arriving in late 2020 and included Tiger Lake mobile processors for Intel Evo notebooks and 8000-series standalone Thunderbolt controllers (codenamed Goshen Ridge for devices and Maple Ridge for hosts). On desktop native Thunderbolt 4 was added in Intel Arrow Lake PCIe 4 lanes directly on the CPU, and by doing so removed data limitation of 25 Gbit/s, now all 40 Gbit/s could be used for data.

## Thunderbolt 5 (2023)

On September 12, 2023, Intel previewed Thunderbolt 5 (codenamed Barlow Ridge), aligned to the USB Implementers Forum's (USB-IF) USB4 2.0 specification. It provides symmetric bandwidth of 80 Gbit/s, e.g., for mass-storage devices, double that of Thunderbolt 4, and unidirectional bandwidth of 120 Gbit/s for displays (three times that of Thunderbolt 3 and 4), supporting dual 8K displays at 60 Hz. The minimum required bandwidth remains unchanged from Thunderbolt 4: 32 Gbit/s for the PCIe link.

The full specifications cover:

- Supporting the latest version of USB4 2.0 80 Gbit/s specification

- Doubling the total bandwidth of Thunderbolt 4 to 80 Gbit/s symmetric, or 120 Gbit/s in either direction and 40 the other, generally for video-intensive uses
- Support for DisplayPort 2.1
- Two times (64 Gbit/s) the PCI Express data-throughput using PCI Express Gen. 4 × 4, for faster storage and external graphics
- Up to 240 W of charging power downstream
- Works with existing passive cables up to 1 m (3.3 ft) via PAM-3
- Compatible with previous versions of Thunderbolt, USB, and DisplayPort
- Supported by Intel's enabling and certification programs

Intel announced that computers and accessories compatible with Thunderbolt 5 will come out starting in 2024.[105]

In October 2024, Apple announced the Mac Mini (M4 Pro, 2024)[106] and the 14-inch and 16-inch MacBook Pro (M4 Pro/Max, 2024)[107] with three Thunderbolt 5 ports. In March 2025, Apple announced the release of the Mac Studio (M4 Max/M3 Ultra, 2025), with the Max model having four Thunderbolt 5/USB4 USB-C ports and the Ultra having six Thunderbolt 5/USB4 USB-C ports.[108] All these devices operate at up to 120Gb/s in both Thunderbolt 5 or USB4 mode.[109]

## Royalty situation (2019–)

On 24 May 2017, Intel announced that Thunderbolt 3 would become a royalty-free standard to OEMs and chip manufacturers in 2018, as part of an effort to boost the adoption of the protocol.[110] The Thunderbolt 3 specification was later released to the USB-IF on 4 March 2019, making it royalty-free, to be used to form USB4.[84][111][112] Intel says it will retain control over the certification of all Thunderbolt 3 devices.[113] Intel also states it employs "mandatory certification for all Thunderbolt products".[114]

Before March 2019, there were no AMD chipsets or computers with Thunderbolt support released or announced due to the certification requirements (Intel did not certify non-Intel platforms). However, the YouTuber Wendell Wilson from Level1Techs was able to get Thunderbolt 3 support on an AMD computer with a Threadripper CPU and Titan Ridge add-in card working by modifying the firmware, indicating that the lack of Thunderbolt support on non-Intel systems is not due to any hardware limitations.[115][116] As of May 2019, it is possible to have Thunderbolt 3 support on AMD using add-in cards without any problems,[117] and motherboards like ASRock X570 Creator already have Thunderbolt 3 ports.[118]

In January 2020, Intel certified[119] ASRock X570 Phantom Gaming ITX/TB3, and now vendors are freely allowed to produce Thunderbolt controller silicon (even though those ASRock motherboards used Intel Titan Ridge).[120]

Asus currently supports Thunderbolt 3 on AMD with the add-in card Thunderboltex 3-TR, being compatible with AMD motherboards and Ryzen 3, 5 (56xx): ROG Strix B550-E Gaming, ROG Strix B550-F Gaming, Prime B550-PLUS, TUF Gaming B550-Plus.[121] The ASUS ProArt B550-Creator has 2 Thunderbolt 4 ports.[122]

GIGABYTE also has a pair of certified motherboards, B550 VISION D-P and B550 VISION D,[123][124] with an Intel Thunderbolt 3 controller.

## Copper vs. optical

Though Thunderbolt was originally conceived as an optical technology, Intel switched to copper-based electrical connections to reduce costs and to supply up to 10 watts of power to connected devices.[125]

In 2009, Intel officials said the company was "working on bundling the optical fiber with copper wire so Light Peak can be used to power devices plugged into the PC."[126] In 2010, Intel said the original intent was "to have one single connector technology" that would let "electrical USB 3.0 ... and piggyback on USB 3.0 or 4.0 DC power."[127] Light Peak aimed to make great strides in consumer-ready optical technology, by then having achieved "[connectors rated] for 7,000 insertions, which matches or exceeds other PC connections ... cables [that were tied] in multiple knots to make sure it didn't break and the loss is acceptable," and, "You can almost get two people pulling on it at once and it won't break the fibre." They predicted that "Light Peak cables will be no more expensive than HDMI."[128]

In January 2011, Intel's David Perlmutter told *Computerworld* that initial Thunderbolt implementations would be based on copper wires.[127] "The copper came out very good, surprisingly better than what we thought," he said.[129] A major advantage of copper is its ability to carry power. The final Thunderbolt standard specifies 10 W DC on every port. *See comparison section below.*

Intel and industry partners are still developing optical Thunderbolt hardware and cables.[130] The optical fiber cables would run "tens of meters" but would not supply power, at least not initially.[12][131][132] The version from Corning contains four 80/125 μm VSDN (Very Short Distance Network) fibers to transport an infrared signal up to 190 m (600 ft).[133] The conversion of an electrical signal to optical is embedded into the cable itself, so the current MDP connector is forward-compatible. Eventually, Intel hopes for a purely optical transceiver assembly embedded in the PC.[132]

The first such optical Thunderbolt cable was introduced by Sumitomo Electric Industries in January 2013.[134] It is available in lengths of 10 m (30 ft), 20 m (70 ft), and 30 m (100 ft). However, those cables are retailed almost exclusively in Japan, and the price is 20 to 30 times that of copper Thunderbolt cables.

German company DeLock also released optical Thunderbolt cables in lengths of 10 m (30 ft), 20 m (70 ft), and 30 m (100 ft) in 2013, priced similarly to the Sumitomo ones, and retailed only in Germany.[135]

In September 2013, glass company Corning Inc. released the first range of optical Thunderbolt cables available in the Western marketplace, along with optical USB 3.0 cables, both under the brand name "Optical Cables".[18] Half the diameter and a fifth the mass of comparable copper Thunderbolt cables, they work with the 10 Gbit/s Thunderbolt protocol and the 20 Gbit/s Thunderbolt 2 protocol, and thus are able to work with all self-powered Thunderbolt devices (unlike copper cables, optical cables cannot provide power).[18] The cables extend the current 30 m (100 ft) maximum length offered by copper to a maximum of 60 m (200 ft).

Before 2020, there were no optical Thunderbolt 3 cables on the market. However, optical Thunderbolt 1 and 2 cables could be used at the time with Apple's Thunderbolt 3 (USB-C) to Thunderbolt 2 adapters on each end of the cable. This achieves connections up to the 60 m (200 ft) maximum offered by previous versions of the standard.[136]

In April 2019, Corning showed an optical Thunderbolt 3 cable at the 2019 NAB Show in Las Vegas.[137] Just over a year later, in September 2020, Corning released their optical Thunderbolt 3 cables in lengths of 5 m (20 ft), 10 m (30 ft), 15 m (50 ft), 25 m (80 ft), and 50 m (160 ft).[138] In the meantime, Taiwanese company Areca released optical Thunderbolt 3 cables in April 2020 in lengths of 10 m (30 ft), 20 m (70 ft), and 30 m (100 ft).[139]

Copper versions of Thunderbolt 4 cables offer full 40 Gbit/s speed and support backward compatibility with all versions of USB (up to USB4), DisplayPort Alternate Mode (DP 1.4 HBR3), and Thunderbolt 3. Released in early 2021, they were also to be available in three specified lengths: 0.2 m (0.66 ft), 0.8 m (2.6 ft), and 2 m (6.6 ft) – with many companies initially offering 0.8 m (2.6 ft) ones.

Copper Thunderbolt 4 cables up to 1.0 m (3.3 ft) are passive cables, while longer cables must integrate active signal conditioning circuitry. 2 m (6.6 ft) maximum is the length of active cables available from most brands, including CalDigit,[140] Cable Matters,[141] et al., while Apple are currently the only company that offers a 3 m (9.8 ft) active copper cable.

Optical Thunderbolt 4 cables were targeting lengths from ≈5 m (16 ft) to 50 m (160 ft),[142] although this may not happen, instead jumping to Thunderbolt 5 optical cables, sometime after the arrival of that standard in late 2024.

The 80 Gbps mode of USB4, used in Thunderbolt 5, uses three-level PAM3 instead of two-level NRZ encoding used by all older versions of Thunderbolt and USB. As a result, active (including optical) cables that have worked for previous generations will not allow this new speed to be achieved. Due to how the link is negotiated, Thunderbolt 3 cables will not work *at all* while Thunderbolt 4 cables will work at 40 Gbps speed.[143]

# Compatibility

## Before Thunderbolt 3

### Peripheral devices

The first Thunderbolt peripheral devices appeared in retail stores only in late 2011, following Apple's release of its first Thunderbolt-equipped computer in early 2011 with MacBook Pro, with the relatively expensive *Pegasus R4* (4-drive) and *Pegasus R6* (6-drive) RAID enclosures by Promise Technology aimed at the prosumer and professional market, initially offering up to 12 TB of storage, later increased to 18 TB. Sales of these units were hurt by the 2011 floods in Thailand (who manufacture much of the world's supply of hard-drives) resulting in a cut to worldwide hard-drive production and a subsequent driving-up of storage costs, hence the retail price of these Promise units increased in response, contributing to a slower take-up of the devices.

A multiple hard disk storage device that attaches to a computer through a Thunderbolt connection

It also took some time for other storage manufacturers to release products: most were smaller devices aimed at the professional market, and focused on speed rather than high capacity. Many storage devices were under a terabyte in size, with some featuring SSDs for faster external-data access rather than standard hard-drives.

Other companies have offered interface products that can route multiple older, usually slower, connections through a single Thunderbolt port. In July 2011, Apple released its Apple Thunderbolt Display, whose gigabit Ethernet and other older connector types made it the first hub of its type. Later, companies such as Belkin, CalDigit (http://www.caldigit.com/us/), Other World Computing, Matrox, StarTech, and Elgato have all released Thunderbolt docks.

As of late 2012, few other storage devices with capacities of ten terabytes or more had appeared. Exceptions included Sonnet Technologies' highly priced professional units, and Drobo's four- and five-drive enclosures, the latter featuring their own BeyondRAID proprietary data-handling system.

Backwards compatibility with non-Thunderbolt-equipped computers was a problem, as most storage devices featured only two Thunderbolt ports, for daisy-chaining up to six devices from each one. In mid-2012, LaCie, Drobo, and other device makers started to swap out one of the two Thunderbolt ports for a USB 3.0 connection on some of their low-to-mid end products. Later models had the USB 3.0 added *in addition* to the two Thunderbolt ports, including those from LaCie on their *2big* range.

### Apple devices before Thunderbolt 3

Apple released its first Thunderbolt-equipped computer in early 2011 with MacBook Pro, and have continued to immediately update their devices with newer generations of Thunderbolt as soon as available.

List of Apple devices featuring Thunderbolt ports:[144]

- MacBook Pro (Retina, 13-inch, Late 2012 to Early 2013)
- MacBook Pro (Retina, 15-inch, Mid 2012 to Early 2013)
- MacBook Pro (17-inch, Early 2011 to Late 2011)
- MacBook Pro (15-inch, Early 2011 to Mid 2012)
- MacBook Pro (13-inch, Early 2011 to Mid 2012)
- MacBook Air (13-inch, Mid 2011 to Early 2014)
- MacBook Air (11-inch, Mid 2011 to Early 2014)
- Mac Mini (Mid 2011 to Late 2012)
- iMac (27-inch, Mid 2011 to Late 2013)
- iMac (21.5-inch, Mid 2011 to Mid 2014)

The late 2013 Retina MacBook Pro was the first product to have Thunderbolt 2 ports, following which manufacturers started to update their model offerings to those featuring the newer, faster, 20 Gbit/s connection throughout 2014.[145] Again, among the first was Promise Technology, who released updated *Pegasus 2* versions of their R4 and R6 models along with an even larger R8 (8-drive) RAID unit, offering up to 32 TBs of storage. Later, other brands similarly introduced high capacity models with the newer connection type, including SanDisk Professional (with their *G-RAID Studio* models offering up to 24 TB) and LaCie (with their *5big*, and rack mounted *8big* models, offering up to 48 TB). LaCie also offering updated designed versions of their *2big* mainstream consumer models, up to 12 TB, using new 6 TB hard-drives.

List of Apple devices featuring Thunderbolt 2 ports:[144]

- MacBook Pro (Retina, 15-inch, Late 2013 to Mid 2015)
- MacBook Pro (Retina, 13-inch, Late 2013 to Early 2015)
- MacBook Air (13-inch, Early 2015 to 2017)
- MacBook Air (11-inch, Early 2015)
- Mac Mini (Late 2014)
- iMac (Retina 4K, 21.5-inch, Late 2015)
- iMac (21.5-inch, Late 2015)
- iMac (Retina 5K, 27-inch, Late 2014 to Late 2015)
- Mac Pro (Late 2013)

## After Thunderbolt 3

Although Thunderbolt initially had poor hardware support outside of Apple devices, and had been relegated to a niche gadget port, adoption of Thunderbolt 3, which uses the USB-C connector standard, meant wider market acceptance, especially as it later became part of the USB4 standard.

Thunderbolt 3 was introduced in late 2015, with several motherboard manufacturers and OEM laptop manufacturers including Thunderbolt 3 with their products. Gigabyte and MSI, large computer component manufacturers, entered the market for the first time with Thunderbolt 3 compatible components.[146][147]

Dell was the first to include Thunderbolt 3 ports in laptops with their XPS Series and their Dell Alienware range.[148]

Details on compatibility are available from the Thunderbolt Technology Community Web site.[6]

A single Thunderbolt 3 or later port provides data transfer, support for two 4K 60 Hz displays, and quick notebook charging up to 100W with a single cable. Any Thunderbolt or USB dock can connect to a Thunderbolt 3 computer. USB devices can be connected to a Thunderbolt 3 or later port. DisplayPort and Mini DisplayPort devices are supported.

Some limited functionality may be available if a Thunderbolt device is connected to an ordinary USB-C port; this is implementation-dependent, and not guaranteed.

### Peripherals since Thunderbolt 3

Several vendors such as Dell, HP Inc, Lenovo have released docks for their devices offering power button capabilities so that laptops can be powered on even if lid is closed and device power button is not accessible.
Some docks are equipped with two distinc NICs, each supporting a different bus HP Thunderbolt G4 Dock specifications (https://support.hp.com/us-en/document/ish_6213504-6213991-16) ThinkPad Universal Thunderbolt 4 Dock (https://www.lenovo.com/us/en/p/accessories-and-software/docking/docking_thunderbolt-docks/40b00135us). One of the NICs uses a USB interface for compatibility while the other NIC uses a PCIe interface for higher performance. The USB NIC supports 1Gbit/s Ethernet while the PCIe NIC supports 2.5Gbit/s Ethernet. When the dock is connected to a Thunderbolt/USB4 port the PCIe NIC is activated. When the dock is connected to a non Thunderbolt USB port the USB NIC is activated. This does not apply to Lenovo ThinkPad Universal Thunderbolt 4 Dock (40B0) since the vendor will keep the PCIe NIC disabled if the PC is based on a non vPro Intel CPU or an AMD CPU. This is not mandated by some hardware limitation, it is purely a marketing decision. In contrast, the HP Thunderbolt G4 dock will activate the PCIe NIC on all Thunderbolt/USB4 capable PCs based on Intel or AMD CPUs.

### Backwards compatibility

The signaling for thunderbolt 1 and 2 devices are electrically compatible with Thunderbolt 3/4/5 ports. As a result, TB1/2 devices only require a passive adapter to connect to a later-version port if the device does not use bus power. Thunderbolt 3 devices can also be used with Thunderbolt 1 or 2 hosts using an adapter.[149] Apple makes an adapter that works "bidirectionally".[150]

However, newer Intel firmware on Windows has disabled Thunderbolt 1 and 2 support due to security concerns; older firmware allowed their use. Apple provides backwards support for Thunderbolt 1 and 2 devices on their Thunderbolt 4-enabled systems.

Thunderbolt 4/5 ports supports Thunderbolt 3 devices. Thunderbolt 4/5 devices do not work with Thunderbolt 1/2 hosts even with an adapter. Thunderbolt 4/5 devices work with Thunderbolt 3 ports on Apple devices, but compatibility is not guaranteed on the Thunderbolt 3 ports on other devices due to a smaller subset of Thunderbolt 3 being implemented.[150]

### Apple devices (since TB3)

Apple first included Thunderbolt 3 on Mac in 2016.

List of Apple devices featuring Thunderbolt 3 ports:[144]

- MacBook Pro (13-inch, Two Thunderbolt 3 ports, 2016 to 2020)
- MacBook Pro (13-inch, Four Thunderbolt 3 ports, 2016 to 2020)
- MacBook Pro (15-inch, 2016 to 2019)
- MacBook Pro (16-inch, 2019)
- MacBook Air (Retina, 13-inch, 2018 to 2020)
- iMac (Retina 5K, 27-inch, 2017 to 2020)
- iMac (Retina 4K, 21.5-inch, 2017 to 2019)
- iMac (21.5-inch, 2017)
- iMac Pro (2017)
- Mac Pro (2019 + Rack, 2019)
- Mac Mini (2018)
- Pro Display XDR (2019)
- Studio Display (2022)

List of Apple devices featuring Thunderbolt 3/USB4 ports include:[144]

- MacBook Pro (13-inch, M1, 2020 to M2, 2022)
- MacBook Pro (14-inch, M3, 2023)
- MacBook Air (13-inch, M1, 2020 to M3, 2024)
- MacBook Air (15-inch, M2, 2023 to M3, 2024)
- Mac Mini (M1, 2020)
- iMac (24-inch, M1, 2021 to M3, 2023)
- iPad Pro 11-inch (3rd generation, 2021 to 4th generation, 2022)
- iPad Pro 12.9-inch (5th generation, 2021 to 6th generation, 2022)
- iPad Pro 11-inch (M4, 2024)
- iPad Pro 13-inch (M4, 2024)
- iPad Pro 11-inch (M5, 2025)
- iPad Pro 13-inch (M5, 2025)

Apple started to include Thunderbolt 4 on some of their devices, starting in 2021 with the MacBook Pro.

List of Apple devices featuring Thunderbolt 4 ports include:[144]

- MacBook Pro (14-inch, M1 Pro/Max, 2021 to M3 Pro/Max, 2023)
- MacBook Pro (16-inch, M1 Pro/Max, 2021 to M3 Pro/Max, 2023)
- Mac Studio (2022 to 2023)
- Mac Mini (2023)
- Mac Pro (2023 + Rack, 2023)
- iMac (24-inch, M4, 2024)
- Mac Mini (M4, 2024)

- MacBook Pro (14-inch, M4, 2024)
- MacBook Air (13-inch, M4, 2025)
- MacBook Air (15-inch, M4, 2025)
- MacBook Pro (14-inch, M5, 2025)

Apple started to include Thunderbolt 5 on some of their devices, starting in 2024 with the Mac Mini (M4 Pro) and 14-inch/16-inch MacBook Pro (M4 Pro/Max).

List of Apple devices featuring Thunderbolt 5 ports include:

- Mac Mini (M4 Pro, 2024)
- MacBook Pro (14-inch, M4 Pro/Max, 2024)
- MacBook Pro (16-inch, M4 Pro/Max, 2024)
- Mac Studio (M3 Ultra/M4 Max, 2025)
- MacBook Pro (14-inch, M5 Pro/Max, 2026)
- MacBook Pro (16-inch, M5 Pro/Max, 2026)

# Security vulnerabilities

## Vulnerability to DMA attacks

Thunderbolt 3 – like many high-speed expansion buses, including PCI Express, PC Card, ExpressCard, FireWire, PCI, and PCI-X — is potentially vulnerable to a direct memory access (DMA) attack. If users extend the PCI Express bus (the most common high-speed expansion bus in systems as of 2018) with Thunderbolt, it allows very low-level access to the computer. An attacker could physically attach a malicious device, which, through its direct and unimpeded access to system memory and other devices, would be able to bypass almost all security measures of the operating system, allowing the attacker to read and write system memory, potentially exposing encryption keys or installing malware.[151] Such attacks have been demonstrated by modifying inexpensive commodity Thunderbolt hardware.[152][153] The IOMMU virtualization, if present, and configured by the BIOS and the operating system, can close a computer's vulnerability to DMA attacks,[152] but only if the IOMMU can block the DMA access of malicious device. As of 2019, the major OS vendors had not taken into account the variety of ways in which a malicious device could take advantage of complex interactions between multiple emulated peripherals, exposing subtle bugs and vulnerabilities.[154] Some motherboard and UEFI implementations offer Kernel DMA Protection. Intel VT-d-based direct memory access (DMA) protection is a mandatory requirement for Thunderbolt 4 Host Certification.[155]

This vulnerability is not present when Thunderbolt is used as a system interconnection (IPoTB supported on OS X Mavericks), because the IP implementation runs on the underlying Thunderbolt low-latency packet-switching fabric, and the PCI Express protocol is not present on the cable. That means that if IPoTB networking is used between a group of computers, there is no threat of such DMA attack between them.[151][152][156][157]

## Vulnerability to Option ROM attacks

When a system with Thunderbolt boots, it loads and executes Option ROMs from attached devices. A malicious Option ROM can allow malware to execute before an operating system is started. It can then invade the kernel, log keystrokes, or steal encryption keys.[158] The ease of connecting Thunderbolt devices to portable computers makes them ideal for evil-maid attacks.[159]

Some systems load Option ROMs during firmware updates, allowing the malware in a Thunderbolt device's Option ROM to potentially overwrite the SPI flash ROM containing the system's boot firmware.[160][161] In February 2015, Apple issued a Security Update to Mac OS X to eliminate the vulnerability of loading Option ROMs during firmware updates, although the system is still vulnerable to Option ROM attacks during normal boots.[162]

Firmware-enforced boot security measures, such as UEFI Secure Boot (which specifies the enforcement of signatures or hash allowlists of Option ROMs) are designed to mitigate this kind of attack.

### Vulnerability to data exposure attacks (Thunderspy)

In May 2020, seven major security flaws were discovered in the Thunderbolt protocol, collectively named Thunderspy. They allow a malicious party to access all data stored in a computer, even if the device is locked, password-protected, and has an encrypted hard drive. These vulnerabilities affect all Thunderbolt 1, 2 and 3 ports.[153] The attack requires the computer to be in sleep mode and have a Thunderbolt controller with a writable firmware chip. A well-trained attacker with physical access to the computer ("evil maid") can perform the required steps in 5 minutes. With a malicious firmware, the attacker can covertly disable Thunderbolt security, clone device identities, and proceed to use DMA to extract data.[163] Thunderspy vulnerabilities can largely be mitigated using Kernel DMA Protection, along with traditional anti-intrusion hardware features.[164][165]

## Cables

In June 2011, Apple introduced the first Thunderbolt cable, a 2 m (6.6 ft), 10 Gbit/s, full-duplex, active cable costing US$49.[166]

Thunderbolt Ethernet adapter

In June 2012, Apple began selling a Thunderbolt-to-gigabit Ethernet adapter for US$29.[167] In the third quarter of 2012, other manufacturers started shipping Thunderbolt cables, including cables reaching the 3-meter (9.8-foot) length limit, while some storage-enclosure builders began bundling Thunderbolt cables with their devices, rather than making customers buy them separately, as had been standard practice.

In January 2013, Apple reduced the price of their 2 m (6.6 ft) length cable to US$39 and added a half-meter cable for US$29.[168]

In Thunderbolt 3's introduction, Intel announced passive USB-C cables would connect Thunderbolt devices at speeds greater than USB 3.1 (though less than active Thunderbolt cables), thereby eliminating the adoption barrier of Thunderbolt active cable costs.[169]

In mid-2016, copper Thunderbolt 3 cables became available at lengths up to 2 m (6.6 ft). However, 40 Gbit/s on copper required either *active* cables, or short (initially 0.5 m (1.6 ft), later 0.8 m (3 ft)) *passive* cables. Passive copper cables exceeding 0.8 m (3 ft) are limited to 20 Gbit/s. Despite that limit, passive cables provide USB 3 (20 Gbit/s) backward compatibility, while active cables support only USB 2.0 (480 Mbit/s). In April 2020, optical Thunderbolt 3 cables debuted (see *Copper vs. optical*).

Copper versions of Thunderbolt 4 cables offer full 40 Gbit/s speed and backward compatibility with all versions of USB (up to USB4), DisplayPort Alternate Mode (DP 1.4 HBR3), and Thunderbolt 3. Released in early 2021, they are also all to be available in three *specified* lengths: 0.2 m (0.66 ft), 0.8 m (2.6 ft), and 2 m (6.6 ft) – with many companies initially offering 0.8 m (2.6 ft) lengths. Copper Thunderbolt 4 cables up to 1 m (3.3 ft) are passive cables, while longer cables must integrate active signal conditioning circuitry. Apple are currently the only company that offers a 3 m (9.8 ft) copper cable, whilst other companies maximum length of copper cables are 2 m (6.6 ft). Optical Thunderbolt 4 cables were targeting lengths from ≈5 m (16 ft) to 50 m (160 ft),[142] although this may not happen, instead jumping to Thunderbolt 5 optical cables, sometime after the arrival of that standard in late 2024.

## Intel controllers

Intel Thunderbolt controllers

| Ver. | Model | Aimed | Ch. | Size (mm) | Power (W) | Family | Release date | Features |
|---|---|---|---|---|---|---|---|---|
| 1[170] | 82523EF | ? | 4 | 15 × 15 | 3.8 | Light Ridge | 2010 Q4 | |
| | 82523EFL | ? | | | 3.2 | | | |
| | DSL2510 | ? | 2 | | ? | Eagle Ridge | 2011 Q1 | |
| | DSL2310 | ? | | 8 × 9 | 1.85 | | | SFF |
| | DSL2210 | ? | 1 | 5 × 6 | 0.7 | Port Ridge | 2011 Q4 | Device only |
| | DSL3510H | ? | 4 | 12 × 12 | 3.4 | Cactus Ridge | — | *Cancelled* |
| | DSL3510L | ? | | | 2.8 | | 2012 Q2 | |
| | DSL3310 | ? | 2 | | 2.1 | | | Host only |
| | DSL4510 | ? | 4 | | ? | Redwood Ridge | 2013 | |
| | DSL4410 | ? | 2 | 10 × 10 | ? | | | Host only |
| 2[171] | DSL5520 | ? | 4 | ? | ? | Falcon Ridge | 2013 Q3 | 20 Gbit/s speed, DisplayPort 1.2 |
| | DSL5320 | ? | 2 | ? | ? | | | |
| 3[172] | DSL6540 | ? | | 10.7 × 10.7 | 2.2 | Alpine Ridge[173] | 2015 Q4 | 40 Gbit/s, DisplayPort 1.2, PCIe 3.0, HDMI 2.0 LSPCon (DP Protocol Converter), USB 3.1, 100 W (compatible with USB Power Delivery)[174][175] |
| | DSL6340 | ? | 1 | | 1.7 | | 2015 Q1 | 40 Gbit/s speed, DisplayPort 1.2 |
| | JHL6240 | Computers/Peripheral | | | 1.2 | | 2016 Q2 | 40 Gbit/s speed, DisplayPort 1.2, lead-free |
| | JHL6340 | Computers/Peripheral | | | 1.7 | | | |
| | JHL6540 | Computers/Peripheral | 2 | | 2.2 | | | |
| | JHL7340 | Computers | 1 | | 1.9 | Titan Ridge | 2018 Q1 | 40 Gbit/s speed, DisplayPort 1.4 |
| | JHL7540 | Computers | 2 | | 2.4 | | | |
| | JHL7440 | Peripheral | | | 2.4 | | 2018 Q1 | 40 Gbit/s speed, DisplayPort 1.4, optional USB-C port compatibility, backwards compatibility when a Thunderbolt 3 docking station is connected to a non-Thunderbolt 3 computer |

<table>
<tr><td rowspan="6">4[176]</td><td>JHL8140[177]</td><td>Peripheral</td><td>1</td><td>?</td><td>2.0</td><td>Hoover Ridge</td><td>2021 Q1?</td><td>TB4/USB4 endpoint, three downstream USB-C ports, two DisplayPort-out adapters, USB 3 hub, compatible with Thunderbolt 3. Essentially Goshen Ridge without downstream Thunderbolt or USB4 support.</td></tr>
<tr><td>JHL8340</td><td>Computers</td><td>1</td><td rowspan="3">10.7 × 10.7</td><td>?</td><td rowspan="2">Maple Ridge</td><td>2020 Q4</td><td rowspan="2">40 Gbit/s speed, USB4 compliant</td></tr>
<tr><td>JHL8540</td><td>Computers</td><td>2</td><td>?</td><td>2020 Q4</td></tr>
<tr><td>JHL8440</td><td>Peripheral</td><td>4</td><td>?</td><td>Goshen Ridge</td><td>2020 Q3</td><td>40 Gbit/s speed, USB4 compliant (peripheral only), with four Thunderbolt 4 ports for branching hub topology. Tunnelling of DisplayPort 1.4, USB 3 (10 Gbit/s), PCIe (32 Gbit/s). Has PCIe 3.0 × 1 and USB 3 (10 Gbit/s) native interfaces</td></tr>
<tr><td>JHL9540</td><td>Computers</td><td>2</td><td>13 × 13</td><td>3.25–4</td><td rowspan="4">Barlow Ridge</td><td rowspan="4">2024 Q3</td><td rowspan="2">40 Gbit/s speed, USB4 v2, three DisplayPort 2.1,[a] PCIe Gen 4 × 4</td></tr>
<tr><td>JHL9440</td><td>Peripheral</td><td>4</td><td>?</td><td>?</td></tr>
<tr><td rowspan="2">5[178]</td><td>JHL9580</td><td>Computers</td><td>2</td><td>13 × 13</td><td>3.25–4</td><td rowspan="2">80 Gbit/s speed (also 120 ⇄ 40 Gbit/s: asymmetric), USB4 v2, three DisplayPort 2.1,[a] PCIe Gen 4 × 4</td></tr>
<tr><td>JHL9480</td><td>Peripheral</td><td>4</td><td>?</td><td>?</td></tr>
<tr><td colspan="9">Sources:[179]</td></tr>
</table>

a. All speeds up to HBR3 plus UHBR10 and UHBR20. No support for UHBR13.5.

## See also

- Apple Thunderbolt Display
- Computer bus
- DisplayPort / Mini DisplayPort
- IEEE 1394 (FireWire)
- Interconnect bottleneck
- Lightning
- DockPort
- OCuLink
- List of interface bit rates
- List of computer peripheral bus bit rates

- MacBook Pro
- Optical communication
- Fiber-optic cable
- Parallel optical interface
- eGPU
- USB 3.0
- USB4
- USB-C

# References


1. "Apple Updates MacBook Pro with Next Generation Processors, Graphics & Thunderbolt I/O Technology" (https://web.archive.org/web/20110710012629/https://www.apple.com/pr/library/2011/02/24Apple-Updates-MacBook-Pro-with-Next-Generation-Processors-Graphics-Thunderbolt-I-O-Technology.html) (Press release). Apple. 24 February 2011. Archived from the original (https://www.apple.com/pr/library/2011/02/24Apple-Updates-MacBook-Pro-with-Next-Generation-Processors-Graphics-Thunderbolt-I-O-Technology.html) on 10 July 2011. Retrieved 17 August 2011.
2. "Thunderbolt – Technology Brief" (http://www.intel.com/content/www/us/en/architecture-and-technology/thunderbolt/thunderbolt-technology-brief.html). Intel. Retrieved 1 October 2012.
3. "OWC Thunderbolt Hub" (https://eshop.macsales.com/shop/owc-thunderbolt-hub). Retrieved 19 November 2020.
4. "Thunderbolt 4 Press deck" (https://thunderbolttechnology.net/sites/default/files/intel-thunderbolt4-announcement-press-deck.pdf) (PDF). *Thunderbolt technology*. Retrieved 19 November 2020.
5. "Thunderbolt Device Driver Programming Guide" (https://developer.apple.com/library/mac/#documentation/HardwareDrivers/Conceptual/ThunderboltDevGuide/Introduction/Introduction.html#//apple_ref/doc/uid/TP40011138-CH1-SW1). Apple. Retrieved 21 December 2011.
6. "Frequently Asked Questions (FAQs)" (https://www.thunderbolttechnology.net/tech/faq). *Thunderbolt Technology Community*. Retrieved 14 January 2024.
7. Shamah, David. "Thunderbolt 3: How USB cooperation could lead to 100 million connected computers soon" (https://www.zdnet.com/home-and-office/networking/thunderbolt-3-how-usb-cooperation-could-lead-to-100-million-connected-computers-soon/). *ZDNet*. Retrieved 26 August 2021.
8. Shah, Agam (6 June 2013). "Intel shows 'world's fastest thumb drive' " (https://www.computerworld.com/article/2711416/intel-shows--world-s-fastest-thumb-drive-.html). *Computerworld*. Retrieved 26 August 2021.
9. Frakes, Dan (24 February 2011). "What you need to know about Thunderbolt" (https://www.macworld.com/article/1158145/thunderbolt_what_you_need_to_know.html). *MacWorld*. Retrieved 17 April 2019.
10. Cunningham, Andrew. "USB 3.1 and Type-C: The only stuff at CES that everyone is going to use" (https://arstechnica.com/gadgets/2015/01/usb-3-1-and-type-c-the-only-stuff-at-ces-that-everyone-is-going-to-use). *Ars Technica*.
11. Foresman, Chris (24 February 2011). "Thunderbolt smokes USB, FireWire with 10 Gb/s throughput" (https://arstechnica.com/apple/news/2011/02/thunderbolt-smokes-usb-firewire-with-10Gbps-throughput.ars). *Ars Technica*. Condé Nast Digital. Retrieved 24 February 2011.
12. Dilger, Daniel Eran (24 February 2011). "Intel details Thunderbolt, says Apple has a full year's head start" (http://www.appleinsider.com/articles/11/02/24/intel_details_thunderbolt_as_exclusive_to_apple_until_2012.html). *AppleInsider*. Retrieved 25 February 2011.
13. Nilsson, LG (25 February 2011). "Intel announces Thunderbolt" (https://web.archive.org/web/20110226040913/http://vr-zone.com/articles/intel-announces-thunderbolt/11333.html). *VR-Zone*. VR Media. Archived from the original (http://vr-zone.com/articles/intel-announces-thunderbolt/11333.html) on 26 February 2011. Retrieved 27 February 2011.
14. "Light Peak: Overview" (http://techresearch.intel.com/spaw2/uploads/files/Intel%20Light%20Peak%20White%20Paper_0910.pdf) (PDF). Intel. Retrieved 29 June 2011.
15. "White Paper: The 50G Silicon Photonics Link" (http://download.intel.com/pressroom/pdf/photonics/Intel_SiliconPhotonics50gLink_WhitePaper.pdf?iid=pr_smrelease_vPro_materials2) (PDF). Intel. Retrieved 29 June 2011.

16. "Thunderbolt trademark rights will be transferred from Apple to Intel" (http://www.appleinsider.com/articles/11/05/20/thunderbolt_trademark_rights_will_be_transferred_from_apple_to_intel.html). AppleInsider. 20 May 2011. Retrieved 16 June 2013.
17. "Sumitomo Electric Starts Selling Optical Thunderbolt Cable at Amazon Online" (http://global-sei.com/news/press/13/prs004_s.html). Global Sei.
18. "Optical Cables by Corning launched as the first Thunderbolt all-optical fiber cables" (http://www.corning.com/news_center/news_releases/2013/2013091101.aspx). Corning. 11 September 2013. Retrieved 22 September 2013.
19. Patel, Nilay (24 September 2009). "Video: Intel's Light Peak running an HD display while transferring files... on a hackintosh" (https://www.engadget.com/2009/09/24/video-intels-light-peak-running-an-hd-display-while-transferri). *Engadget*. AOL. Retrieved 25 February 2011.
20. Ziller, Jason (26 January 2010). *Intel Light Peak Interconnect Technology Update* (https://web.archive.org/web/20110323095802/http://www.youtube.com/watch?v=nfGevFIVKw4) (YouTube). Intel. Event occurs at 1:20. Archived from the original (https://www.youtube.com/watch?v=nfGevFIVKw4) on 23 March 2011. Retrieved 23 February 2011.
21. Light Peak to Connect Consumer Devices at Record Speed (https://www.youtube.com/watch?v=izNoF1SWtSg) on YouTube
22. "Light Peak to succeed USB 3.0" (https://www.theregister.co.uk/2010/04/15/light_peak_to_succeed_usb_3_0). UK: The Register. 15 April 2010.
23. Shiels, Maggie (25 September 2009). "Future is TV-shaped, says Intel" (http://news.bbc.co.uk/2/hi/technology/8272003.stm). BBC News. Retrieved 27 September 2009.
24. Collins, Barry (4 May 2010). "Intel shows off first Light Peak laptop" (https://web.archive.org/web/20130617062352/http://www.pcpro.co.uk/news/357688/intel-shows-off-first-light-peak-laptop). *PC Pro*. Dennis Publishing. Archived from the original (http://www.pcpro.co.uk/news/357688/intel-shows-off-first-light-peak-laptop) on 17 June 2013. Retrieved 5 May 2010.
25. Hollister, Sean (14 September 2010). "Intel's Light Peak optical interconnect shrinks slightly, LaCie, WD, Compal and Avid begin prototyping" (https://www.engadget.com/2010/09/14/intels-light-peak-optical-interconnect-shrinks-slightly-while). *Engadget*. AOL. Retrieved 28 November 2010.
26. Crothers, Brooke (19 February 2011). "New high-speed connection tech due from Apple" (https://news.cnet.com/8301-13924_3-20033940-64.html). *CNet News*. CBS Interactive. Retrieved 25 February 2011.
27. Kim, Arnold (19 February 2011). "Apple to Introduce Light Peak (High Speed Connection Technology) Soon?" (http://www.macrumors.com/2011/02/19/apple-to-introduce-light-peak-high-speed-connection-technology-soon). *MacRumors*. Retrieved 25 February 2011.
28. Smith, Mat (21 May 2012). "ASUS and MSI launch Thunderbolt motherboards, tie for first place" (https://www.engadget.com/2012/05/21/asus-and-msi-launch-thunderbolt-motherboards). *Engadget*.
29. "Thunderbolt: Next-Generation high-speed I/O technology" (https://web.archive.org/web/20110226054716/http://www.apple.com/thunderbolt). Apple. 24 February 2011. Archived from the original (https://www.apple.com/thunderbolt) on 26 February 2011. Retrieved 25 February 2011.
30. "MacBook Pro (13-inch, Early 2011) – Technical Specifications" (https://support.apple.com/en-us/112600). *Apple Support*. Retrieved 12 June 2025.
31. "MacBook Pro (15-inch, Early 2011) – Technical Specifications" (https://support.apple.com/en-us/112599). *Apple Support*. Retrieved 12 June 2025.
32. "MacBook Pro (17-inch, Early 2011) – Technical Specifications" (https://support.apple.com/en-us/112598). *Apple Support*. Retrieved 12 June 2025.
33. "iMac (21.5-inch, Mid 2011) – Technical Specifications" (https://support.apple.com/en-us/111983). *Apple Support*. Retrieved 12 June 2025.
34. "iMac (27-inch, Mid 2011) – Technical Specifications" (https://support.apple.com/en-us/112569). *Apple Support*. Retrieved 12 June 2025.
35. "Mac mini (Mid 2011) – Technical Specifications" (https://support.apple.com/en-us/112007). *Apple Support*. Retrieved 12 June 2025.
36. "MacBook Air (11-inch, Mid 2011) – Technical Specifications" (https://support.apple.com/en-us/112439). *Apple Support*. Retrieved 12 June 2025.

37. "MacBook Air (13-inch, Mid 2011) – Technical Specifications" (https://support.apple.com/en-us/112038). *Apple Support*. Retrieved 12 June 2025.
38. "Apple Thunderbolt Display – Technical Specifications" (https://support.apple.com/en-us/112597). *Apple Support*. Retrieved 12 June 2025.
39. "Apple Announces New iMac With Next Generation Quad-Core Processors, Graphics & Thunderbolt I/O Technology" (https://www.apple.com/pr/library/2011/05/03imac.html). Apple. Retrieved 10 May 2011.
40. "iMac (Mid 2011): Target Display Mode does not accept video over a Mini DisplayPort cable" (https://support.apple.com/en-us/HT204592). Apple. 14 July 2011. Retrieved 17 July 2011.
41. Shah, Agam (12 April 2011). "Intel to Open up Thunderbolt Development This Quarter" (https://www.pcworld.com/article/224995/article.html). *PC World*. PCWorld Communications. Retrieved 13 April 2011.
42. Holland, Colin (12 April 2011). "LeCroy lines up armada for Thunderbolt testing" (https://www.embedded.com/electronics-products/electronic-product-reviews/debug-and-optimization/4215081/LeCroy-lines-up-armada-for-Thunderbolt-testing). Retrieved 18 April 2011.
43. "Registration required for developer information" (https://thunderbolttechnology.net/developer-application). Intel. Retrieved 22 August 2012.
44. "Thunderbolt smokes USB, FireWire with 10Gbps throughput" (https://arstechnica.com/gadgets/2011/02/thunderbolt-smokes-usb-firewire-with-10gbps-throughput/). 24 February 2011.
45. "Video Creation Bolts Ahead – Intel's Thunderbolt 2 Doubles Bandwidth, Enabling 4K Video Transfer & Display" (http://blogs.intel.com/technology/2013/06/video-creation-bolts-ahead-%E2%80%93-intel%E2%80%99s-thunderbolt%E2%84%A2-2-doubles-bandwidth-enabling-4k-video-transfer-display-2). Intel. 4 June 2013.
46. "Mac Pro (Late 2013) – Technical Specifications" (https://support.apple.com/en-us/112025). *Apple Support*. Retrieved 12 June 2025.
47. "MacBook Pro (Retina, 13-inch, Late 2013) – Technical Specifications" (https://support.apple.com/en-us/111946). *Apple Support*. Retrieved 12 June 2025.
48. "MacBook Pro (Retina, 15-inch, Late 2013) – Technical Specifications" (https://support.apple.com/en-us/111971). *Apple Support*. Retrieved 12 June 2025.
49. "Mac mini (Late 2014) - Technical Specifications" (https://support.apple.com/en-us/111931). *Apple Support*. Retrieved 12 June 2025.
50. "iMac (Retina 5K, 27-inch, Late 2014) – Technical Specifications" (https://support.apple.com/en-us/112436). *Apple Support*. Retrieved 12 June 2025.
51. "MacBook Air (11-inch, Early 2015) – Technical Specifications" (https://support.apple.com/en-us/112441). *Apple Support*. Retrieved 12 June 2025.
52. "MacBook Air (13-inch, Early 2015) – Technical Specifications" (https://support.apple.com/en-us/111956). *Apple Support*. Retrieved 12 June 2025.
53. Apple (2013) "Thunderbolt" (https://www.apple.com/thunderbolt/). *Apple*. Retrieved 4 February 2014.
54. "Video Creation Bolts Ahead – Intel's Thunderbolt 2 Doubles Bandwidth, Enabling 4K Video Transfer & Display" (https://blogs.intel.com/technology/2013/06/video-creation-bolts-ahead-%E2%80%93-intel%E2%80%99s-thunderbolt%E2%84%A2-2-doubles-bandwidth-enabling-4k-video-transfer-display-2). *intel.com*. Retrieved 1 January 2016.
55. Walton, Jarred (19 August 2013). "ASUS Introduces Z87-Deluxe/Quad: World's First Thunderbolt 2 Certified Motherboard" (https://web.archive.org/web/20130820021451/http://www.anandtech.com/show/7230/asus-introduces-z87deluxequad-worlds-first-thunderbolt-2-certified-motherboard). *AnandTech*. Retrieved 28 December 2013.
56. Torres, Edwin (28 January 2013) [1] (http://buyersguide.macrumors.com/#Retina_MacBook_Pro). *MacRumors*. Retrieved 28 January 2014.
57. "Intel" (https://www.facebook.com/IntelIsrael/photos/a.172984516091615.45124.131422793581121/911323402257719/?type=1&theater). *Facebook*.
58. Byrne, Seamus (2 June 2015). "One port to rule them all: Thunderbolt 3 and USB Type-C join forces" (https://www.cnet.com/news/thunderbolt-3-and-usb-type-c-join-forces-for-one-port-to-rule-them-all). CNET. Retrieved 12 March 2018.

59. "Thunderbolt 3 is twice as fast and uses reversible USB Type-C" (https://www.engadget.com/2015/06/02/thunderbolt-3-usb-c). *Engadget*. 2 June 2015.
60. "Thunderbolt 3 embraces USB Type-C connector, doubles bandwidth to 40Gbps" (https://arstechnica.com/gadgets/2015/06/thunderbolt-3-embraces-usb-type-c-connector-doubles-bandwidth-to-40gbps). *arstechnica.co.uk*. 2 June 2015.
61. "Intel® 7000 Series Thunderbolt™ 3 Controllers" (https://www.thunderbolttechnology.net/sites/default/files/18-241_Thunder7000Controller_Brief_FIN_HI.pdf) (PDF). Intel.
62. "Thunderbolt 3 Technology Brief" (https://thunderbolttechnology.net/sites/default/files/Thunderbolt3_TechBrief_FINAL.pdf) (PDF). Intel.
63. "Thunderbolt 3 – The USB-C That Does It All" (https://thunderbolttechnology.net/blog/thunderbolt-3-usb-c-does-it-all). Thunderbolt Technology Community. Retrieved 24 November 2015.
64. "Leaked Info on Third-Generation Thunderbolt Points to 40Gbps Transfer Speeds" (https://www.macrumors.com/2014/04/21/thunderbolt-third-generation-details). MacRumors. 21 April 2014. Retrieved 19 November 2014.
65. "Next-gen Thunderbolt details: 40Gbps, PCIe 3.0, HDMI 2.0, and 100W power delivery for single-cable PCs" (https://www.extremetech.com/computing/181099-next-gen-thunderbolt-details-40gbps-pcie-3-0-hdmi-2-0-and-100w-power-delivery-for-single-cable-pcs). Extreme Tech. 22 April 2014. Retrieved 19 November 2014.
66. "Next-gen Thunderbolt doubles speeds but changes the connector" (https://arstechnica.com/gadgets/2014/04/next-gen-thunderbolt-doubles-speeds-but-changes-the-connector). *Ars Technica*. 22 April 2014. Retrieved 19 November 2014.
67. "Day 1 of Thunderbolt peripheral training Q4 15 final v1.0" (https://thunderbolttechnology.net/sites/default/files/Day%201%20of%202_Thunderbolt%E2%84%A2%203%20Peripheral%20Device%20Training_Q4_15_Final_v1_0.pdf) (PDF), *Thunderbolt technology*
68. "Thunderbolt 3 devices" (https://thunderbolttechnology.net/products?tid=All&field_company_nid=All&field_prod_os_value_many_to_one=All&field_prod_tb_version_value_many_to_one=tbv3). *Thunderbolt Technology*. January 2016. Retrieved 9 January 2016.
69. "MacBook Pro (13-inch, 2016, Two Thunderbolt 3 ports) – Technical Specifications" (https://support.apple.com/en-us/111999). *Apple Support*. Retrieved 12 June 2025.
70. "MacBook Pro (13-inch, 2016, Four Thunderbolt 3 ports) – Technical Specifications" (https://support.apple.com/en-us/112003). *Apple Support*. Retrieved 12 June 2025.
71. "MacBook Pro (15-inch, 2016) – Technical Specifications" (https://support.apple.com/en-us/111975). *Apple Support*. Retrieved 12 June 2025.
72. "iMac (21.5-inch, 2017) – Technical Specifications" (https://support.apple.com/en-us/111921). *Apple Support*. Retrieved 12 June 2025.
73. "iMac (Retina 4K, 21.5-inch, 2017) – Technical Specifications" (https://support.apple.com/en-us/112026). *Apple Support*. Retrieved 12 June 2025.
74. "iMac (Retina 5K, 27-inch, 2017) – Technical Specifications" (https://support.apple.com/en-us/111969). *Apple Support*. Retrieved 12 June 2025.
75. "iMac Pro (2017) – Technical Specifications" (https://support.apple.com/en-us/111995). *Apple Support*. Retrieved 12 June 2025.
76. "MacBook Air (Retina, 13-inch, 2018) – Technical Specifications" (https://support.apple.com/en-us/111933). *Apple Support*. Retrieved 12 June 2025.
77. "Mac mini (2018) – Technical Specifications" (https://support.apple.com/en-us/111912). *Apple Support*. Retrieved 12 June 2025.
78. "Mac Pro (2019) – Technical Specifications" (https://support.apple.com/en-us/118461). *Apple Support*. Retrieved 12 June 2025.
79. "Mac Pro (Rack, 2019) – Technical Specifications" (https://support.apple.com/en-us/111907). *Apple Support*. Retrieved 12 June 2025.
80. "Pro Display XDR – Technical Specifications" (https://www.apple.com/pro-display-xdr/specs/). *Apple*. Retrieved 6 September 2022.
81. "Studio Display – Technical Specifications" (https://www.apple.com/studio-display/specs/). *Apple*. Retrieved 6 September 2022.

82. Ziller, Jason. "New Intel Thunderbolt 3 controllers offer DisplayPort 1.4, and basic peripheral compatibility with USB-C computer ports" (https://thunderbolttechnology.net/blog/new-intel®-thunderbolt-3-controllers-offer-displayport-14-and-basic-peripheral-compatibility-u). *Thunderbolt Technology*. Intel. Retrieved 15 January 2018.
83. "USB Promoter Group USB4 Specification" (https://www.usb.org/sites/default/files/USB4%20Specification.zip). *USB implementers forum*. 29 August 2019.
84. Bright, Peter (4 March 2019). "Thunderbolt 3 becomes USB4, as Intel's interconnect goes royalty-free" (https://arstechnica.com/gadgets/2019/03/thunderbolt-3-becomes-usb4-as-intels-interconnect-goes-royalty-free). *Ars Technica*. Retrieved 4 March 2019.
85. Grunin, Lori (4 March 2019). "USB4 marries Thunderbolt 3 for faster speeds and smarter transfers" (https://www.cnet.com/news/usb4-marries-thunderbolt-3-for-faster-speeds-and-smarter-transfers). *CNET*. Retrieved 4 March 2019.
86. Brant, Tom (4 March 2019). "Thunderbolt 3 Merges With USB to Become USB4" (https://www.pcmag.com/news/366931/thunderbolt-3-merges-with-usb-to-become-usb4). *PC Magazine*. Retrieved 4 March 2019.
87. "VESA Releases Updated DisplayPort Alt Mode Spec to Bring DisplayPort 2.0 Performance to USB4 and New USB Type-C® Devices" (https://vesa.org/featured-articles/vesa-releases-updated-displayport-alt-mode-spec-to-bring-displayport-2-0-performance-to-usb4-and-new-usb-type-c-devices/). *VESA - Interface Standards for The Display Industry*. 29 April 2020. Retrieved 22 November 2021.
88. Smith, Ryan. "DisplayPort Alt Mode 2.0 Spec Released: Defining Alt Mode for USB4" (https://web.archive.org/web/20200430080010/https://www.anandtech.com/show/15752/displayport-alt-mode-20-spec-released). *www.anandtech.com*. Archived from the original (https://www.anandtech.com/show/15752/displayport-alt-mode-20-spec-released) on 30 April 2020. Retrieved 22 November 2021.
89. "VESA Releases Updated DisplayPort Alt Mode Spec to Bring DisplayPort 2.0 Performance to USB4 and New USB Type-C® Devices" (https://vesa.org/featured-articles/vesa-releases-updated-displayport-alt-mode-spec-to-bring-displayport-2-0-performance-to-usb4-and-new-usb-type-c-devices/). *VESA – Interface Standards for The Display Industry*. 29 April 2020. Retrieved 22 November 2021.
90. "MacBook Air (M1, 2020) – Technical Specifications" (https://support.apple.com/en-us/111883). *support.apple.com*. Retrieved 6 September 2022.
91. "MacBook Pro (13-inch, M1, 2020) – Technical Specifications" (https://support.apple.com/en-us/111893). *support.apple.com*. Retrieved 6 September 2022.
92. "Mac mini (M1, 2020) – Technical specifications" (https://support.apple.com/en-us/111894). *support.apple.com*. Retrieved 6 September 2022.
93. *USB type C system overview* (https://www.usb.org/sites/default/files/D1T1-2%20-%20USB%20Type-C%20System%20Overview.pdf) (PDF), USB
94. "USB4 Systems PCIe Tunneling Support" (https://docs.microsoft.com/en-us/windows-hardware/test/hlk/testref/7d627bf0-25f3-4564-b554-b2a3450e2bcf). Microsoft.
95. "USB4 is Coming! Here is What You Need to Know" (https://www.blackbox.com/en-it/blog/technology/detail/technology/2020/10/29/usb4-is-coming!-here-is-what-you-need-to-know). Black box.
96. "DisplayPort Alt Mode 2.0 Spec Released: Defining Alt Mode for USB4" (https://web.archive.org/web/20200430080010/https://www.anandtech.com/show/15752/displayport-alt-mode-20-spec-released). *Anand tech*. Archived from the original (https://www.anandtech.com/show/15752/displayport-alt-mode-20-spec-released) on 30 April 2020.
97. "USB4 vs. USB C" (https://linuxhint.com/usb4-vs-usb-c/). *Linux hint*.
98. Owen, Malcolm (7 January 2020). "Intel confirms Thunderbolt 4 on way with four-times USB 3 speed" (https://appleinsider.com/articles/20/01/07/intel-confirms-thunderbolt-4-on-way-with-four-times-usb-3-speed). *Apple Insider*. Retrieved 7 January 2020.
99. "Introducing Thunderbolt 4 universal cable connectivity for everyone" (https://newsroom.intel.com/news/introducing-thunderbolt-4-universal-cable-connectivity-everyone/), *Intel Newsroom*, Intel
100. Introducing Thunderbolt 4: Universal Cable Connectivity for Everyone (https://newsroom.intel.com/news/introducing-thunderbolt-4-universal-cable-connectivity-everyone). Intel. July 8, 2020

101. "What is Thunderbolt 4?" (https://www.cablematters.com/blog/Thunderbolt/what-is-thunderbolt-4). *Cable Matters*.
102. "Razer Thunderbolt 4 Dock" (https://www.razer.com/gaming-pc-accessories/razer-thunderbolt-4-dock-chroma/RC21-01690100-R3U1). Retrieved 5 February 2021.
103. "Sonnet Thunderbolt 4 Dock Compatibility" (https://web.archive.org/web/20210219085402/https://www.sonnettech.com/support/downloads/manuals/Thunderbolt_4_Dock_Compatiblity.pdf) (PDF). Archived from the original (https://www.sonnettech.com/support/downloads/manuals/Thunderbolt_4_Dock_Compatiblity.pdf) (PDF) on 19 February 2021. Retrieved 19 February 2021.
104. S, Ganesh T. "Intel Teases Thunderbolt 4, Light on Details" (https://web.archive.org/web/20200107201523/https://www.anandtech.com/show/15345/intel-teases-thunderbolt-4-light-on-details). *Anandtech*. Archived from the original (https://www.anandtech.com/show/15345/intel-teases-thunderbolt-4-light-on-details) on 7 January 2020. Retrieved 16 May 2020.
105. "Intel Introduces Thunderbolt 5 Connectivity Standard" (https://www.intel.com/content/www/us/en/newsroom/news/intel-introduces-thunderbolt-5-standard.html). Intel. 19 October 2022. Retrieved 12 September 2023.
106. "Apple's new Mac mini is more mighty, more mini, and built for Apple Intelligence" (https://www.apple.com/newsroom/2024/10/apples-new-mac-mini-is-more-mighty-more-mini-and-built-for-apple-intelligence). *Apple Newsroom*. Retrieved 30 October 2024.
107. "New MacBook Pro features M4 family of chips and Apple Intelligence" (https://www.apple.com/newsroom/2024/10/new-macbook-pro-features-m4-family-of-chips-and-apple-intelligence). *Apple Newsroom*. Retrieved 30 October 2024.
108. "Apple Announces New Mac Studio With M4 Max and M3 Ultra Chips, Thunderbolt 5, and More" (https://www.macrumors.com/2025/03/05/new-mac-studio-with-m4-max-and-m3-ultra). *MacRumors*. 5 March 2025. Retrieved 30 March 2025.
109. "Mac Mini - Technical Specifications - Apple" (https://www.apple.com/mac-mini/specs). *Apple Inc.* Retrieved 30 March 2025.
110. "Intel has a grand plan to bring Thunderbolt 3 ports to every laptop" (https://www.techradar.com/news/intel-has-a-grand-plan-to-bring-thunderbolt-3-ports-to-every-laptop). *Techradar*. 24 May 2017. Retrieved 24 May 2017.
111. "USB Promoter Group Announces USB4 Specification" (https://usb.org/sites/default/files/2019-03/USB_PG_USB4_DevUpdate_Announcement_FINAL_20190226.pdf) (PDF). *USB*. Retrieved 7 July 2019.
112. "USB4 Specification" (https://www.usb.org/sites/default/files/USB4%20Specification.zip). *USB*. Retrieved 3 September 2019.
113. "The new USB4 spec promises a lot: Thunderbolt 3 support, 40Gbps bandwidth, and less confusion" (https://www.pcworld.com/article/3347403/the-new-usb4-spec-promises-a-lot-thunderbolt-3-support-40gbps-bandwidth-and-less-confusion.html). *PCWorld*. 5 March 2019. Retrieved 7 July 2019.
114. "Introducing Thunderbolt 4: Universal Cable Connectivity for Everyone" (https://newsroom.intel.com/news/introducing-thunderbolt-4-universal-cable-connectivity-everyone). *PCWorld*. 8 July 2020. Retrieved 11 July 2020.
115. "Where is Thunderbolt on Threadripper?! Here it is, but…" (https://www.youtube.com/watch?v=uOlQbP63lDQ), *Level1Techs*, 29 July 2018, archived (https://ghostarchive.org/varchive/youtube/20211103/uOlQbP63lDQ) from the original on 3 November 2021, retrieved 20 February 2019 – via You tube
116. Lilly, Paul (30 July 2018). "Complex Threadripper hack gets Intel's Thunderbolt 3 working on AMD hardware" (https://www.pcgamer.com/complex-threadripper-hack-gets-intels-thunderbolt-3-working-on-amd-hardware). *PC Gamer*. Retrieved 20 February 2019.
117. "Thunderbolt 3 AMD Threadripper 1950X + RTX 2080@32Gbps-TB3" (https://egpu.io/forums/builds/thunderbolt-3-on-amd-x399-threadripper-rtx-208032gbps-tb3-razer-core-x-win10-1803-theitsage). *eGPU.io*. Retrieved 26 May 2019.
118. "ASRock X570 Creator" (https://www.asrock.com/mb/AMD/X570%20Creator/index.asp). *Asrock*. Retrieved 9 November 2019.

119. "X570 Phantom Gaming–ITX/TB3" (https://www.thunderbolttechnology.net/product/x570-phantom-gaming%E2%80%93itxtb3). *Thunderbolt Technology Community*. Retrieved 4 March 2020.
120. Broekhuijsen, Niels (6 February 2020). "Intel Finally Certified an AMD Thunderbolt Motherboard: Here's Why That Matters (Updated)" (https://www.tomshardware.com/news/amd-motherboard-asrock-first-thunderbolt-certification-intel). *Tom's Hardware*. Retrieved 3 March 2020.
121. "THUNDERBOLTEX 3-TR｜Motherboards｜ASUS Global" (https://www.asus.com/Motherboards-Components/Motherboards/Accessories/THUNDERBOLTEX-3-TR/HelpDesk_Manual/). *www.asus.com*. Retrieved 21 July 2022.
122. "ProArt B550-CREATOR｜Motherboards｜ASUS Global" (https://www.asus.com/Motherboards-Components/Motherboards/ProArt/ProArt-B550-CREATOR/). *ASUS Global*. Retrieved 21 July 2022.
123. B550 VISION D (https://www.thunderbolttechnology.net/product/b550-vision-d)
124. B550 VISION D-P (https://www.thunderbolttechnology.net/product/b550-vision-d-p)
125. Hachman, Mark (24 February 2011). "Intel Thunderbolt Rollout Won't Be Lightning Fast" (https://www.pcmag.com/article2/0,2817,2380890,00.asp). *PC Mag*. Ziff Davis. Retrieved 26 February 2011.
126. Shankland, Stephen (23 September 2009). "Intel's Light Peak: One PC cable to rule them all" (https://news.cnet.com/8301-30685_3-10360047-264.html). *CNet News*. CBS Interactive. Retrieved 28 November 2010.
127. Crothers, Brooke (29 September 2009). "Sources: 'Light Peak' technology not Apple idea" (https://news.cnet.com/8301-13924_3-10363956-64.html). *CNet News*. CNet. Retrieved 23 February 2011.
128. Branscombe, Mary (5 August 2010). "Intel Light Peak: a tech guide" (https://www.zdnet.com/article/intel-light-peak-a-tech-guide/). *ZDNet*. Ziff Davis. Retrieved 22 October 2012.
129. Shah, Agam (8 January 2011). "Intel says Light Peak interconnect technology is ready" (http://www.computerworld.com/s/article/9204158/Intel_says_Light_Peak_interconnect_technology_is_ready). *Computerworld*. International Data. Retrieved 23 February 2011.
130. "IPtronics Develops Components for Light Peak Technology" (http://www.design-reuse.com/news/21682/light-peak-intel-iptronics.html) (Press release). IPtronics. 1 October 2009. Retrieved 5 April 2011.
131. Clarke, Peter (1 October 2009). "IPtronics, Avago chip in to Intel's optical interconnect" (http://www.eetimes.com/news/latest/showArticle.jhtml?articleID=220300670). *EE Times*. Retrieved 1 October 2009.
132. Metz, Cade (24 February 2011). "Intel: 'PC makers took the light out of Light Peak' " (https://www.theregister.co.uk/2011/02/24/intel_thunderbolt_copper_embrace_explained). *The Register*. Situation Publishing. Retrieved 25 February 2011.
133. Ziller, Jason (23 January 2010). *Light Peak to Connect Consumer Devices at Record Speed* (https://www.youtube.com/watch?v=izNoF1SWtSg) (YouTube). Intel. Event occurs at 1:13. Retrieved 23 February 2011.
134. "Sumitomo Electric Starts Selling Optical Thunderbolt Cable at Amazon Online" (https://global-sei.com/news/press/13/prs004_s.html) (Press release). Global SEI. 2013. Retrieved 16 June 2013.
135. "Delock Cable Thunderbolt optical male/male 30 m black" (https://www.delock.com/produkte/G_83259/merkmale.html?setLanguage=en). DeLock. Retrieved 15 October 2013.
136. "PSA: For Thunderbolt 3 over distance; use *OPTICAL Thunderbolt 1/2 cables!*" (https://forums.macrumors.com/threads/psa-for-thunderbolt-3-over-distance-use-optical-thunderbolt-1-2-cables.2112230). Macrumors. 28 March 2018. Retrieved 28 January 2019.
137. "Optical thunderbolt 3 cables from Corning. Up to 60m. #NAB2019 #NABShow" (https://twitter.com/macvfx/status/1115746090177818624). *Twitter*. Mat X. 9 April 2019. Retrieved 11 June 2019.
138. "Corning's Optical Thunderbolt 3 Cables Now Available in Lengths From 5 to 50 Meters" (https://forums.macrumors.com/threads/cornings-optical-thunderbolt-3-cables-now-available-in-lengths-from-5-to-50-meters.2257908). Macrumors. 30 September 2020. Retrieved 8 October 2020.

139. "Optical Thunderbolt 3 Cables Begin Rolling Out in Lengths Up to 50 Meters" (https://forums.macrumors.com/threads/optical-thunderbolt-3-cables-begin-rolling-out-in-lengths-up-to-50-meters.2228158). Macrumors. 26 March 2020. Retrieved 28 April 2020.
140. "Thunderbolt 4 / USB 4 Cable (2m) Active 40Gb/s, 100W, 20V, 5A" (http://shop.caldigit.com/us/index.php?route=product/product&product_id=207). CalDigit.
141. "Cable Matters Active Thunderbolt 4 Cable" (https://www.cablematters.com/pc-1283-132-40gbps-thunderbolt-4-usb-c-cable-with-100w-charging.aspx#DescriptionTitle). Cable Matters.
142. "Intel Thunderbolt 4 announcement press deck" (https://thunderbolttechnology.net/sites/default/files/intel-thunderbolt4-announcement-press-deck.pdf) (PDF). Intel Thunderbolt. 8 July 2020. Retrieved 23 April 2021.
143. "USB Type-C® Cable and Connector Specification Release 2.3 | USB-IF" (https://usb.org/document-library/usb-type-cr-cable-and-connector-specification-release-23). *usb.org*. USB-IF. p. 261. Retrieved 5 August 2024.
144. "Identify the ports on your Mac" (https://support.apple.com/en-us/HT201736). *Apple Support*. Retrieved 6 September 2022.
145. Butler, Stacey (7 November 2023). "What Exactly Is Thunderbolt and How Is It Different From USB-C? • macReports" (https://macreports.com/what-exactly-is-thunderbolt-and-how-is-it-different-from-usb-c/). *macReports*. Retrieved 8 November 2023.
146. "Gigabyte Unveils the Z170X-UD5 TH Thunderbolt 3 Certified Motherboard" (http://www.techpowerup.com/215773/gigabyte-unveils-the-z170x-ud5-th-thunderbolt-3-certified-motherboard.html). 4 September 2015. Retrieved 30 September 2015.
147. "MSI embraces Skylake and Thunderbolt 3.0 for new gaming laptops" (http://www.pcworld.com/article/2978318/laptop-computers/msi-embraces-skylake-and-thunderbolt-3-0-for-new-gaming-laptops.html). Retrieved 30 September 2015.
148. "New Alienware laptops pack Thunderbolt 3 and prettier screens, but oddly lack Skylake" (http://www.pcworld.com/article/2976593/laptop-computers/new-alienware-laptops-pack-thunderbolt-3-and-better-screens-but-oddly-lack-skylake.html). Retrieved 30 September 2015.
149. "Frequently Asked Questions about Thunderbolt 3 | Seagate US" (https://www.seagate.com/support/kb/lacie/frequently-asked-questions-about-thunderbolt-3-007771en/). *Seagate.com*.
150. Larry, O. W. C. (21 November 2024). "The Simple Guide to Thunderbolt Forwards and Backwards Compatibility" (https://eshop.macsales.com/blog/96867-the-simple-guide-to-thunderbolt-forwards-and-backwards-compatibility/). *Rocket Yard*.
151. Graham, Robert (24 February 2011). "Thunderbolt: Introducing a new way to hack Macs" (https://blog.erratasec.com/2011/02/thunderbolt-introducing-new-way-to-hack.html). *Errata Security*. Retrieved 5 March 2011.
152. Sevinsky, Russ (1 August 2013). *Funderbolt: Adventures in Thunderbolt DMA Attacks* (https://media.blackhat.com/us-13/US-13-Sevinsky-Funderbolt-Adventures-in-Thunderbolt-DMA-Attacks-Slides.pdf) (PDF). Black Hat Briefings (https://www.blackhat.com). Las Vegas.
153. Porter, Jon (11 May 2020). "Thunderbolt flaw allows access to a PC's data in minutes" (https://www.theverge.com/2020/5/11/21254290/thunderbolt-security-vulnerability-thunderspy-encryption-access-intel-laptops). *The Verge*. Retrieved 11 May 2020.
154. "Thunderclap: Exploring Vulnerabilities in Operating System IOMMU Protection via DMA from Untrustworthy Peripherals – NDSS Symposium" (https://www.ndss-symposium.org/ndss-paper/thunderclap-exploring-vulnerabilities-in-operating-system-iommu-protection-via-dma-from-untrustworthy-peripherals). Retrieved 21 January 2020.
155. "Thunderbolt 3 vs. Thunderbolt 4 Technology: What's the Difference?" (https://www.intel.com/content/www/us/en/architecture-and-technology/thunderbolt/thunderbolt-3-vs-4.html). Retrieved 23 February 2023.
156. "Thunderbolt how it works" (https://web.archive.org/web/20141029185956/https://thunderbolttechnology.net/tech/how-it-works). Intel. 2014. Archived from the original (https://thunderbolttechnology.net/tech/how-it-works) on 29 October 2014. Retrieved 1 May 2018.

157. Sevinsky, Russ (1 October 2013). *Black Hat USA 2013 – Funderbolt: Adventures in Thunderbolt DMA Attacks* (https://www.youtube.com/watch?v=q0HthE3qDMw). Archived (https://ghostarchive.org/varchive/youtube/20211103/q0HthE3qDMw) from the original on 3 November 2021. Retrieved 16 July 2014.
158. Heasman, John (2007). "Hacking the Extensible Firmware Interface" (https://www.blackhat.com/presentations/bh-usa-07/Heasman/Presentation/bh-usa-07-heasman.pdf) (PDF). Black Hat.
159. Snare (2012). "Mac EFI rootkits" (http://ho.ax/downloads/De_Mysteriis_Dom_Jobsivs_Black_Hat_Slides.pdf) (PDF). Black Hat.
160. Hudson, Trammell (27 December 2014). "Thunderstrike: EFI firmware rootkits for MacBooks" (https://trmm.net/Thunderstrike). Chaos Communication Congress.
161. US-CERT/NIST (30 January 2015). "CVE-2014-4498: The Thunderstrike issue" (https://web.nvd.nist.gov/view/vuln/detail?vulnId=CVE-2014-4498).
162. "About the security content of OS X Yosemite v10.10.2 and Security Update 2015-001" (https://support.apple.com/en-us/HT204244). Apple. 4 February 2015.
163. Ruytenberg, Björn (2020). "Thunderspy: When Lightning Strikes Thrice: Breaking Thunderbolt 3 Security" (https://thunderspy.io/). *Thunderspy.io*. Retrieved 11 May 2020.
164. Ruytenberg, Björn. "Thunderspy 2: Kernel DMA Protection for Unpatched Thunderbolt Systems" (https://thunderspy.io/ts2.html).
165. "Kernel DMA Protection (Windows 10) – Microsoft 365 Security" (https://docs.microsoft.com/en-us/windows/security/information-protection/kernel-dma-protection-for-thunderbolt). *docs.microsoft.com*.
166. Foresman, Chris (30 June 2011). "The technology inside Apple's $50 Thunderbolt cable" (https://arstechnica.com/gadgets/2011/06/why-apples-2m-thunderbolt-cable-costs-a-whopping-50). Retrieved 2 July 2011.
167. van Beijnum, Iljitsch (20 June 2012). "Hands-on: Apple's Thunderbolt Gigabit Ethernet adapter" (https://arstechnica.com/gadgets/2012/06/hands-on-apples-thunderbolt-gigabit-ethernet-adapter). *Ars Technica*.
168. Gurman, Mark (10 January 2013). "Apple slashes price on Thunderbolt cable, releases additional shorter model" (https://9to5mac.com/2013/01/09/apple-slashes-price-on-thunderbolt-cable-releases-additional-shorter-model). *9to5Mac*.
169. Anthony, Sebastian (2 June 2015). "Thunderbolt 3 embraces USB Type-C connector, doubles bandwidth to 40Gbps" (https://arstechnica.com/gadgets/2015/06/thunderbolt-3-embraces-usb-type-c-connector-doubles-bandwidth-to-40gbps). *Ars Technica UK*.
170. "Thunderbolt Controllers" (https://ark.intel.com/content/www/us/en/ark/products/series/67021/thunderbolt-controllers.html). *intel.com*.
171. "Thunderbolt 2 Controllers" (https://ark.intel.com/content/www/us/en/ark/products/series/79640/thunderbolt-2-controllers.html). *intel.com*.
172. "Thunderbolt 3 Controllers" (https://ark.intel.com/content/www/us/en/ark/products/series/87742/thunderbolt-3-controllers.html). *intel.com*.
173. "Next-Gen Intel "Alpine Ridge" Thunderbolt Controller Detailed" (https://www.techpowerup.com/200064/next-gen-intel-alpine-ridge-thunderbolt-controller-detailed). *techpowerup.com*. 21 April 2014.
174. "Intel 6000 Series Thunderbolt 3 Controllers" (https://thunderbolttechnology.net/sites/default/files/18-241_ThunderboltController_Brief_HI.pdf) (PDF).
175. "Alpine Ridge 頻寬可達 40Gb/s，新一代 Thunderbolt 晶片將具更多功能 – VR-Zone 中文版" (https://chinese.vrzone.com/109297/bandwidth-upto-40gbs-new-gen-thunderbolt-alpine-ridge-spec-shown-19042014). *vr-zone.com*. 19 April 2014.
176. "Thunderbolt 4 Controllers" (https://ark.intel.com/content/www/us/en/ark/products/series/197981/thunderbolt-4-controllers.html). *intel.com*.
177. "Intel Hoover Ridge JHL8140 dock" (https://dancharblog.wordpress.com/2023/09/11/sparkle-td-8140-usb4-travel-dock-review-teardown-intel-hoover-ridge-jhl8140-dock/). 11 September 2023.
178. "Thunderbolt 5 Controllers" (https://ark.intel.com/content/www/us/en/ark/products/series/225929/thunderbolt-5-controllers.html). *intel.com*.

179. 
- Nilsson, LG. "Intel cans 3.4W TDP Cactus Ridge Thunderbolt chip" (https://web.archive.org/web/20120325021733/http://vr-zone.com/articles/intel-cans-3.4w-tdp-cactus-ridge-thunderbolt-chip/15297.html). VR-Zone. Archived from the original (http://vr-zone.com/articles/intel-cans-3.4w-tdp-cactus-ridge-thunderbolt-chip/15297.html) on 25 March 2012. Retrieved 30 March 2012.
- Nilsson, LG. "Thunderbolt roadmap unveiled, new features coming to Apple et al" (https://web.archive.org/web/20120507043306/http://vr-zone.com/articles/thunderbolt-roadmap-unveiled-new-features-coming-to-apple-et-al/15554.html). VR-Zone. Archived from the original (http://vr-zone.com/articles/thunderbolt-roadmap-unveiled-new-features-coming-to-apple-et-al/15554.html) on 7 May 2012. Retrieved 24 July 2012.
- Chen, Monica (20 July 2012). "Intel to release new Thunderbolt chip in 2Q13" (http://www.digitimes.com/news/a20120720PD210.html). Digitimes. Retrieved 24 July 2012.
- Nilsson, LG. "Intel finally shipping 2nd gen Thunderbolt controllers, just in time for new Macs" (https://web.archive.org/web/20121020001549/http://vr-zone.com/articles/intel-finally-shipping-2nd-gen-thunderbolt-controllers-just-in-time-for-new-macs/15539.html). VR-Zone. Archived from the original (http://vr-zone.com/articles/intel-finally-shipping-2nd-gen-Thunderbolt-controllers-just-in-time-for-new-macs/15539.html) on 20 October 2012. Retrieved 14 August 2012.
- Shimpi, Anand. "New Thunderbolt Controllers (DSL4510/4410) and Future 20 Gbps Falcon Ridge TB Controller Announced" (http://www.anandtech.com/show/6885/new-thunderbolt-controllers-dsl45104410-and-future-20Gbps-falcon-ridge-tb-controller-announced). *AnandTech*. Retrieved 8 April 2013.
- Verry, Tim (15 September 2013). "IDF 2013: Products With 20Gbps Thunderbolt 2 Appear At IDF 2013" (http://www.pcper.com/news/General-Tech/IDF-2013-Products-20Gbps-Thunderbolt-2-Appear-IDF-2013). *PC Perspective*. Retrieved 14 October 2013.
- "Products (Formerly Falcon Ridge)" (https://web.archive.org/web/20131014231312/http://ark.intel.com/products/codename/60141/Falcon-Ridge). *Intel*. Archived from the original (http://ark.intel.com/products/codename/60141/Falcon-Ridge) on 14 October 2013. Retrieved 14 October 2013.
- "New Intel Thunderbolt 3 controllers offer DisplayPort 1.4, and basic peripheral compatibility with USB-C computer ports" (https://thunderbolttechnology.net/blog/new-intel%C2%AE-thunderbolt%E2%84%A2-3-controllers-offer-displayport-14-and-basic-peripheral-compatibility-u).

# Further reading

- Ziller, Jason (2 June 2015). "Thunderbolt 3 – The USB-C That Does It All" (https://thunderbolttechnology.net/blog/thunderbolt-3-usb-c-does-it-all). *thunderbolttechnology.net*. Intel.
- "Adapters for the Thunderbolt 3 or USB-C port on your Mac" (https://support.apple.com/en-us/HT207443). *apple.com*. Apple. 11 August 2021.

# External links

- Thunderbolt (https://thunderbolttechnology.net) – official site
- Thunderbolt May Not Be As Fast As You Think (https://larryjordan.com/articles/thunderbolt-may-not-be-as-fast-as-you-think/) – Larry Jordan

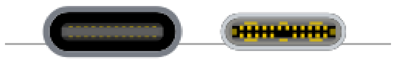
# USB-C

**USB-C**, or **USB Type-C**, is a 24-pin reversible connector (not a protocol) that supersedes all previous USB connectors, which were designated *legacy* in 2014. This connector also supersedes Mini DisplayPort and Lightning[3] connectors. USB-C is used for multiple purposes: exchanging data with peripheral devices, such as external drives, mobile phones, keyboards, track-pads, and mice, or between hosts; transferring A/V-data to displays and speakers; or powering peripheral devices and getting powered by power adapters, either through directly wired connectors or indirectly via hubs and docking stations. This connector type can be used for other data transfer protocols besides USB, such as Thunderbolt, PCIe, DisplayPort, and HDMI. It is considered extensible, allowing the support of future protocols.

The design for the USB-C connector was initially developed in 2012 by members of the USB Implementers Forum, with Intel, HP Inc., and Texas Instruments being distinguished on the design document, though Apple Inc. and Microsoft among others also contributed.The Type-C Specification 1.0 was published by the USB Implementers Forum (USB-IF) on August 11, 2014.[4] In 2016 it was adopted by the International Electrotechnical Commission as "IEC 62680-1-3".

The USB Type-C connector has 24 pins and is reversible.[5][6] The designation *C* distinguishes it from the various USB connectors it replaced, all termed either *Type-A* or *Type-B*. Whereas earlier USB cables had a host end *A* and a peripheral device end *B*, a USB-C cable connects either way; and for interoperation with older equipment, there are cables with a Type-C plug at one end and either a Type-A (host) or a Type-B (peripheral device) plug at the other.

**USB Type-C connector**



Illustrations of the Full-Featured Type-C connectors (receptacle left, plug right)

| | |
|---|---|
| **Type** | Digital audio/video/data/power – connector |
| **Production history** | |
| **Designer** | USB Implementers Forum |
| **Designed** | 11 August 2014 (published)[1] |
| **Produced** | 12 August 2014–present[2] |
| **Superseded** | All earlier USB connectors (Type-A, -B, and -AB, and their different sizes: Standard, Mini, and Micro)<br>DisplayPort<br>Mini DisplayPort<br>Lightning[3] |
| **General specifications** | |
| **Pins** | 24 |

The designation *C* refers only to the connector's physical configuration, or form factor, not to be confused with the connector's specific capabilities and performance, such as Thunderbolt 3, DisplayPort 2.0, USB 3.2 Gen 2×2. While USB-C is the single modern connector for all USB protocols, there are valid uses of the connector that do not involve any USB protocol. Based on the protocols supported by all, host, intermediate devices (hubs), and peripheral devices, a USB-C connection normally provides much higher data rates, and often more electrical power, than anything using the superseded connectors.

USB-C plug

A device with a Type-C connector does not necessarily implement any USB transfer protocol, USB Power Delivery, or any of the Alternate Modes: the Type-C connector is common to several technologies while mandating only a few of them.[7]

USB 3.2, released in September 2017, fully replaced the USB 3.1 (and therefore also USB 3.0) specifications. It preserves the former USB 3.1 SuperSpeed and SuperSpeed+ data transfer modes and introduces two additional data transfer modes by newly applying two-lane operations, with signalling rates of 5 Gbit/s (resulting speed of USB 10Gbps with two lanes; raw data rate: 1.212 GB/s) and 10 Gbit/s (USB 20Gbps speed; raw data rate: 2.422 GB/s). They are only applicable with Full-Featured USB-C cables and connectors and hosts, hubs, and peripheral devices that use them.

USB-C (USB 5Gbps) receptacle on a laptop with the legacy trident port logo

USB4 specification, released in 2019, is the first USB transfer protocol standard that is applicable exclusively via USB-C.

## Ease of use

The USB-C standard simplifies usage by specifying cables having identical plugs on both ends; orientation is moot. The plugs are flat, but will work if inserted right-side-up or upside-down. Furthermore, when connecting two devices the user can plug either end of the cable into either device.

### Symmetry

- Mechanically, the USB-C receptacles have two-fold rotational symmetry because a plug may be inserted into a receptacle in either of two orientations. The ends are also mechanically interchangeable, because both end have the same type of plug.
- Electrically, USB-C plugs are not symmetric, as can be seen in the tables of pin layouts. Also, the two ends of the USB-C are electrically different, as can be seen in the table of cable wiring.
- Logically, the plugs are completely symmetrical. The illusion of symmetry results from how devices respond to the cable. Software makes the plugs and cables behave as though they are symmetric. According to the specifications, "Determination of this host-to-device relationship is accomplished through a Configuration Channel (CC) that is connected through the cable."[8]

### Universal application

The USB-C standard attempts to eliminate the need to have different cables for other communication technologies, such as Thunderbolt, PCIe, HDMI, DisplayPort and more. Over the past decade since 2014, many companies including Samsung Electronics, Apple Inc. and Transsion have adopted the USB-C standard into their products.[3] USB-C cables can contain circuit boards and processors giving them much more capability than simple circuit connections.

## Overview

USB-C cables interconnect hosts and peripheral devices, replacing various other electrical cables and connectors, including all earlier (legacy) USB connectors, HDMI connectors, DisplayPort ports, and 3.5 mm audio jacks.[9][10]

### Name

USB Type-C and USB-C are trademarks of the USB Implementers Forum.[11]

### Connectors

The 24-pin double-sided connector is slightly larger than the non-SuperSpeed, USB 2.0 Micro connectors, with a USB-C receptacle opening measuring 8.34 mm × 2.56 mm, 6.20 mm deep.


USB-C plug and receptacle on an iPad Pro

### Cables

Type-C cables can be split among various categories and subcategories. The first one is USB 2.0 vs Full-Featured. USB 2.0 Type-C cables have very limited wires and are only good for USB 2.0 communications and power delivery. They are also called charging cables colloquially. Conversely, Full-Featured cables need to have all wires populated and in general support Alternate Modes and are further distinguished by their speed rating.

Full-Featured cables exist in different speed grades. Their technical names, the operation mode, use the "Gen A" notation, each higher number increasing capabilities in terms of bit rates. The marketing names are based on the theoretically highest bit rate a connection should fulfill – but in reality never can –, namely "USB 5Gbps", "USB 10Gbps", "USB 20Gbps", "USB 40Gbps", and so on.

A Gen 1 (signalling rate of 5 Gbit/s) cable supports that rate on every one of its two lanes (2*2 twisted data wire pairs). So it can be used to establish a USB 3.2 Gen 1x2 operation mode with nominally 10 Gbit/s (2*5 Gbit/s) between two USB 3.2, or USB4 or newer capable hosts connected by a Fully-Featured USB-C cable.

According the specification USB4 and USB4  2.0, the implementation of the USB 3.2 Gen 2x2 operation mode, resulting in nominally 20 Gbit/s, is only optional; indeed most USB4 controllers do not implement it. In other words, the "USB 20Gbps" will only be achieved by a USB4 or more modern version of an operation mode, such as USB4 Gen 2x2, with USB4 capable devices and Fully-Featured USB-C cables (and intermediate hubs) of a rate of "USB 20Gbps" (I.e. all 4 twisted data wire pairs are wired) or higher between a host and a peripheral device.

The USB Implementers Forum certifies valid cables so they can be marked accordingly with the official logos and users can distinguish them from non-compliant products.[12] There have been simplifications in the logos.[13] Previous logos and names also referenced specific USB protocols like SuperSpeed for the USB 3 family of connections or USB4 directly. The current official names and logos have removed those references as most full-featured cables can be used for USB4 connections as well as USB 3 connections.

In order to achieve longer cable lengths, cable variants with active electronics to amplify the signals also exist. The Type-C standard mostly mandates these active cables to behave similarly to passive cables with vast backwards compatibility, but they are not mandated to support all possible features and typically have no forward compatibility to future standards. Optical cables are even allowed to further reduce the backwards compatibility. For example, an active cable may not be able to use all high speed wire-pairs in the same direction (as used for DisplayPort connections), but only in the symmetric combinations expected by classic USB connections. Passive cables have no such limitations.

### Power delivery

Every USB-C cable must support at least 3 amps of current and up to 20 volts for up to 60 watts of power according to the USB PD specification. Cables were also allowed to support up to 5 A while retaining the 20 V limit, allowing up to 100 W of power; however, the 20 V limit for 5 A cables has been deprecated in favor of 48 V. The combination of higher voltage support and 5 A current support is called Extended Power Range (EPR) and allows for up to 240 W (48 V, 5 A) of power according to the USB PD specification.

### E-Marker

All Type-C cables except the minimal combination of USB 2.0 and only 3 A must contain E-Marker chips that identify the cable and its capabilities via the USB PD protocol. This identification data includes information about product/vendor, cable connectors, USB signalling protocol (2.0, Gen speed rating , Gen 2), passive/active construction, use of $V_{CONN}$ power, available $V_{BUS}$ current, latency, RX/TX directionality, SOP controller mode, and hardware/firmware version.[14] It also can include further vendor-defined messages (VDM) that detail support for Alt modes or vendor-specific functionality outside of the USB standards.

## Hosts and peripheral devices

For any two pieces of equipment connecting over USB, one is a host (with a downstream-facing port, DFP) and the other is a peripheral device (with an upstream-facing port, UFP). Some products, such as mobile phones, can take either role, whichever is opposite that of the connected equipment. Such equipment is said to have Dual-Role-Data (DRD) capability, which was known as USB On-The-Go in the previous specification.[15] With USB-C, when two such devices are connected, the roles are first randomly assigned, but a swap can be commanded from either end, although there are optional path and role detection methods that would allow equipment to select a preference for a specific role. Furthermore, Dual-Role equipment that implements USB Power Delivery may swap data and power roles independently using the Data Role Swap or Power Role Swap processes. This allows for charge-through hub or docking station applications such as a portable computer acting as a *host* to connect to peripherals but being powered by the dock, or a computer being powered by a display, through a single USB-C cable.[7]

USB-C devices may optionally provide or consume bus power currents of 1.5 A and 3.0 A (at 5 V) in addition to baseline bus power provision; power sources can either advertise increased USB current through the configuration channel or implement the full USB Power Delivery specification using both the BMC-coded configuration line and the legacy BFSK-coded $V_{BUS}$ line.[7][16]

All older USB connectors (all Type-A and Type-B) are designated legacy. Connecting legacy and modern, USB-C equipment requires either a legacy cable assembly (a cable with any Type-A or Type-B plug on one end and a Type-C plug on the other) or, in very specific cases, a legacy adapter assembly.

An older device can connect to a modern (USB-C) host by using a legacy cable, with a Standard-B, Mini-B, or Micro-B plug on the device end and a USB-C plug on the other. Similarly, a modern device can connect to a legacy host by using a legacy cable with a USB-C plug on the device end and a Standard-A plug on the host end. Legacy adapters with USB-C receptacles are "not defined or allowed" by the specification because they can create "many invalid and potentially unsafe" cable combinations (being any cable assembly with two *A* ends or two *B* ends). However, exactly two types of USB adapters with Type-C plugs are defined: An adapter with a Standard-A receptacle (for connecting a legacy device to a modern host, and supporting up to 10 Gbit/s), and one with a Micro-B receptacle (for connecting a modern device to a legacy host or power supply, and supporting up to USB 2.0).[17]

## Non-USB modes

### Liquid Corrosion Mitigation Mode

This is an optional mode that aims to reduces the risk of corrosion within the Type-C port by driving voltages down as close as possible to 0V.

### Debug Accessory Mode

This mode can be used for both high-level and low-level debugging purposes. For embedded devices this can be used to allow access to e.g. JTAG Test Access Port without having to open the casing of the device. It is designed for both usage within LAB as well as within production environments. Basic debug requirements are defined as a standard feature and should therefore be present on all compliant devices. A vendor may add additional debug features as required. The spec demands the assurance by the vendor that an explicit user authorization and the actual vendor specific implementation do not compromise system security and user privacy. Also noteworthy is that detecting the orientation of the cable is considered optional for this mode. Meaning it may only work when the cable is plugged in one way but not when it is plugged in upside down.

To enter this mode the device needs to detect both CC pins being terminated using pull-up or pull-down resistors each (either both pins using pull-up or both using pull-down resistors). Because this mode uses both CC Pins a receptacle to receptacle connection using compliant Type-C to Type-C cables cannot be detected. This is because the spec does not allow USB-C cables to connect both CC wires through and demands CC2 aka $V_{CONN}$ to be isolated instead. Therefore either a non-compliant cable that connects this pin anyway, a captive cable (that is one where one side is either hard-wired or has a proprietary connector), or a directly attaching device (aka something that connects without a cable) is needed.

#### Alternate modes

An Alternate Mode dedicates some of the physical wires in a USB-C cable for direct device-to-host transmission using non-USB data protocols, such as DisplayPort or Thunderbolt. The four high-speed lanes, two side-band pins, and (for dock, detachable device and permanent-cable applications only) five additional pins can be used for Alternate Mode transmission. The modes are configured using vendor-defined messages (VDM) through the configuration channel.

#### Analog Audio Adapter Accessory Mode (deprecated)

Deprecated in October 2024, with the Type-C Cable and Connector Specification version 2.3,[18] to allow for the new Liquid Corrosion Mitigation Mode, this mode allowed a device with a Type-C port to drive analog headsets directly through an audio adapter with a 3.5 mm jack, providing three analog audio channels (left and right output and a monaural microphone input). Unlike superficially similar Lightning adapters, which handle all analog conversion and audio amplification internally, the adapters that used this Accessory Mode contained no electronics and required that the host device have all the additional components to handle analog audio – digital-to-analog converters and amplifiers for audio output and an analog-to-digital converter to handle the analog microphone signal. Such an adapter could optionally include a USB-C charge-through port to allow 500 mA device charging. The engineering specification states that an analog headset shall not use a USB-C plug instead of a 3.5 mm plug. In other words, a headset with a USB-C plug must always support digital audio (but optionally could support the Accessory Mode).[19]

Analog signals used the USB 2.0 differential pair contacts (Dp and Dn for right and left) and the two side-band use contacts for microphone and ground. The presence of the audio accessory was signaled through the configuration channel and $V_{CONN}$.

With the deprecation of Analog Audio mode, the Type-C specification strongly recommends using USB Audio Device Class 4.0 while also recommending version 2.0.[20]

## Specifications

### USB Type-C cable and connector specifications

The USB Type-C specification 1.0 was published by the USB Implementers Forum (USB-IF) and was finalized in August 2014.[10]

It defines requirements for cables and connectors.

- Rev 1.1 was published 2015-04-03.[21]
- Rev 1.2 was published 2016-03-25.[22]
- Rev 1.3 was published 2017-07-14.[23]
- Rev 1.4 was published 2019-03-29.[23]
- Rev 2.0 was published 2019-08-29.[24]
- Rev 2.1 was published 2021-05-25 (USB PD Extended Power Range: 240 W as 48 V × 5 A).[25]
- Rev 2.2 was published 2022-10-18, primarily for enabling USB 80Gbps (USB4 Version 2.0 specification) over USB Type-C connectors and cables.[17]
- Rev 2.3 was published 2023-10-31.[26]
- Rev 2.4 was published 2024-10-21.[27]

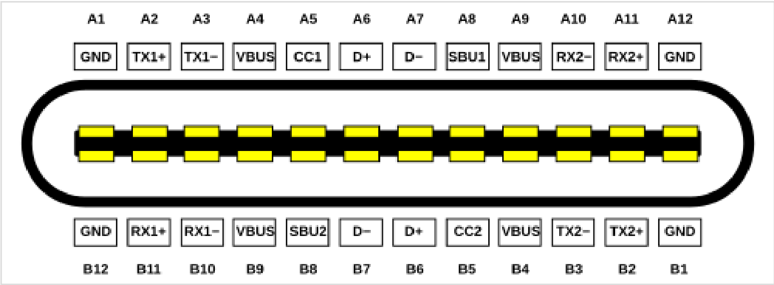
Adoption as IEC specification:

- IEC 62680-1-3:2016 (2016-08-17, edition 1.0) "Universal serial bus interfaces for data and power – Part 1-3: Universal Serial Bus interfaces – Common components – USB Type-C cable and connector specification"[28][29]

- IEC 62680-1-3:2017 (2017-09-25, edition 2.0) "Universal serial bus interfaces for data and power – Part 1-3: Common components – USB Type-C Cable and Connector Specification"[30]
- IEC 62680-1-3:2018 (2018-05-24, edition 3.0) "Universal serial bus interfaces for data and power – Part 1-3: Common components – USB Type-C Cable and Connector Specification"[31]

### Receptacles

The receptacle features four power and four ground pins, two differential pairs (connected together on devices) for legacy USB 2.0 high-speed data, four shielded differential pairs for Enhanced SuperSpeed data (two transmit and two receive pairs), two Sideband Use (SBU) pins, and two Configuration Channel (CC) pins.



Type-C receptacle pinout (end-on view)

Type-C receptacle A pin layout

| Pin | Name | Description |
|---|---|---|
| **A1** | GND | Ground return |
| **A2** | SSTXp1 ("TX1+") | SuperSpeed differential pair #1, transmit, positive |
| **A3** | SSTXn1 ("TX1−") | SuperSpeed differential pair #1, transmit, negative |
| **A4** | $V_{BUS}$ | Bus power |
| **A5** | CC1 | Configuration channel |
| **A6** | D+ | USB 2.0 differential pair, position 1, positive |
| **A7** | D− | USB 2.0 differential pair, position 1, negative |
| **A8** | SBU1 | Sideband use (SBU) |
| **A9** | $V_{BUS}$ | Bus power |
| **A10** | SSRXn2 ("RX2−") | SuperSpeed differential pair #4, receive, negative |
| **A11** | SSRXp2 ("RX2+") | SuperSpeed differential pair #4, receive, positive |
| **A12** | GND | Ground return |

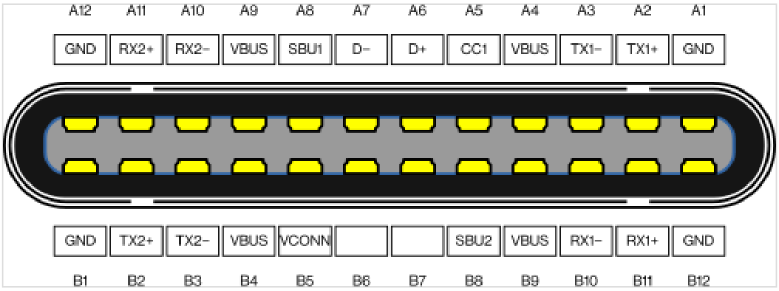
Type-C receptacle B pin layout

| Pin | Name | Description |
|---|---|---|
| **B12** | GND | Ground return |
| **B11** | SSRXp1 ("RX1+") | SuperSpeed differential pair #2, receive, positive |
| **B10** | SSRXn1 ("RX1−") | SuperSpeed differential pair #2, receive, negative |
| **B9** | $V_{BUS}$ | Bus power |
| **B8** | SBU2 | Sideband use (SBU) |
| **B7** | D− | USB 2.0 differential pair, position 2, negative[a] |
| **B6** | D+ | USB 2.0 differential pair, position 2, positive[a] |
| **B5** | CC2 | Configuration channel |
| **B4** | $V_{BUS}$ | Bus power |
| **B3** | SSTXn2 ("TX2−") | SuperSpeed differential pair #3, transmit, negative |
| **B2** | SSTXp2 ("TX2+") | SuperSpeed differential pair #3, transmit, positive |
| **B1** | GND | Ground return |

a. In USB-C cables, only one of the differential pairs for non-SuperSpeed are wired. Therefore only either receptacle's A6/A7 or B6/B7 pin pairs are active for non-SuperSpeed connections.

### Plugs

The plug has only one USB 2.0 high-speed differential pair, and one of the CC pins (CC2) is replaced by $V_{CONN}$, to power optional electronics in the cable, and the other is used to actually carry the Configuration Channel (CC) signals. These signals are used to determine the orientation of the cable, as well as to carry USB Power Delivery communications.



Type-C plug pinout (end-on view)

### Cables

Although plugs have 24 pins, cables commonly have only 18 wires. In the following table, the "**No.**" column shows the wire number as assigned within the spec. It is allowed to use multiple wires instead of a single wire. The spec does not demand having two GND and $V_{BUS}$ wires even though it allocated a wire number for them. Note that within the plugs all of the $V_{BUS}$ wires must be joined together. The same is true for all of the GND wires (including shielding).

Full-Featured USB 3.2 and 2.0 Type-C cable wiring

| Plug 1, USB Type-C | | USB Type-C cable | | | | | Plug 2, USB Type-C | |
|---|---|---|---|---|---|---|---|---|
| **Pin** | **Name** | **Wire color** | **No.** | **Name** | **Description** | **2.0**[a] | **Pin** | **Name** |
| Shell[b] | Shield | Braid | Braid | Shield | Cable external braid | ✓ | Shell[b] | Shield |
| A1, B12, B1, A12[b] | GND | Tin-plated | 1 | GND_PWRrt1 | Ground for power return | ✓ | A1, B12, B1, A12[b] | GND |
| | | | 16 | GND_PWRrt2 (**optional**) | | | | |
| A4, B9, B4, A9[c] | $V_{BUS}$ | Red | 2 | PWR_$V_{BUS}$1 | $V_{BUS}$ power | ✓ | A4, B9, B4, A9[c] | $V_{BUS}$ |
| | | | 17 | PWR_$V_{BUS}$2 (**optional**) | | | | |
| B5 | $V_{CONN}$ | Yellow | 18 | PWR_$V_{CONN}$ (**optional**) | $V_{CONN}$ power, for powered cables[d] | ✓ | B5 | $V_{CONN}$ |
| A5 | CC | Blue | 3 | CC | Configuration channel | ✓ | A5 | CC |
| A6 | D+ | Green | 4 | UTP_Dp[e] | Unshielded twisted pair, positive | ✓ | A6 | D+ |
| A7 | D− | White | 5 | UTP_Dn[e] | Unshielded twisted pair, negative | ✓ | A7 | D− |
| A8 | SBU1 | Red | 14 | SBU_A | Sideband use A | ✗ | B8 | SBU2 |
| B8 | SBU2 | Black | 15 | SBU_B | Sideband use B | ✗ | A8 | SBU1 |
| A2 | SSTXp1 | Yellow[f] | 6 | SDPp1 | Shielded differential pair #1, positive | ✗ | B11 | SSRXp1 |
| A3 | SSTXn1 | Brown[f] | 7 | SDPn1 | Shielded differential pair #1, negative | ✗ | B10 | SSRXn1 |
| B11 | SSRXp1 | Green[f] | 8 | SDPp2 | Shielded differential pair #2, positive | ✗ | A2 | SSTXp1 |
| B10 | SSRXn1 | Orange[f] | 9 | SDPn2 | Shielded differential pair #2, negative | ✗ | A3 | SSTXn1 |
| B2 | SSTXp2 | White[f] | 10 | SDPp3 | Shielded differential pair #3, positive | ✗ | A11 | SSRXp2 |
| B3 | SSTXn2 | Black[f] | 11 | SDPn3 | Shielded differential pair #3, negative | ✗ | A10 | SSRXn2 |
| A11 | SSRXp2 | Red[f] | 12 | SDPp4 | Shielded differential pair #4, positive | ✗ | B2 | SSTXp2 |
| A10 | SSRXn2 | Blue[f] | 13 | SDPn4 | Shielded differential pair #4, negative | ✗ | B3 | SSTXn2 |

a. USB 2.0 Type-C cables do not include wires for SuperSpeed or sideband use.
b. All GND wires and shielding must be connected together
c. All $V_{BUS}$ wires must be connected together
d. $V_{CONN}$ must not traverse end-to-end through the cable. Some isolation method must be used. Can also be terminated directly within the plugs at both ends without a connecting wire in between. There are exceptions for cases where the USB-C wire is directly connected to a device at one end to allow these devices to be powered through $V_{CONN}$. Thereby it is basically the active component within the cable which allows to workaround the restriction of USB-C cables not being allowed to Backfeed power $V_{CONN}$ into connected devices.
e. There is only a single differential pair for non-SuperSpeed data in the cable, which is connected to A6 and A7. Contacts B6 and B7 should not be present in the plug.
f. Wire colors for differential pairs are not mandated.

## Related USB-IF specifications

**USB Type-C Locking Connector Specification**

The USB Type-C Locking Connector Specification was published 2016-03-09. It defines the mechanical requirements for USB-C plug connectors and the guidelines for the USB-C receptacle mounting configuration to provide a standardized screw lock mechanism for USB-C connectors and cables.[32]

**USB Type-C Port Controller Interface Specification**

The USB Type-C Port Controller Interface Specification was published 2017-10-01. It defines a common interface from a USB-C Port Manager to a simple USB-C Port Controller.[33]

**USB Type-C Authentication Specification**

Adopted as IEC specification: IEC 62680-1-4:2018 (2018-04-10) "Universal Serial Bus interfaces for data and power – Part 1-4: Common components – USB Type-C Authentication Specification"[34]

**USB 2.0 Billboard Device Class Specification**

USB 2.0 Billboard Device Class is defined to communicate the details of supported Alternate Modes to the computer host OS. It provides user readable strings with product description and user support information. Billboard messages can be used to identify incompatible connections made by users. They optionally appear to negotiate multiple Alternate Modes and must appear when negotiation fails between the host (source) and device (sink).

**USB Audio Device Class 3.0 Specification**

USB Audio Device Class 3.0 defines powered digital audio headsets with a USB-C plug.[7] The standard supports the transfer of both digital and analog audio signals over the USB port.[35]

**USB Power Delivery Specification**

While it is not necessary for USB-C compliant devices to implement USB Power Delivery, for USB-C DRP/DRD (Dual-Role-Power/Data) ports, USB Power Delivery introduces commands for altering a port's power or data role after the roles have been established when a connection is made.[36]

**USB 3.2 Specification**

USB 3.2, released in September 2017, replaces the USB 3.1 specification. It preserves existing USB 3.1 SuperSpeed and SuperSpeed+ data modes and introduces two new SuperSpeed+ transfer modes over the USB-C connector using two-lane operation, doubling the signalling rates to 10 and 20 Gbit/s (raw data rate 1 and ~2.4 GB/s).

**USB4 Specification**

The USB4 specification released in 2019 is the first USB data transfer specification to be exclusively applicable by the Type-C connector.

## Alternate Mode partner specifications

As of 2018, five system-defined Alternate Mode partner specifications exist. Additionally, vendors may support proprietary modes for use in dock solutions. Alternate Modes are optional; Type-C features and devices are not required to support any specific Alternate Mode, nor are they required to support USB (though some standards using Alternate Modes, such as Thunderbolt, require that all compatible ports support USB communications as well). The USB Implementers Forum is working with its Alternate Mode partners to make sure that ports are properly labelled with respective logos.[37]

List of Alternate Mode partner specifications

| Logo | Name | Date | Protocol | Status |
|---|---|---|---|---|
| | Thunderbolt Alternate Mode | Announced in June 2015[38] | USB-C is the native (and only) connector for Thunderbolt 3 and later<br><br>Thunderbolt 3 (also carries 4× PCI Express 3.0, DisplayPort 1.2, DisplayPort 1.4, USB 3.1 Gen 2),[38][39][40][41]<br>Thunderbolt 4 (also carries 4× PCI Express 3.0, DisplayPort 2.0, USB4),<br>Thunderbolt 5 (also carries 4× PCI Express 4.0, DisplayPort 2.1, USB4) | Current |
| | DisplayPort Alternate Mode | Published in September 2014 | DisplayPort 1.2, DisplayPort 1.4,[42][43] DisplayPort 2.0[44] | Current |
| | Mobile High-Definition Link (MHL) Alternate Mode | Announced in November 2014[45] | MHL 1.0, 2.0, 3.0 and superMHL 1.0[46][47][48][49] | Current |
| | HDMI Alternate Mode | Announced in September 2016[50] | HDMI 1.4b[51][52][53][54] | Not being updated |
| | VirtualLink Alternate Mode | Announced in July 2018[55] | VirtualLink 1.0[56] | Abandoned |

Other protocols, like Ethernet,[57] have been proposed, although Thunderbolt 3 and later are also capable of 10 Gigabit Ethernet networking.[58]

All Thunderbolt 3 controllers support both Thunderbolt Alternate Mode and DisplayPort Alternate Mode.[59] Because Thunderbolt can encapsulate DisplayPort data, every Thunderbolt controller can either output DisplayPort signals directly over DisplayPort Alternative Mode or encapsulated within Thunderbolt in Thunderbolt Alternate Mode. Low-cost peripherals mostly connect via DisplayPort Alternate Mode while some docking stations tunnel DisplayPort over Thunderbolt.[60]

DisplayPort Alternate Mode does not support DisplayPort Dual-Mode (DP++), which allows DisplayPort sources to output HDMI-compatible signals. As a result, USB Type-C-to-HDMI adapters or cables which use DisplayPort Alternate Mode must incorporate active conversion circuitry.[61] DisplayPort Alternate Mode 2.0: DisplayPort 2.0 can run directly over USB-C alongside USB4. DisplayPort 2.0 can support 8K resolution at 60 Hz with HDR10 color and can use up to 80 Gbps, which is double the amount available to USB data.[62]

As of 2023, there were no known USB type-C to HDMI adapters using HDMI Alternate Mode, according to the HDMI Licensing Association.[63]

The USB SuperSpeed protocol is similar to DisplayPort and PCIe/Thunderbolt, in using packetized data transmitted over differential LVDS lanes with embedded clock using comparable bit rates, so these Alternate Modes are easier to implement in the chipset.[42]

Alternate Mode hosts and peripheral devices can be connected with either regular Full-Featured Type-C cables, or with converter cables or adapters:

**USB 3.1 Type-C to Type-C Full-Featured cable**

DisplayPort, Mobile High-Definition Link (MHL), HDMI and Thunderbolt (20 Gbit/s, or 40 Gbit/s with cable length up to 0.5 m) Alternate Mode Type-C ports can be interconnected with standard passive Full-Featured USB Type-C cables. These cables are only marked with standard "trident" SuperSpeed USB logo (for Gen 1 mode only) or the SuperSpeed+ USB 10 Gbit/s logo on both ends.[64] Cable length should be 2.0 m or less for Gen 1 and 1.0 m or less for Gen 2.

**Thunderbolt Type-C to Type-C active cable**

Thunderbolt 3 (40 Gbit/s) Alternate Mode with cables longer than 0.8 m requires active Type-C cables that are certified and electronically marked for high-speed Thunderbolt 3 transmission, similarly to high-power 5 A cables.[38][41] These cables are marked with a Thunderbolt logo on both ends. They do not support USB 3 backwards compatibility, only USB 2 or Thunderbolt. Cables can be marked for both Thunderbolt and 5 A power delivery at the same time.[65]

Active cables and adapters contain powered electronics to allow for longer cables or to perform protocol conversion. The adapters for video Alternate Modes may allow conversion from native video stream to other video interface standards (e.g., DisplayPort, HDMI, VGA or DVI).

Using Full-Featured Type-C cables for Alternate Mode connections provides some benefits. Alternate Mode does not employ USB 2.0 lanes and the configuration channel lane, so USB 2.0 and USB Power Delivery protocols are always available. In addition, DisplayPort and MHL Alternate Modes can transmit on one, two, or four SuperSpeed lanes, so two of the remaining lanes may be used to simultaneously transmit USB 3.1 data.[66]

Alternate Mode protocol support matrix for Type-C cables and adapters

| Mode | USB 3.1 Type-C cable[a] | | | | | | | Adapter cable or adapter | | | | | Construction |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | USB[b] | DisplayPort | | Thunderbolt | | superMHL | HDMI | HDMI | | DVI-D | | Component video | |
| | 3.1 | 1.2 | 1.4 | 20 Gbit/s | 40 Gbit/s | | 1.4b | 1.4b | 2.0b | Single-link | Dual-link | (YPbPr, VGA/DVI-A) | |
| **DisplayPort** | Yes | Yes | | — | | | | No | | | | | Passive |
| | — | Optional | | | | | | Yes | | Yes | | Yes | **Active** |
| **Thunderbolt** | Yes[c] | Yes[c] | | Yes | Yes[d] | — | | No | | | | | Passive |
| | — | Optional | | Optional | Yes | | | Yes | | Yes | | Yes | **Active** |
| **MHL** | Yes | — | | | | Yes | — | Yes | No | Yes | No | No | Passive |
| | — | | | | | Optional | | — | Yes | — | | Yes | **Active** |
| **HDMI** | — | | | | | | Yes | Yes | No | Yes | No | No | Passive |
| | | | | | | | Optional | — | | | | Yes | **Active** |

a. USB 2.0 and USB Power Delivery are available at all times in a Type-C cable
b. USB 3.1 can be transmitted simultaneously when the video signal bandwidth requires two or fewer lanes.
c. Is only available in Thunderbolt 3 DisplayPort mode
d. Thunderbolt 3 40 Gbit/s passive cables are only possible <0.8 m due to limitations of current cable technology.

# USB-C receptacle pin usage in different modes

The diagrams below depict the pins of a USB-C receptacle in different use cases.

## USB 2.0/1.1

A simple USB 2.0/1.1 device mates using one pair of D+/D− pins. Hence, the source (host) does not require any connection management circuitry, but it lacks the same physical connector so therefore USB-C is not backward compatible. $V_{BUS}$ and GND provide 5 V up to 500 mA of current.

However, to connect a USB 2.0/1.1 device to a USB-C host, use of pull-down resistors Rd[67] on the CC pins is required, as the source (host) will not supply $V_{BUS}$ until a connection is detected through the CC pins.

This means many USB-A–to–USB-C cables will only work in the A to C direction (connecting to a USB-C devices, e.g. for charging) as they do not include the termination resistors needed to work in the C to A direction (from a USB-C host). Adapters or cables from USB-C to a USB-A receptacle usually do work as they include the required termination resistor.

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **GND** | TX1+ | TX1− | $\mathbf{V_{BUS}}$ | CC1 | **D+** | **D−** | SBU1 | $\mathbf{V_{BUS}}$ | RX2− | RX2+ | **GND** |
| **GND** | RX1+ | RX1− | $\mathbf{V_{BUS}}$ | SBU2 | **D−** | **D+** | CC2 | $\mathbf{V_{BUS}}$ | TX2− | TX2+ | **GND** |

## USB Power Delivery

The USB Power Delivery specification uses one of CC1 or CC2 pins for power negotiation between source device and sink device, up to 20 V at 5 A. It is transparent to any data transmission mode, and can therefore be used together with any of them as long as the CC pins are intact.

An extension to the specification has added 28 V, 36 V and 48 V to support up to 240 W of power for laptops, monitors, hard disks and other peripherals.[68]

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **GND** | TX1+ | TX1− | $\mathbf{V_{BUS}}$ | **CC1** | D+ | D− | SBU1 | $\mathbf{V_{BUS}}$ | RX2− | RX2+ | **GND** |
| **GND** | RX1+ | RX1− | $\mathbf{V_{BUS}}$ | SBU2 | D− | D+ | **CC2** | $\mathbf{V_{BUS}}$ | TX2− | TX2+ | **GND** |

## USB 3.2

In the USB 3.2 mode, two or four high speed links are used in TX/RX pairs to provide 5, 10, or 20 Gbit/s (only by USB 3.2 x2 two-lane operations) signalling rates respectively. One of the CC pins is used to negotiate the mode.

$V_{BUS}$ and GND provide 5 V up to 900 mA, in accordance with the USB 3.1 specification. A specific USB-C mode may also be entered, where 5 V at nominal either 1.5 A or 3 A is provided.[69] A third alternative is to establish a USB Power Delivery (USB-PD) contract.

In single-lane mode, only the differential pairs closest to the CC pin are used for data transmission. For dual-lane data transfers, all four differential pairs are in use.

The D+/D− link for USB 2.0/1.1 is *typically* not used when a USB 3.x connection is active, but devices like hubs open simultaneous 2.0 and 3.x uplinks in order to allow operation of both types of devices connected to it. Other devices may have the ability to fall back to 2.0, in case the 3.x connection fails. For this, it is important that SS and HS lanes are correctly aligned so that i.e. operating system messages indicating overcurrent conditions report the correct shared USB plug.

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **GND** | **TX1+** | **TX1−** | $\mathbf{V_{BUS}}$ | **CC1** | **D+** | **D−** | SBU1 | $\mathbf{V_{BUS}}$ | **RX2−** | **RX2+** | **GND** |
| **GND** | **RX1+** | **RX1−** | $\mathbf{V_{BUS}}$ | SBU2 | **D−** | **D+** | **CC2** | $\mathbf{V_{BUS}}$ | **TX2−** | **TX2+** | **GND** |

## Alternate Modes

In Alternate Modes one of up to four high speed links are used in whatever direction is needed. SBU1, SBU2 provide an additional lower speed link. If two high speed links remain unused, then a USB 3.2 link can be established concurrently to the Alternate Mode.[43] One of the CC pins is used to perform all the negotiation. An additional low band bidirectional channel (other than SBU) may share that CC pin as well.[43][51] USB 2.0 is also available through D+/D− pins.

In regard to power, the devices are supposed to negotiate a Power Delivery contract before an Alternate Mode is entered.[70]

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **GND** | **TX1+** | **TX1−** | **$V_{BUS}$** | **CC1** | **D+** | **D−** | **SBU1** | **$V_{BUS}$** | **RX2−** | **RX2+** | **GND** |
| **GND** | **RX1+** | **RX1−** | **$V_{BUS}$** | **SBU2** | **D−** | **D+** | **CC2** | **$V_{BUS}$** | **TX2−** | **TX2+** | **GND** |

## Debug Accessory Mode

The external device test system (DTS) signals to the target system (TS) to enter debug accessory mode via CC1 and CC2 both being pulled down with an Rd resistor value or pulled up as Rp resistor value from the test plug (Rp and Rd defined in Type-C specification).

After entering debug accessory mode, optional orientation detection via the CC1 and CC2 is done via setting CC1 as a pullup of Rd resistance and CC2 pulled to ground via Ra resistance (from the test system Type-C plug). While optional, orientation detection is required if USB Power Delivery communication is to remain functional.

In this mode, all digital circuits are disconnected from the connector, and the 14 bold pins can be used to expose debug related signals (e.g. JTAG interface). USB IF requires for certification that security and privacy consideration and precaution has been taken and that the user has actually requested that debug test mode be performed.

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| GND | **TX1+** | **TX1−** | $V_{BUS}$ | CC1 | **D+** | **D−** | **SBU1** | $V_{BUS}$ | **RX2−** | **RX2+** | GND |
| GND | **RX1+** | **RX1−** | $V_{BUS}$ | **SBU2** | **D−** | **D+** | CC2 | $V_{BUS}$ | **TX2−** | **TX2+** | GND |

If a reversible Type-C cable is required but Power Delivery support is not, the test plug will need to be arranged as below, with CC1 and CC2 both being pulled down with an Rd resistor value or pulled up as Rp resistor value from the test plug:

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| GND | **TS1** | **TS2** | $V_{BUS}$ | CC1 | **TS6** | **TS7** | **TS5** | $V_{BUS}$ | **TS4** | **TS3** | GND |
| GND | **TS3** | **TS4** | $V_{BUS}$ | **TS5** | **TS7** | **TS6** | CC2 | $V_{BUS}$ | **TS2** | **TS1** | GND |

This mirroring of test signals will only provide 7 test signals for debug usage instead of 14, but with the benefit of minimizing extra parts count for orientation detection.

## Audio Adapter Accessory Mode

In this mode, all digital circuits are disconnected from the connector, and certain pins become reassigned for analog outputs or inputs. The mode, if supported, is entered when both CC pins are shorted to GND. D− and D+ become audio output left L and right R, respectively. The SBU pins become a microphone pin MIC, and the analog ground AGND, the latter being a return path for both outputs and the microphone. Nevertheless, the MIC and AGND pins must have automatic swap capability, for two reasons: firstly, the USB-C plug may be inserted either side; secondly, there is no agreement, which TRRS rings shall be GND and MIC, so devices equipped with a headphone jack with microphone input must be able to perform this swap anyway.[71]

This mode also allows concurrent charging of a device exposing the analog audio interface (through $V_{BUS}$ and GND), however only at 5 V and 500 mA, as CC pins are unavailable for any negotiation.

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **GND** | TX1+ | TX1− | **$V_{BUS}$** | **CC1** | **R** | **L** | **MIC** | **$V_{BUS}$** | RX2− | RX2+ | **GND** |
| **GND** | RX1+ | RX1− | **$V_{BUS}$** | **AGND** | **L** | **R** | **CC2** | **$V_{BUS}$** | TX2− | TX2+ | **GND** |

Plug insertions detection is performed by the TRRS plug's physical plug detection switch. On plug insertions, this will pull down both CC and VCONN in the plug (CC1 and CC2 in the receptacle). This resistance must be less than 800 ohms which is the minimum "Ra" resistance specified in the USB Type-C specification). This is essentially a direct connection to USB digital ground.

TRRS rings wiring to Type-C plug (Figure A-2 of USB Type-C Cable and Connector Specification Release 1.3)

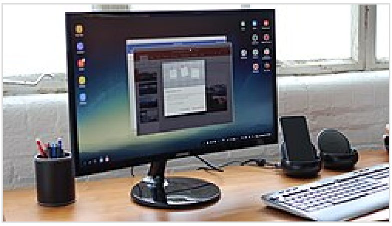
| TRRS socket | Analog audio signal | USB Type-C plug |
|---|---|---|
| Tip | L | D− |
| Ring 1 | R | D+ |
| Ring 2 | Microphone/ground | SBU1 or SBU2 |
| Sleeve | Microphone/ground | SBU2 or SBU1 |
| DETECT1 | Plug presence detection switch | CC, VCONN |
| DETECT2 | Plug presence detection switch | GND |

## Software support

- Android from version 6.0 "Marshmallow" onwards works with USB 3.1 and USB-C.[72]
- ChromeOS, starting with the Chromebook Pixel 2015, supports USB 3.1, USB-C, Alternate Modes, Power Delivery, and USB Dual-Role support.[73]
- FreeBSD released the Extensible Host Controller Interface, supporting USB 3.0, with release 8.2[74]
- iOS first supported USB-C with version 12.1-12.4.1 on iPad Pro (3rd generation). Support returned with version 17.0 or later with iPhone 15 or later.
- iPadOS supports USB-C on iPad Pro (3rd generation) or later, iPad Air (4th generation) or later, iPad Mini (6th generation) or later, and iPad (10th generation) or later.
- NetBSD began supporting USB 3.0 with release 7.2[75]
- Linux has supported USB 3.0 since kernel version 2.6.31 and USB version 3.1 since kernel version 4.6.
- OpenBSD began supporting USB 3.0 in version 5.7[76]
- macOS supports USB-C with USB 3.1, USB-C, Alternate Modes, and Power Delivery[77] on OS X Yosemite 10.10.2 or later on MacBook (Early 2015) or later, MacBook Air (2018) or later, MacBook Pro (2016) or later, Mac mini (2018) or later, iMac (2017) or later, iMac Pro, Mac Studio, and Mac Pro (2019) or later.
- Windows 8.1 added USB-C and billboard support in an update.[78]
- Windows 10 and Windows 10 Mobile support USB 3.1, USB-C, alternate modes, billboard device class, Power Delivery and USB Dual-Role.[79][80]

## Hardware support

### USB-C devices



A Samsung Galaxy S8 plugged into a DeX docking station: The monitor is displaying the PowerPoint and Word Android applications.

An increasing number of motherboards, notebooks, tablet computers, smartphones, hard disk drives, USB hubs and other devices released from 2014 onwards include the USB-C receptacles. However, the initial adoption of USB-C was limited by the high cost of USB-C cables[81] and the wide use of Micro-USB chargers.

#### Video output

A USB-C multiport adapter converts the device's native video stream to DisplayPort/HDMI/VGA, allowing it to be displayed on an external display, such as a television set or computer monitor.

It is also used on USB-C docks designed to connect a device to a power source, external display, USB hub, and optional extra (such as a network port) with a single cable. These functions are sometimes implemented directly into the display instead of a separate dock,[82] meaning a user connects their device to the display via USB-C with no other connections required.

### Compatibility issues

#### Power issues with cables

Many cables claiming to support USB-C are actually not compliant to the standard. These cables can, potentially, damage a device.[83][84][85] There are reported cases of laptops being destroyed due to the use of non-compliant cables.[86]

Some non-compliant cables with a USB-C connector on one end and a legacy USB-A plug or Micro-B receptacle (receptacles also usually being invalid on cables, but see known exceptions in the sections on Hosts and peripheral devices and Audio adapter accessory mode above) on the other end fail to be, or incorrectly terminate the Configuration Channel (CC) with a 10 kΩ pull-up to $V_{BUS}$ instead of the specification-mandated 56 kΩ pull-up,[87] causing a device connected to the cable to incorrectly determine the amount of power it is permitted to draw from the cable. Cables with this issue may not work properly with certain products, including Apple and Google products, and may even damage power sources such as chargers, hubs, or PC USB ports.[88][89]

A defective USB-C cable or power source can cause a USB-C device to see an incorrect and different "declared" voltage than what the source will actually deliver. This may result in an overvoltage on the VBUS pin.

Also due to the fine pitch of the USB-C receptacle, the VBUS pin from the cable may contact with the CC pin of the USB-C receptacle resulting in a short-to-VBUS electrical issue due to the fact that the VBUS pin is rated up to 20 V while the CC pins are rated up to 5.5 V.

To overcome these issues, USB Type-C port protection must be used between a USB-C connector and a USB-C Power Delivery controller.[90]

### Compatibility with audio adapters

The USB-C port can be used to connect wired accessories such as headphones.

There are two modes of audio output from devices: digital and analog. There are primarily two types of USB-C audio adapters: active, e.g. those with digital-to-analog converters (DACs), and passive, without electronics.[91][92]

When an active set of USB-C headphones or adapter is used, digital audio is sent through the USB-C port. The conversion by the DAC and amplifier is done inside of the headphones or adapter, instead of on the phone. The sound quality is dependent on the headphones/adapter's DAC. Active adapters with a built-in DAC have near-universal support for devices that output digital and analog audio, adhering to the **Audio Device Class 3.0** and **Audio Adapter Accessory Mode** specifications.

Examples of such active adapters include external USB sound cards and DACs that do not require special drivers,[93] and USB-C to 3.5 mm headphone jack adapters by Apple, Google, Essential, Razer, HTC, and Samsung.[94]

On the other hand, when a passive adapter is used, digital-to-analog conversion is done on the host device and analog audio is sent through the USB-C port. The sound quality is dependent on the phone's onboard DAC. Passive adapters are only compatible with devices that output analog audio, adhering to the **Audio Adapter Accessory Mode** specification.

USB-C to 3.5 mm audio adapters and USB sound cards compatibility

| Output mode | Specification | Devices | USB-C adapters | |
|---|---|---|---|---|
| | | | **Active** | **Passive, without DACs** |
| **Digital audio** | Audio Device Class 3.0 (digital audio) | Apple iPhone 15, Google Pixel 2, HTC U11, Essential Phone, Razer Phone, Samsung Galaxy Note 10, Samsung Galaxy S10 Lite, Sharp Aquos S2, Asus ZenFone 3, Bluedio T4S, Lenovo Tab 4, GoPro, MacBook | No conversion | Conversion unavailable |
| **Analog audio** | Audio Device Class 3.0 (digital audio)<br>Audio Adapter Accessory Mode (analog audio) | Apple iPhone 15, Moto Z/Z Force, Moto Z2/Z2 Force/Z2 Play, Moto Z3/Z3 Play, Sony Xperia XZ2, Huawei Mate 10 Pro, Huawei P20/P20 Pro, Honor Magic2, LeEco, Xiaomi phones, OnePlus 6T, OnePlus 7/7 Pro/7T/7T Pro, Oppo Find X/Oppo R17/R17 Pro, ZTE Nubia Z17/Z18 | Conversion by adapter | Pass-through |

### Compatibility with other fast-charging technology

In 2016, Benson Leung, an engineer at Google, pointed out that Quick Charge 2.0 and 3.0 technologies developed by Qualcomm are not compatible with the USB-C standard.[95] Qualcomm responded that it is possible to make fast-charge solutions fit the voltage demands of USB-C and that there are no reports of problems; however, it did not address the standard compliance issue at that time.[96] Later in the year, Qualcomm released Quick Charge 4, which it claimed was – as an advancement over previous generations – "USB Type-C and USB PD compliant".[97]

### Regulations for compatibility

In 2021, the European Commission proposed the use of USB-C as a universal charger.[98][99][100] On 4 October 2022, the European Parliament voted in favor of the new law, Radio Equipment Directive 2022/2380, with 602 votes in favor, 13 against and 8 abstentions.[101] The regulation requires that all new mobile phones, tablets, cameras, headphones, headsets, handheld video game consoles, portable speakers, e-readers, keyboards, mice, portable navigation systems, and earbuds sold in the European Union and supporting wired charging, would have to be equipped with a USB-C port and charge with a standard USB-C cable by the end of 2024. Additionally, if these devices support fast charging, they must support USB Power Delivery. These regulations will extend to laptops by early 2026.[102] To comply with these regulations, Apple Inc. replaced its proprietary Lightning connector with USB-C beginning with the iPhone 15 and AirPods Pro second generation, released in 2023.[103] A first modified iPhone having USB-C connector was the result of a hack by Ken Pillonel.[104]

In late December 2024, new EU regulations took effect, mandating USB-C charging ports for all small and medium-sized electronic devices sold in the EU, with laptops following on 28 April 2026[105]. These rules were aimed at reducing waste and saving €250 million annually for consumers. Apple, which initially opposed the changes, had since adopted USB-C for its products. Additionally, consumers can opt not to receive a new charger with their device.[106]

# Security

## Authentication

USB Type-C Authentication is an extension to the USB-C protocol which can add security to the protocol.[107][108][109]

## Vulnerabilities

# See also

- GPMI – Digital audiovisual data interface
- HDMI – Digital audiovisual data interface
  - HDMI § Version 2.1
- Thunderbolt (interface) – Computer hardware interface
- USB hardware – Communication connector using the USB protocol
  - USB hardware § Host and device interface receptacles

# References


1. *Universal Serial Bus Type-C Cable and Connector Specification* Revision 1.3 (14 July 2017), Revision History, page 14.
2. *Universal Serial Bus Type-C Cable and Connector Specification* Revision 1.3 (14 July 2017), Revision History, page 14.
3. "Apple to complete its USB-C transition for AirPods and other accessories by 2025" (https://web.archive.org/web/20250109133551/https://superchargednews.com/2023/09/17/apple-usb-c-transition-done-2025/). 2023-09-17. Archived from the original (https://superchargednews.com/2023/09/17/apple-usb-c-transition-done-2025/) on 2025-01-09. Retrieved 2023-09-21.
4. "Universal Serial Bus Type-C Cable and Connector Specification" (https://www.those.ch/designtechnik/wp-content/uploads/2014/08/USB-Type-C-Specification-Release-1.0.pdf) (PDF). 1.0. USB 3.0 Promoter Group. 2014-08-11. Archived (https://web.archive.org/web/20250213014452/https://www.those.ch/designtechnik/wp-content/uploads/2014/08/USB-Type-C-Specification-Release-1.0.pdf) (PDF) from the original on 2025-02-13. Retrieved 2025-02-26.
5. "USB Type-C Cable and Connector: Language Usage Guidelines from USB-IF" (https://www.usb.org/sites/default/files/usb_type-c_language_product_and_packaging_guidelines_final.pdf) (PDF). Usb.org. Archived (https://web.archive.org/web/20181105062225/https://www.usb.org/sites/default/files/usb_type-c_language_product_and_packaging_guidelines_final.pdf) (PDF) from the original on 2018-11-05. Retrieved 2018-12-15.
6. Hruska, Joel (2015-03-13). "USB-C vs. USB 3.1: What's the difference?" (http://www.extremetech.com/computing/197145-reversible-usb-type-c-finally-on-its-way-alongside-usb-3-1s-10gbit-performance). ExtremeTech. Archived (https://web.archive.org/web/20150411081838/http://www.extremetech.com/computing/197145-reversible-usb-type-c-finally-on-its-way-alongside-usb-3-1s-10gbit-performance) from the original on 2015-04-11. Retrieved 2015-04-09.
7. "USB Type-C Overview" (https://web.archive.org/web/20161220102924/http://www.usb.org/developers/presentations/USB_DevDays_Hong_Kong_2016_-_USB_Type-C.pdf) (PDF). *usb.org*. USB-IF. 2016-10-20. Archived from the original (http://www.usb.org/developers/presentations/USB_DevDays_Hong_Kong_2016_-_USB_Type-C.pdf) (PDF) on 2016-12-20.
8. "Universal Serial Bus Type-C Cable and Connector Specification" (https://www.usb.org/sites/default/files/USB%20Type-C%20Spec%20R2.0%20-%20August%202019.pdf) (PDF). August 2019. p. 30. Retrieved 2023-12-23.
9. Ngo, Dong (2016-10-26). "USB Type-C and Thunderbolt 3: One port to connect them all" (https://www.cnet.com/how-to/usb-type-c-thunderbolt-3-one-cable-to-connect-them-all/). CNET. Archived (https://web.archive.org/web/20170511200535/https://www.cnet.com/how-to/usb-type-c-thunderbolt-3-one-cable-to-connect-them-all/) from the original on 2017-05-11. Retrieved 2015-06-18.
10. Howse, Brett (2014-08-12). "USB Type-C Connector Specifications Finalized" (https://web.archive.org/web/20141228122529/http://www.anandtech.com/show/8377/usb-typec-connector-specifications-finalized). Archived from the original (https://www.anandtech.com/show/8377/usb-typec-connector-specifications-finalized) on 2014-12-28. Retrieved 2014-12-28.
11. "USB Type-C Cable and Connector Specification" (https://www.usb.org/usb-type-ctm-cable-and-connector-specification). USB Implementers Forum, Inc. Archived (https://web.archive.org/web/20181103092132/https://www.usb.org/usb-type-ctm-cable-and-connector-specification) from the original on 2018-11-03. Retrieved 2019-12-19.
12. "USB Compliance and Certification" (https://web.archive.org/web/20161220102836/http://www.usb.org/developers/presentations/USB_DevDays_Hong_Kong_2016_-_USB_Compliance_and_Certification.pdf) (PDF). *usb.org*. USB-IF. 2016-10-20. Archived from the original (http://www.usb.org/developers/presentations/USB_DevDays_Hong_Kong_2016_-_USB_Compliance_and_Certification.pdf) (PDF) on 2016-12-20.
13. "USB Trademark Requirements Chart from USB-IF" (https://usb.org/sites/default/files/trademark_license_agreement_licensed_mark_requirements_final_as_of_september_20_2023.pdf) (PDF). *usb.org*. Retrieved 2024-08-15.

14. "USB Power Delivery" (https://usb.org/document-library/usb-power-delivery). *usb.org*. p187, tab 6.42, p190, tab6.43. Retrieved 2024-08-15.
15. "Transition existing products from USB 2.0 OTG to USB Type-C" (http://www.ti.com/lit/wp/slly017/slly017.pdf) (PDF). Archived (https://web.archive.org/web/20171024095538/http://www.ti.com/lit/wp/slly017/slly017.pdf) (PDF) from the original on 2017-10-24. Retrieved 2017-10-23.
16. "USB Power Delivery" (https://web.archive.org/web/20170816061107/http://www.usb.org/developers/presentations/USB_DevDays_Hong_Kong_2016_-_USB_PD.pdf) (PDF). *usb.org*. USB-IF. 2016-10-20. Archived from the original (http://www.usb.org/developers/presentations/USB_DevDays_Hong_Kong_2016_-_USB_PD.pdf) (PDF) on 2017-08-16. Retrieved 2018-01-03.
17. "Universal Serial Bus Type-C Cable and Connector Specification, Release 2.2, October 2022" (https://usb.org/document-library/usb-type-cr-cable-and-connector-specification-release-22). Archived (https://web.archive.org/web/20231002011738/https://usb.org/document-library/usb-type-cr-cable-and-connector-specification-release-22) from the original on 2023-10-02. Retrieved 2023-10-04.
18. USB Type-C® Cable and Connector Specification Release 2.4 (https://www.usb.org/document-library/usb-type-cr-cable-and-connector-specification-release-24) `USB Type-C 2.4 Release 202410.zip` archive: `USB Type-C Spec R2.4 - October 2024.pdf`, p. 27: *Revision History* table: row *2.3 – October 2023*: *"New release primarily for deprecating the Audio Adapter Accessory Mode and replacing it with the Liquid Corrosion Mitigation Mode, and for updating the Multi-port Charger Shared Capacity definition and behaviors. […]"*
19. *Universal Serial Bus Type-C Cable and Connector Specification* Release 1.3 (14 July 2017), section A.1, page 213.
20. "C. USB Type-C Digital Audio". USB Type-C® Cable and Connector Specification (https://www.usb.org/document-library/usb-type-cr-cable-and-connector-specification-release-24) (Report). Release 2.4. USB Implementers Forum. 2024-10-28. p. 378. Retrieved 2025-08-21.
21. "USB Type-C Specification Release 1.1" (https://github.com/vi117/ppkos/blob/master/extdoc/usb_31_060115/USB%20Type-C/USB%20Type-C%20Specification%20Release%201.1.pdf) (PDF). *GitHub*. 2015-07-13. Archived (https://web.archive.org/web/20190711152956/https://github.com/vi117/ppkos/blob/master/extdoc/usb_31_060115/USB%20Type-C/USB%20Type-C%20Specification%20Release%201.1.pdf) (PDF) from the original on 2019-07-11. Retrieved 2018-11-03.
22. "USB Type-C Connectors and Cable Assemblies Compliance Document, v1.2 | USB-IF" (https://usb.org/document-library/usb-type-c-connectors-and-cable-assemblies-compliance-document-v12). *usb.org*. Archived (https://web.archive.org/web/20200319163422/https://www.usb.org/document-library/usb-type-c-connectors-and-cable-assemblies-compliance-document-v12) from the original on 2020-03-19. Retrieved 2019-05-02.
23. "USB Type-C Cable and Connector Specification Revision 1.4, March 29, 2019" (https://cavousb.it/wp-content/uploads/2020/03/USB-Type-C-Spec-R2.0-August-2019.pdf) (PDF). Archived (https://web.archive.org/web/20200320135754/https://cavousb.it/wp-content/uploads/2020/03/USB-Type-C-Spec-R2.0-August-2019.pdf) (PDF) from the original on 2020-03-20. Retrieved 2020-03-20.
24. "Universal Serial Bus Type-C Cable and Connector Specification, September 21, 2019" (https://www.usb.org/sites/default/files/USB%20Type-C%20Spec%20R2.0%20-%20August%202019.pdf) (PDF). Archived (https://web.archive.org/web/20190914105709/https://www.usb.org/sites/default/files/USB%20Type-C%20Spec%20R2.0%20-%20August%202019.pdf) (PDF) from the original on 2019-09-14. Retrieved 2019-09-21.
25. "USB Type-C Cable and Connector Specification Revision 2.1, May, 2021" (https://usb.org/document-library/usb-type-cr-cable-and-connector-specification-revision-21). Archived (https://web.archive.org/web/20210527135133/https://usb.org/document-library/usb-type-cr-cable-and-connector-specification-revision-21) from the original on 2021-05-27. Retrieved 2021-05-27.
26. "Universal Serial Bus Type-C Cable and Connector Specification" (https://www.usb.org/sites/default/files/USB%20Type-C%202.3%20Release%20202310.zip). *USB Implementers Forum*. 2.3. October 2023. Archived (https://web.archive.org/web/20241009163214/https://www.usb.org/document-library/usb-type-cr-cable-and-connector-specification-release-23) from the original on 2024-10-09. Retrieved 2025-06-23.
27. "Universal Serial Bus Type-C Cable and Connector Specification" (https://usb.org/sites/default/files/USB%20Type-C%202.4%20Release%20202410.zip). *USB Implementers Forum*. 2.4. October 2024. Archived (https://web.archive.org/web/20241221182741/https://usb.org/sites/default/files/USB%20Type-C%202.4%20Release%20202410.zip) from the original on 2024-12-21. Retrieved 2025-02-28.
28. "IEC Formally Adopts USB Type-C, USB Power Delivery and USB 3.1 Specifications" (https://www.businesswire.com/news/home/20160713005046/en/IEC-Formally-Adopts-USB-Type-C%E2%84%A2-USB-Power-Delivery-and-USB-3.1-Specifications) (Press release). 2016-07-13. Archived (https://web.archive.org/web/20210130134100/https://www.businesswire.com/news/home/20160713005046/en/IEC-Formally-Adopts-USB-Type-C%E2%84%A2-USB-Power-Delivery-and-USB-3.1-Specifications) from the original on 2021-01-30. Retrieved 2022-09-10.
29. "IEC 62680-1-3:2016 | IEC Webstore | energy, multimedia, cable, USB, LVDC" (https://webstore.iec.ch/publication/25676). *webstore.iec.ch*. Archived (https://web.archive.org/web/20181104010137/https://webstore.iec.ch/publication/25676) from the original on 2018-11-04. Retrieved 2018-11-03.
30. "IEC 62680-1-3:2017 | IEC Webstore | energy, multimedia, cable, USB, LVDC" (https://webstore.iec.ch/publication/32071). *webstore.iec.ch*. Archived (https://web.archive.org/web/20181104010054/https://webstore.iec.ch/publication/32071) from the original on 2018-11-04. Retrieved 2018-11-03.
31. "IEC 62680-1-3:2018 | IEC Webstore | energy, multimedia, cable, USB, LVDC" (https://webstore.iec.ch/publication/61599). *webstore.iec.ch*. Archived (https://web.archive.org/web/20220501104932/https://webstore.iec.ch/publication/61599) from the original on 2022-05-01. Retrieved 2018-11-03.
32. "USB Type-C Locking Connector Specification | USB-IF" (https://www.usb.org/document-library/usb-type-ctm-locking-connector-specification). *www.usb.org*. Archived (https://web.archive.org/web/20181103092049/https://www.usb.org/document-library/usb-type-ctm-locking-connector-specification) from the original on 2018-11-03. Retrieved 2018-11-03.

33. "USB Type-C Port Controller Interface Specification | USB-IF" (https://www.usb.org/document-library/usb-type-ctm-port-controller-interface-specification). *www.usb.org*. Archived (https://web.archive.org/web/20181103092117/https://www.usb.org/document-library/usb-type-ctm-port-controller-interface-specification) from the original on 2018-11-03. Retrieved 2018-11-03.
34. "IEC 62680-1-4:2018 | IEC Webstore" (https://webstore.iec.ch/publication/60748). *webstore.iec.ch*. Archived (https://web.archive.org/web/20220501104955/https://webstore.iec.ch/publication/60748) from the original on 2022-05-01. Retrieved 2018-11-03.
35. Shilov, Anton. "USB-IF Publishes Audio over USB Type-C Specifications" (https://web.archive.org/web/20181209140540/https://www.anandtech.com/show/10719/usbif-publishes-audio-over-usb-typec-specifications). Archived from the original (https://www.anandtech.com/show/10719/usbif-publishes-audio-over-usb-typec-specifications) on 2018-12-09. Retrieved 2018-11-03.
36. *Universal Serial Bus Type-C Cable and Connector Specification* Release 1.3 (14 July 2017), section 4.5.2, page 144.
37. Cunningham, Andrew (2015-01-09). "USB 3.1 and Type-C: The only stuff at CES that everyone is going to use | Ars Technica UK" (http://arstechnica.co.uk/gadgets/2015/01/usb-3-1-and-type-c-the-only-stuff-at-ces-that-everyone-is-going-to-use/). *ArsTechnica.co.uk*. Archived (https://web.archive.org/web/20150618214443/http://arstechnica.co.uk/gadgets/2015/01/usb-3-1-and-type-c-the-only-stuff-at-ces-that-everyone-is-going-to-use/) from the original on 2015-06-18. Retrieved 2015-06-18.
38. "Thunderbolt 3 – The USB-C That Does It All | Thunderbolt Technology Community" (https://thunderbolttechnology.net/blog/thunderbolt-3-usb-c-does-it-all). Thunderbolttechnology.net. Archived (https://web.archive.org/web/20150905185714/https://thunderbolttechnology.net/blog/thunderbolt-3-usb-c-does-it-all) from the original on 2015-09-05. Retrieved 2015-06-18.
39. "One port to rule them all: Thunderbolt 3 and USB Type-C join forces" (https://www.cnet.com/news/thunderbolt-3-and-usb-type-c-join-forces-for-one-port-to-rule-them-all/). Archived (https://web.archive.org/web/20150602195337/http://www.cnet.com/news/thunderbolt-3-and-usb-type-c-join-forces-for-one-port-to-rule-them-all/) from the original on 2015-06-02. Retrieved 2015-06-02.
40. "Thunderbolt 3 is twice as fast and uses reversible USB-C" (https://www.engadget.com/2015/06/02/thunderbolt-3-usb-c/). 2015-06-02. Archived (https://web.archive.org/web/20150603000428/http://www.engadget.com/2015/06/02/thunderbolt-3-usb-c/) from the original on 2015-06-03. Retrieved 2015-06-02.
41. Anthony, Sebastian (2015-06-02). "Thunderbolt 3 embraces USB Type-C connector, doubles bandwidth to 40Gbps" (http://arstechnica.co.uk/gadgets/2015/06/thunderbolt-3-embraces-usb-type-c-connector-doubles-bandwidth-to-40gbps). *Ars Technica UK*. Archived (https://web.archive.org/web/20150603045124/http://arstechnica.co.uk/gadgets/2015/06/thunderbolt-3-embraces-usb-type-c-connector-doubles-bandwidth-to-40gbps/) from the original on 2015-06-03. Retrieved 2015-06-02.
42. "VESA Brings DisplayPort to New USB Type-C Connector" (http://www.displayport.org/pr/vesa-brings-displayport-new-usb-type-c-connector/). DisplayPort. 2014-09-22. Archived (https://web.archive.org/web/20151005182809/http://www.displayport.org/pr/vesa-brings-displayport-new-usb-type-c-connector/) from the original on 2015-10-05. Retrieved 2015-06-18.
43. "DisplayPort Alternate Mode on USB-C – Technical Overview" (https://web.archive.org/web/20161220080944/http://www.usb.org/developers/presentations/USB_DevDays_Hong_Kong_2016_-_DisplayPort_Alternate_Mode.pdf) (PDF). *usb.org*. USB-IF. 2016-10-20. Archived from the original (http://www.usb.org/developers/presentations/USB_DevDays_Hong_Kong_2016_-_DisplayPort_Alternate_Mode.pdf) (PDF) on 2016-12-20.
44. "VESA Releases Updated DisplayPort Alt Mode Spec to Bring DisplayPort 2.0 Performance to USB4 and New USB Type-C Devices" (https://vesa.org/featured-articles/vesa-releases-updated-displayport-alt-mode-spec-to-bring-displayport-2-0-performance-to-usb4-and-new-usb-type-c-devices/). *VESA – Interface Standards for The Display Industry*. 2020-04-29. Archived (https://web.archive.org/web/20211123012349/https://vesa.org/featured-articles/vesa-releases-updated-displayport-alt-mode-spec-to-bring-displayport-2-0-performance-to-usb4-and-new-usb-type-c-devices/) from the original on 2021-11-23. Retrieved 2020-10-01.
45. "MHL – Expand Your World" (http://mhltech.org/usbtype-c.aspx). *MHLTech.org*. Archived (https://web.archive.org/web/20151002053612/http://mhltech.org/usbtype-c.aspx) from the original on 2015-10-02. Retrieved 2015-06-18.
46. "MHL Alternate Mode reference design for superMHL over USB Type-C" (https://web.archive.org/web/20170604024210/http://www.anandtech.com/show/10146/lattice-and-mediatek-to-enable-4k-outputs-on-smartphones-using-usbc). *AnandTech.com*. 2016-03-15. Archived from the original (http://www.anandtech.com/show/10146/lattice-and-mediatek-to-enable-4k-outputs-on-smartphones-using-usbc) on 2017-06-04. Retrieved 2015-06-18.
47. "MHL Releases Alternate Mode for New USB Type-C Connector" (http://mhltech.org/news/PressReleaseDetail.aspx?id=2147501807). *MHLTech.org*. 2014-11-17. Archived (https://web.archive.org/web/20141129022228/http://www.mhltech.org/news/PressReleaseDetail.aspx?id=2147501807) from the original on 2014-11-29. Retrieved 2015-06-18.
48. "MHL Alternate Mode over USB Type-C to support superMHL" (http://www.mhltech.org/PressReleaseDetail.aspx?id=5581). *www.mhltech.org*. 2015-01-06. Archived (https://web.archive.org/web/20161115193331/http://www.mhltech.org/PressReleaseDetail.aspx?id=5581) from the original on 2016-11-15. Retrieved 2016-11-15.
49. "MHL Alt Mode: Optimizing Consumer Video Transmission" (https://web.archive.org/web/20160914061910/http://www.usb.org/developers/presentations/USB_DevDays_Taipei_2015_-_MHL_Alt_Mode_on_USB_Type-C.pdf) (PDF). *usb.org*. MHL, LLC. 2015-11-18. Archived from the original (http://www.usb.org/developers/presentations/USB_DevDays_Taipei_2015_-_MHL_Alt_Mode_on_USB_Type-C.pdf) (PDF) on 2016-09-14.
50. "HDMI Press Release: HDMI Releases Alternate Mode for USB Type-C Connector" (http://www.hdmi.org/press/press_release.aspx?prid=144). *hdmi.org*. Archived (https://web.archive.org/web/20181224041848/https://www.hdmi.org/press/press_release.aspx?prid=144) from the original on 2018-12-24. Retrieved 2016-09-10.
51. "HDMI LLC – HDMI Over USB Type-C" (https://web.archive.org/web/20170218212050/http://www.usb.org/developers/presentations/USB_DevDays_Hong_Kong_2016_-_HDMI_Alt_Mode_USB_Type-C.pdf) (PDF). *usb.org*. HDMI LLC. 2016-10-20. Archived from the original (http://www.usb.org/developers/presentations/USB_DevDays_Hong_Kong_2016_-_HDMI_Alt_Mode_USB_Type-C.pdf) (PDF) on 2017-02-18.

52. "HDMI Alt Mode for USB Type-C Announced" (https://web.archive.org/web/20160915180623/http://www.anandtech.com/show/10642/hdmi-alt-mode-for-usb-typec-announced). *anandtech.com*. Archived from the original (http://www.anandtech.com/show/10642/hdmi-alt-mode-for-usb-typec-announced) on 2016-09-15. Retrieved 2016-09-10.
53. "A new standard will allow your USB-C devices to connect to HDMI" (https://www.neowin.net/news/a-new-standard-will-allow-your-usb-c-devices-to-connect-to-hdmi). *neowin.net*. Archived (https://web.archive.org/web/20160924075346/https://www.neowin.net/news/a-new-standard-will-allow-your-usb-c-devices-to-connect-to-hdmi) from the original on 2016-09-24. Retrieved 2016-09-10.
54. "HDMI Alt Mode for USB Type-C Connector" (http://www.hdmi.org/manufacturer/HDMIAltModeUSBTypeC.aspx). *hdmi.org*. Archived (https://web.archive.org/web/20160919043159/http://www.hdmi.org/manufacturer/HDMIAltModeUSBTypeC.aspx) from the original on 2016-09-19. Retrieved 2016-09-10.
55. "New Open Industry Standard Introduced for Connecting Next-Generation VR Headsets to PCs, Other Devices" (http://www.globenewswire.com/news-release/2018/07/17/1538342/0/en/New-Open-Industry-Standard-Introduced-for-Connecting-Next-Generation-VR-Headsets-to-PCs-Other-Devices.html). *GlobeNewswire News Room* (Press release). 2018-07-17. Archived (https://web.archive.org/web/20190220081452/https://globenewswire.com/news-release/2018/07/17/1538342/0/en/New-Open-Industry-Standard-Introduced-for-Connecting-Next-Generation-VR-Headsets-to-PCs-Other-Devices.html) from the original on 2019-02-20. Retrieved 2019-03-12.
56. Smith, Ryan (2018-07-17). "VirtualLink USB-C Alt Mode Announced: Standardized Connector for VR Headsets" (https://web.archive.org/web/20180821160558/https://www.anandtech.com/show/13088/virtuallink-announced-standardized-connector-for-vr-headsets). *AnandTech*. Archived from the original (https://www.anandtech.com/show/13088/virtuallink-announced-standardized-connector-for-vr-headsets) on 2018-08-21. Retrieved 2018-08-21.
57. "[802.3_DIALOG] USB-C Ethernet Alternate Mode" (https://grouper.ieee.org/groups/802/3/email_dialog/msg00262.html). ieee. 2015-03-26. Archived (https://web.archive.org/web/20161003213456/http://grouper.ieee.org/groups/802/3/email_dialog/msg00262.html) from the original on 2016-10-03. Retrieved 2015-11-06.
58. "Thunderbolt – the USB-C that does it all" (https://www.intel.com/content/www/in/en/products/docs/io/thunderbolt/thunderbolt-technology-developer.html). Archived (https://web.archive.org/web/20220910202625/https://www.intel.in/content/www/in/en/homepage.html) from the original on 2022-09-10. Retrieved 2021-01-17.
59. "TECHNOLOGY BRIEF Thunderbolt 3" (https://thunderbolttechnology.net/sites/default/files/Thunderbolt3_TechBrief_FINAL.pdf) (PDF). 2018-09-21. Archived (https://web.archive.org/web/20181224041848/https://thunderbolttechnology.net/sites/default/files/Thunderbolt3_TechBrief_FINAL.pdf) (PDF) from the original on 2018-12-24. Retrieved 2018-09-20.
60. "Node Pro" (https://www.akitio.com/expansion/node-pro). *AKiTiO*. 2018-09-21. Archived (https://web.archive.org/web/20180921034437/https://www.akitio.com/expansion/node-pro) from the original on 2018-09-21. Retrieved 2018-09-20.
61. VESA DisplayPort Alt Mode for USB Type-C Standard, Feature Summary (https://www.vesa.org/wp-content/uploads/2014/09/DP-Alt-Mode-Overview-for-VESA-v1.pdf), Sept 22, 2014, accessed on line July 26, 2025.
62. "VESA Releases Updated DisplayPort Alt Mode Spec to Bring DisplayPort 2.0 Performance to USB4 and New USB Type-C Devices" (https://vesa.org/featured-articles/vesa-releases-updated-displayport-alt-mode-spec-to-bring-displayport-2-0-performance-to-usb4-and-new-usb-type-c-devices/). *VESA*. 2020-04-29. Archived (https://web.archive.org/web/20211123012349/https://vesa.org/featured-articles/vesa-releases-updated-displayport-alt-mode-spec-to-bring-displayport-2-0-performance-to-usb4-and-new-usb-type-c-devices/) from the original on 2021-11-23. Retrieved 2021-08-09.
63. CES 2023 | The demise of HDMI over USB-C (Alt Mode) and more power in cables (https://www.notebookcheck.net/The-demise-of-HDMI-over-USB-C-Alt-Mode-and-more-power-in-cables.680552.0.html), Andreas Sebayang (tr. Zhiwei (Peter) Zhuang), Notebookcheck, Jan. 11, 2023. Accessed on line July 26, 2025.
64. "USB Logo Usage Guidelines" (https://web.archive.org/web/20161220103017/http://www.usb.org/developers/logo_license/USB-IF_TLA_Usage_Guidelines_113016.pdf) (PDF). *usb.org*. USB-IF. 2016-03-11. Archived from the original (http://www.usb.org/developers/logo_license/USB-IF_TLA_Usage_Guidelines_113016.pdf) (PDF) on 2016-12-20.
65. "CalDigit USB-C Cable" (http://www.caldigit.com/Thunderbolt-3-USB-C-Type-C-Charge-Cable/). 2018-09-21. Archived (https://web.archive.org/web/20180921114714/http://www.caldigit.com/Thunderbolt-3-USB-C-Type-C-Charge-Cable/) from the original on 2018-09-21. Retrieved 2018-09-21.
66. "VESA Brings DisplayPort to New USB Type-C Connector" (http://www.vesa.org/news/vesa-brings-displayport-to-new-usb-type-c-connector/). *www.vesa.org*. Archived (https://web.archive.org/web/20170202050221/http://www.vesa.org/news/vesa-brings-displayport-to-new-usb-type-c-connector/) from the original on 2017-02-02. Retrieved 2016-12-11.
67. "Termination Resistors Required for the USB Type-C Connector – KBA97180" (https://community.cypress.com/docs/DOC-10779). 2015-04-17. Archived (https://web.archive.org/web/20190719031522/https://community.cypress.com/docs/DOC-10779) from the original on 2019-07-19. Retrieved 2019-07-19.
68. "USB Charger (USB Power Delivery) | USB-IF" (https://www.usb.org/usb-charger-pd).
69. *Universal Serial Bus Type-C Cable and Connector Specification* Revision 1.3 (14 July 2017), section 2.4, page 26.
70. *Universal Serial Bus Type-C Cable and Connector Specification* Revision 1.3 (14 July 2017), section 5.1.2, page 203.
71. *Universal Serial Bus Type-C Cable and Connector Specification* Revision 1.3 (14 July 2017), section A, page 213.
72. "Android – Marshmallow" (https://www.android.com/versions/marshmallow-6-0/). Archived (https://web.archive.org/web/20190609104214/https://www.android.com/versions/marshmallow-6-0/) from the original on 2019-06-09. Retrieved 2015-10-12.
73. "Charge your Chromebook Pixel (2015)" (https://support.google.com/chromebook/answer/6164519?hl=en%2F). Archived (https://web.archive.org/web/20190711152950/https://support.google.com/chromebook/answer/6164519?hl=en%2F) from the original on 2019-07-11. Retrieved 2015-10-31.

74. "FreeBSD 8.2 Release Notes" (https://www.freebsd.org/releases/8.2R/relnotes-detailed.html). *www.freebsd.org*. 2011-04-22. Archived (https://web.archive.org/web/20180217032915/https://www.freebsd.org/releases/8.2R/relnotes-detailed.html) from the original on 2018-02-17. Retrieved 2018-02-05.
75. "NetBSD 7.2 Released" (https://www.netbsd.org/releases/formal-7/NetBSD-7.2.html). Archived (https://web.archive.org/web/20190115075746/https://www.netbsd.org/releases/formal-7/NetBSD-7.2.html) from the original on 2019-01-15. Retrieved 2019-01-14.
76. "OpenBSD 5.7" (https://www.openbsd.org/57.html). Archived (https://web.archive.org/web/20180927233828/http://www.openbsd.org/57.html) from the original on 2018-09-27. Retrieved 2019-06-27.
77. "Using the USB-C port and adapters on your MacBook (Retina, 12-inch, Early 2015) – Apple Support" (https://support.apple.com/en-us/HT204360). *Support.Apple.com*. 2015-05-28. Archived (https://web.archive.org/web/20150919050144/https://support.apple.com/en-us/HT204360) from the original on 2015-09-19. Retrieved 2015-06-18.
78. Microsoft. "Update for USB Type-C billboard support and Kingston thumb drive is enumerated incorrectly in Windows" (https://support.microsoft.com/en-us/kb/3103696). Archived (https://web.archive.org/web/20151210232920/https://support.microsoft.com/en-us/kb/3103696) from the original on 2015-12-10. Retrieved 2015-12-08.
79. Microsoft. "Windows support for USB Type-C connectors" (https://msdn.microsoft.com/en-us/library/windows/hardware/mt427381.aspx). *Microsoft MSDN*. Archived (https://web.archive.org/web/20170624144842/https://msdn.microsoft.com/en-us/library/windows/hardware/mt427381.aspx) from the original on 2017-06-24. Retrieved 2015-09-30.
80. "USB Dual Role Driver Stack Architecture – Windows drivers" (https://docs.microsoft.com/en-us/windows-hardware/drivers/usbcon/usb-dual-role-driver-stack-architecture). *docs.microsoft.com*. Archived (https://web.archive.org/web/20181101095519/https://docs.microsoft.com/en-us/windows-hardware/drivers/usbcon/usb-dual-role-driver-stack-architecture) from the original on 2018-11-01. Retrieved 2019-06-21.
81. Burke, Steve (2019-03-25). "Why USB 3.1 Type-C Isn't on More Cases & Cable Factory Tour in Dongguan, China" (https://www.gamersnexus.net/guides/3453-why-usb-type-c-isnt-on-more-cases). *Gamers Nexus*. Archived (https://web.archive.org/web/20190626032133/https://www.gamersnexus.net/guides/3453-why-usb-type-c-isnt-on-more-cases) from the original on 2019-06-26. Retrieved 2019-06-26.
82. "DisplayPort over USB-C" (https://www.displayport.org/displayport-over-usb-c/). *DisplayPort*. Archived (https://web.archive.org/web/20191127034723/https://www.displayport.org/displayport-over-usb-c/) from the original on 2019-11-27. Retrieved 2019-10-28.
83. Mills, Chris (2015-11-16). "A Google Engineer Is Publicly Shaming Crappy USB-C Cables" (https://gizmodo.com/a-google-engineer-is-publicly-shaming-crappy-usb-c-cabl-1742719818). Archived (https://web.archive.org/web/20171024043323/https://gizmodo.com/a-google-engineer-is-publicly-shaming-crappy-usb-c-cabl-1742719818) from the original on 2017-10-24. Retrieved 2017-10-23.
84. Opam, Kwame (2015-11-05). "A Google Engineer is Testing USB Type-C Cables So You Don't Have To" (https://www.theverge.com/2015/11/5/9674462/usb-type-c-google-engineer-third-party-cables). *The Verge*. Archived (https://web.archive.org/web/20171024043417/https://www.theverge.com/2015/11/5/9674462/usb-type-c-google-engineer-third-party-cables) from the original on 2017-10-24. Retrieved 2017-10-23.
85. "Be careful about which USB-C cables you buy off the Internet" (https://www.technobuffalo.com/usb-c-cable-safety-what-to-buy). *TechnoBuffalo*. 2015-11-16. Archived (https://web.archive.org/web/20191216011531/https://www.technobuffalo.com/usb-c-cable-safety-what-to-buy) from the original on 2019-12-16. Retrieved 2019-12-16.
86. Bohn, Dieter (2016-02-04). "Laptops are getting destroyed by cheap USB-C cables" (https://www.theverge.com/2016/2/4/10916264/usb-c-russian-roulette-power-cords). *The Verge*. Archived (https://web.archive.org/web/20171024095512/https://www.theverge.com/2016/2/4/10916264/usb-c-russian-roulette-power-cords) from the original on 2017-10-24. Retrieved 2017-10-23.
87. *Universal Serial Bus Type-C Cable and Connector Specification* Revision 1.1 (3 April 2015), page 60, table 3–13, note 1.
88. Leswing, Kif (2015-11-05). "Google Engineer Reviews Defective USB Cables on Amazon – Fortune" (http://fortune.com/2015/11/05/usbc-google-amazon-reviews/). *Fortune*. Archived (https://web.archive.org/web/20170502191502/http://fortune.com/2015/11/05/usbc-google-amazon-reviews/) from the original on 2017-05-02. Retrieved 2015-11-08.
89. "In response to the Type-C cable discussions" (https://forums.oneplus.com/threads/in-response-to-the-type-c-cable-discussions.412344/). *OnePlus Community*. Archived (https://web.archive.org/web/20191216011536/https://forums.oneplus.com/threads/in-response-to-the-type-c-cable-discussions.412344/) from the original on 2019-12-16. Retrieved 2019-12-16.
90. "TCPP01-M12 Type-C Port Protection" (https://www.st.com/content/ccc/resource/sales_and_marketing/presentation/product_presentation/group0/d0/77/b1/1c/7c/7d/4b/66/TCPP01-M12_USB_Type-C_Port_Protection/files/TCPP01-M12_USB_Type-C_Port_Protection.pdf/jcr:content/translations/en.TCPP01-M12_USB_Type-C_Port_Protection.pdf) (PDF). Archived (https://web.archive.org/web/20191203132423/https://www.st.com/content/ccc/resource/sales_and_marketing/presentation/product_presentation/group0/d0/77/b1/1c/7c/7d/4b/66/TCPP01-M12_USB_Type-C_Port_Protection/files/TCPP01-M12_USB_Type-C_Port_Protection.pdf/jcr:content/translations/en.TCPP01-M12_USB_Type-C_Port_Protection.pdf) (PDF) from the original on 2019-12-03. Retrieved 2019-12-03.
91. "USB-C audio: Everything you need to know" (https://www.androidcentral.com/usb-c-audio). *Android Central*. 2018-05-02. Archived (https://web.archive.org/web/20190224231530/https://www.androidcentral.com/usb-c-audio) from the original on 2019-02-24. Retrieved 2019-02-24.
92. "Bring back the headphone jack: Why USB-C audio still doesn't work" (https://www.pcworld.com/article/3284186/bring-back-the-headphone-jack-why-usb-c-audio-still-doesnt-work.html). *PCWorld*. 2018-09-10. Archived (https://web.archive.org/web/20190302233431/https://www.pcworld.com/article/3284186/bring-back-the-headphone-jack-why-usb-c-audio-still-doesnt-work.html) from the original on 2019-03-02. Retrieved 2019-03-12.
93. T, Nick (2015-02-26). "Android 5.0 Lollipop supports USB DAC audio devices, we go ears-on" (https://www.phonearena.com/news/Android-5.0-Lollipop-supports-USB-DAC-audio-devices-we-go-ears-on_id66399). *Phone Arena*. Archived (https://web.archive.org/web/20190225162415/https://www.phonearena.com/news/Android-5.0-Lollipop-supports-USB-DAC-audio-devices-we-go-ears-on_id66399) from the original on 2019-02-25. Retrieved 2019-02-25.

94. Schoon, Ben (2018-11-01). "Hands-on: Apple's new USB-C headphone adapter is your cheapest option for analog audio on Pixel" (https://9to5google.com/2018/11/01/hands-on-apple-usb-c-headphone/). Archived (https://web.archive.org/web/20190225162626/https://9to5google.com/2018/11/01/hands-on-apple-usb-c-headphone/) from the original on 2019-02-25. Retrieved 2019-02-25.
95. "Google engineer warns USB-C, Qualcomm Quick Charge are incompatible – ExtremeTech" (https://www.extremetech.com/mobile/227188-google-engineer-warns-usb-type-c-qualcomm-quick-charge-are-fundamentally-incompatible). 2016-04-25. Archived (https://web.archive.org/web/20171026205646/http://www.extremetech.com/mobile/227188-google-engineer-warns-usb-type-c-qualcomm-quick-charge-are-fundamentally-incompatible) from the original on 2017-10-26. Retrieved 2017-10-23.
96. "Qualcomm says it's fine to fast-charge your phone over USB-C" (https://www.engadget.com/2016/04/24/qualcomm-defends-quick-charge-over-usb-c/). *Engadget*. 2016-04-25. Archived (https://web.archive.org/web/20171024043444/https://www.engadget.com/2016/04/24/qualcomm-defends-quick-charge-over-usb-c/) from the original on 2017-10-24. Retrieved 2017-10-23.
97. "Qualcomm Quick Charge 4: Five minutes of charging for five hours of battery life" (https://www.qualcomm.com/news/onq/2016/11/17/qualcomm-quick-charge-4-five-minutes-charging-five-hours-battery-life). *Qualcomm*. 2016-11-17. Archived (https://web.archive.org/web/20171024095508/https://www.qualcomm.com/news/onq/2016/11/17/qualcomm-quick-charge-4-five-minutes-charging-five-hours-battery-life) from the original on 2017-10-24. Retrieved 2017-10-23.
98. "Apple opposes EU plans to make common charger port for all devices" (https://www.theguardian.com/world/2021/sep/23/apple-opposes-eu-plans-to-make-common-charger-port-for-all-devices). *The Guardian*. 2021-09-23. Archived (https://web.archive.org/web/20211018140109/https://www.theguardian.com/world/2021/sep/23/apple-opposes-eu-plans-to-make-common-charger-port-for-all-devices) from the original on 2021-10-18. Retrieved 2021-10-19.
99. Peltier, Elian (2021-09-23). "In a setback for Apple, the European Union seeks a common charger for all phones" (https://www.nytimes.com/2021/09/23/business/european-union-apple-charging-port.html). *The New York Times*. Archived (https://web.archive.org/web/20220910202629/https://www.nytimes.com/2021/09/23/business/european-union-apple-charging-port.html) from the original on 2022-09-10. Retrieved 2021-10-19.
100. "One common charging solution for all" (https://ec.europa.eu/growth/sectors/electrical-engineering/red-directive/common-charger_en). *Internal Market, Industry, Entrepreneurship and SMEs – European Commission*. 2016-07-05. Archived (https://web.archive.org/web/20211019205753/https://ec.europa.eu/growth/sectors/electrical-engineering/red-directive/common-charger_en) from the original on 2021-10-19. Retrieved 2021-10-19.
101. Riley, Duncan (2022-10-04). "European Parliament votes to compel USB-C charging on all portable devices" (https://siliconangle.com/2022/10/04/european-parliament-votes-compel-usb-c-charging-portable-devices/). *SiliconANGLE*. Retrieved 2022-10-06.
102. "Long-awaited common charger for mobile devices will be a reality in 2024" (https://www.europarl.europa.eu/news/en/press-room/20220930IPR41928/long-awaited-common-charger-for-mobile-devices-will-be-a-reality-in-2024). *European Parliament News*. European Parliament. 2022-10-04. Retrieved 2022-10-06.
103. Porter, Jon (2022-10-04). "When will the iPhone be forced to use USB-C?" (https://www.theverge.com/2022/10/4/23387425/european-parliament-iphone-usb-type-c-radio-equipment-directive-lightning-rip). *The Verge*. The Verge. Retrieved 2022-10-06.
104. Warren, Tom (2022-10-04). "The first USB-C iPhone is here thanks to a mod" (https://www.theverge.com/2021/10/12/22722123/first-iphone-usb-c-port-robotics-engineering-student-custom). *The Verge*. The Verge. Retrieved 2021-10-12.
105. "EU common charger rules: Power all your devices with a single charger - European Commission" (https://commission.europa.eu/news-and-media/news/eu-common-charger-rules-power-all-your-devices-single-charger-2024-12-28_en). *commission.europa.eu*. Retrieved 2026-05-11.
106. Kroet, Cynthia (2024-12-27). "New EU rules on common USB-C chargers for electronics to come into force" (https://www.euronews.com/next/2024/12/27/new-eu-rules-on-common-usb-c-chargers-for-electronics-to-come-into-force). Euronews.
107. Shilov, Anton. "USB-C Authentication Tech to Restrict Usage of Uncertified USB-C Accessories and Cables" (https://web.archive.org/web/20160414072947/http://www.anandtech.com/show/10251/usbc-authentication-announced). *AnandTech*. Archived from the original (https://www.anandtech.com/show/10251/usbc-authentication-announced) on 2016-04-14. Retrieved 2022-06-12.
108. Hollister, Sean (2022-06-06). "Apple will make USB-C accessories ask for your permission to pass data" (https://www.theverge.com/2022/6/6/23157271/apple-macbook-m1-m2-usb-thunderbolt-ports-closed-user-approval). *The Verge*. Retrieved 2022-06-12.
109. Li, Abner (2019-01-02). "USB-C Authentication Program could let OEMs clampdown on 'non-compliant' USB chargers" (https://9to5google.com/2019/01/02/usb-c-authentication-program-chargers/). *9to5Google*. Retrieved 2022-06-12.


## External links

- The *Universal Serial Bus Type-C Cable and Connector Specification* is included in a set of USB documents which can be downloaded from USB.org (https://www.usb.org/documents).
- Introduction to USB Type-C (https://ww1.microchip.com/downloads/en/appnotes/00001953a.pdf), by Andrew Rogers, Microchip Technology, 2015